

The Invisible Trillion

A BOOK FOR CURIOUS MINDS

The Invisible Trillion

A book about chips, data centers, and the things you can't see

MEIRA DAVIDSON



BOOK ONE OF THE CURIOUS FAMILY LIBRARY

The Invisible Trillion

A book about chips, data centers, and the things you can't see

Book One of the Curious Family Library

Copyright © 2026 Meira Davidson.

All rights reserved.

This is a work of nonfiction. Names of public companies, products, and people are used in the ordinary course of reporting and analysis.

First edition, 2026.

*For the Jessica family kids —
be curious, stay open, and be kind.*

A Note to the Reader

A while ago, a kid asked her mom whether data centers really use too much water. It's a question that's been going around online. The mom asked me, and I didn't have a good answer. I use a lot of AI, but the honest reply to a direct question about it was *I'd have to go look it up*. So I went and looked it up. Then I kept going, because once you pull on one of these threads the whole sweater comes along with it.

The water question pulls on the energy question, which pulls on how chips are made, which pulls on where data centers get built, which pulls on what they're built *for*, which pulls on who decides any of it, which pulls on who pays when the deciders get it wrong. I set out to write you a short answer about water. I ended up writing the book you're holding.

I want to tell you how it changed while I was writing it, because the change is part of what the book is now.

When I started, I expected the answer to *is all of this worth it to be yes, mostly, with caveats*. The technology is real. The benefits are real. We are at a genuine inflection point, and some of the people building this will be remembered for a long time. The first several chapters are written mostly from inside that frame, and you can read them on their own if that's the mood you're in.

But the further I got into how the buildout is actually being executed — *where* the data centers go, *who* pays for them, *what* legal tools get used against the towns that host them, and *how much* of it is justified by real, durable demand rather than speculation that may never pay off — the harder it was to keep writing only from that frame. The honest version of the story includes things I hadn't known when I started, and things the industry mostly doesn't volunteer. I had to include them. Leaving them out would have handed you a book that flattered the industry instead of one that

taught you to think clearly about the world you're inheriting. You deserve the second kind.

So the middle of the book isn't comfortable. There are chapters about neighborhoods damaged by infrastructure they never chose to host; about retirement savings — your grandparents', your parents' — being poured into a buildout that may or may not pay off; about the legal machinery used to override ordinary people who object to an industrial site going up next to their homes. I've tried to make those chapters readable. I can't make the situation they describe pleasant, because it isn't, and the people living through it aren't having a pleasant time.

What I can do is tell you, before you start, what the book asks of you. It asks you to hold several things at once. The technology is real, the productivity gains are real, the stakes are real — *and* the costs are real, the communities are real, the people losing things are real. The people who built this industry are real human beings who have done remarkable things, *and* they have made choices that, looked at honestly, have hurt the communities that hosted their decisions. Both are true. The book doesn't ask you to choose between them. It asks you to hold both, because that is what seeing the situation honestly requires.

A few practical notes on how to read it.

If you want the answer to the water question that started all this, it's in the water chapter — chapter six.

If you want to understand why your parents may have been grumbling about the electricity bill — yes, that's real, and yes, it's rising for reasons that have nothing to do with you leaving the lights on — read the energy chapter, chapter five.

If you read only the chapters about people, read the three at the heart of the book — *The Highway in the Bronx*, *The Legal Weapons*, and *How a Neighborhood Dies*. They are the ones I'd most want an adult to read. If you're reading this in 2026, they may be describing a community you or someone you know lives in.

The middle stretch — three chapters together — is the analytical part: how much of the buildout real demand justifies, what the right architecture for AI infrastructure would actually look like, and who is paying the bill. Those are denser than the rest. Read them or skip them; if you read them, you'll close the book able to read the news about AI more clearly than most adults who have opinions about it. If you skip them, come back when you're older.

One chapter steps back to ask what all this is for — the medicines, the discoveries, the access the chips make possible. The last two are about who decides any of it, and what can be done about it.

— *Meira*

Contents

♦ A Note to the Reader	7
1 The Invisible Trillion	13
2 Sand, Light, and the World's Most Precise Factories	23
3 What Chips Actually Do	37
4 Data Centers: The Cathedrals Nobody Visits	49
5 The Energy Question, Honestly	61
6 The Water Question, Honestly	79
7 The Mining and E-Waste Question, Honestly	89
8 The Highway in the Bronx	103
9 The Legal Weapons	115
10 How a Neighborhood Dies	125
11 No Going Back	135
12 How Much Do We Actually Need?	145
13 The Architecture That Works	187
14 The Bill	203
15 The Other Ledger	217
16 The Attention Chapter	231
17 Who Decides	241
18 What You Can Do That Actually Matters	253

CHAPTER 1

The Invisible Trillion

*Why You Have Strong Opinions About a Thing You've Never
Seen*



Look at the phone closest to you. Pick it up if you can.

The chip at the center of that phone — the one doing almost all of the actual thinking — is about the size of a fingernail. Inside that fingernail-sized chip are somewhere between ten and twenty billion transistors, depending on which phone you have.

Ten billion is a number that does not mean anything to humans. Our brains did not evolve for it. So here are some numbers it equals, to give you a feel:

- The number of people on Earth, times one and a half.
- The number of heartbeats in three average human lifetimes.
- The number of seconds in about 300 years.

All of that, in your pocket. Working right now. Perfectly. Every one of those transistors switching billions of times per second, every single one of them smaller than a virus.

Now zoom out. There are a few billion smartphones on Earth, each with a chip like this one. There are also computers, tablets, cars (the newest cars have around 1,500 chips each), traffic lights, hospital monitors, game consoles, satellites, refrigerators, dishwashers, and — importantly for the rest of this book — a few tens of

millions of specialized chips in data centers. All told, humans have manufactured, cumulatively since the 1960s, somewhere around **ten sextillion transistors**.

A sextillion is a one followed by 21 zeros. Ten sextillion is a ten followed by 21 zeros.

For comparison: the number of grains of sand on every beach on Earth is somewhere around seven and a half quintillion — a seven and a half followed by 18 zeros. **We have made, by our best estimates, more than a thousand transistors for every grain of beach sand on Earth.** And we make about a trillion more every second.

This is not a metaphor. This is arithmetic.



The Most Impressive Thing You've Never Looked At

Here is a fact that took me a long time to really absorb, and which might be worth pausing on.

Of all the objects humans make — buildings, cars, planes, bridges, satellites, particle accelerators — the most precisely manufactured thing, by a wide margin, is a modern computer chip.

Not somewhat more precise. Not 10% more precise. *Thousands of times* more precise than anything else we make.

The features on a current high-end chip are about 3 nanometers across. A nanometer is a billionth of a meter. Three nanometers is roughly fifteen silicon atoms in a row. If you enlarged one of these transistors to the size of a grain of rice, the chip it came from would be about the size of Manhattan. Every single one of the fifteen billion transistors on that Manhattan-sized chip would have to be placed correctly, wired correctly, and working correctly, or the chip doesn't work at all.

We make a trillion of these a year.

The factories that make them — “fabs,” short for fabrication plants — are the most expensive buildings humans have ever built for their size. A new one costs around \$20 billion, which is about

twice what it cost to build the tallest building in the world. Inside the fab, the air is about a thousand times cleaner than in a hospital operating room, because a single dust particle landing on a wafer in the wrong place ruins the chip. Workers wear suits that make them look like astronauts, except the suits are to protect the chips from the humans, not the humans from the chips.

The machines inside the fab that actually draw the transistors onto the silicon are, themselves, the most complicated machines humans have ever built. One of them — called an EUV lithography machine — contains 100,000 precisely-machined parts, costs around \$350 million, weighs 180 tons, and is built by exactly one company in the world, in the Netherlands, that has a decade-long waiting list. These machines shoot lasers at drops of molten tin to produce a specific color of light that doesn't exist anywhere in nature, and they use that light to draw the features on your phone's chip.

All of this exists. None of it is science fiction. It is all happening, somewhere, right now, as you read this sentence.

And almost no one has ever seen it.



The Invisibility Problem

Here is what's strange about all of this.

If you grew up in 1925, the most impressive machines in your life were visible. The ocean liners were enormous. The locomotives shook the ground. The dams were on the news. The factories had smokestacks you could see from ten miles away. When your grandmother talked about “modern industry,” she was pointing at something: it was *there*, it was *big*, it *moved* and it *smoked*.

If you grew up in 2012, the most impressive machines in your life were invisible. They were in a building in Oregon, or Ireland, or Singapore, that you have never been to and will never be to. The building does not have a smokestack. It has no windows. It does not move. Most of it is not even labeled on the outside. You cannot take

a field trip there, not because it's secret, but because there is nothing to see — just rows of identical black rectangles in a cold room, humming.

The chips themselves are smaller than your thumbnail. They are sealed inside plastic cases inside bigger metal cases inside cloud racks inside buildings you don't know the location of. The whole stack — from the atoms in the silicon up to the server farm you're talking to when you open an app — is completely and totally invisible.

This matters more than it seems to.

Humans evolved to form opinions about things we can see and smell and hear. Our intuitions about what is powerful, what is dangerous, what is polluting, what is wasteful, what is precious — all of those intuitions were calibrated in a world where the stuff we were judging was right in front of us. A smoking factory *looks* bad. A concrete dam *looks* powerful. A pile of garbage *looks* wasteful. A river on fire — which was an actual thing in the United States in 1969 — *looks* like something you should do something about.

A data center looks like a beige warehouse.

The chip in your phone looks like nothing. It looks like a smooth black rectangle in a plastic shell inside a metal case inside your pocket. It is doing more computation per second than every computer in the world combined could do in 1985, and it looks like nothing.

This is why the arguments about technology — about AI, about data centers, about energy, about privacy, about everything this book is going to get into — are so weirdly disconnected from reality. *We are arguing about things we cannot see.* We have no physical intuitions about them. We have no grandmother-pointing-at-the-factory version of how to think about them. So the arguments fill up with feelings and vibes and TikTok videos, because feelings and vibes and TikTok videos are what we have when we don't have direct experience.

The first job of this book is just to get you to *see* some of this stuff. Not literally — you’re not going to tour a fab, probably, unless you grow up to work in one. But to build up the mental images, the scale comparisons, the physical intuitions that you would have had if this whole industry were visible. Once you have those, the arguments stop being about vibes and start being about something real.



Why This Book Is Worth Reading

You already have opinions about the stuff in this book.

You have opinions about AI. You have opinions about data centers. Probably you have opinions about social media, and about screen time, and about whether “Big Tech” is good or bad for the world. These opinions came from somewhere — from TikTok, from your friends, from a documentary, from a teacher, from a parent. They might be right. They might be completely wrong. You don’t actually know, and that’s not your fault, because the stuff these opinions are *about* is invisible, and you’ve been asked to have opinions about it anyway.

Here is what I am going to ask of you for the next ten chapters.

I am going to ask you to hold those opinions a little more loosely than usual. Not abandon them. Not pretend you don’t have them. Just — loosen your grip. Look at the actual numbers, the actual physics, the actual tradeoffs. See what survives contact with reality.

Some of your opinions will survive. Some of them will probably change. I don’t know which ones, and neither do you, yet. That’s the point.

I am also going to make you a promise in return. I am not going to pretend things are okay when they are not. If the water use is a real problem, I am going to tell you the water use is a real problem. If the mining is a moral disaster, I am going to tell you the mining is a moral disaster. If the companies are being dishonest about their emissions, I am going to say they are being dishonest about their

emissions. You have probably already encountered, in your life, adults who say “don’t worry, it’s fine” about things that are not fine. I am not going to be one of them. The whole point of the book is to tell you what is actually going on, including the parts where what’s going on is genuinely bad.

The deal, in other words: I take your worries seriously and give you the real numbers, and in exchange you give me ten chapters before you conclude anything.

Fair?



A Short Map of What’s Coming

The rest of the book is organized like this.

The next three chapters are about *how chips actually work and get made*. This is the part that will probably be most new to you, and the most fun. You are going to learn about sand and ingots and photolithography and why Taiwan is geopolitically the most important island on Earth. If you don’t care about any of that yet, you will by the end of Chapter 4.

Then there are three chapters about the *real costs*. Energy, water, and mining/e-waste. Each of them follows the same structure: here are the numbers, here is what’s actually a problem, here is what’s been exaggerated, here is what we don’t know. I tried to be careful about not soft-pedaling, because you can tell when you’re being soft-pedaled to, and also not alarmist, because the alarmism version of this story is already everywhere and you don’t need more of it.

Then there is a chapter about how this all compares to the last three hundred years of human history, which is maybe the most important chapter in the book and which also happens to be my favorite.

Then there is a chapter about the *other* side of the ledger — what these chips and data centers and AI systems actually do that is good. (Yes, really. Yes, it’s a lot.) This is the chapter that will bore

people who wanted the book to be about how bad technology is, and it's the chapter that will bore people who wanted the book to be about how great technology is, which is probably a sign it's doing its job.

Then there is a chapter about something almost nobody talks about, which is that the *scarce resource in this whole system isn't the electricity or the water or the silicon*. It's your attention. And the attention economy that's been built around the chips in your phone is, in a lot of ways, more environmentally and psychologically expensive than all the training runs put together. That chapter might make you mad. That's fine.

Then two closing chapters: one about *who actually decides* what happens in this industry — which is a surprisingly small number of people, making decisions you should probably know about — and one about *what you can actually do* that would actually matter, which is also a surprisingly small list, and does not include turning off your wifi router.

Then the book ends. Take what you want from it.



One Last Thing

Before you turn the page, I want you to look at your phone again.

Feel the weight of it. Notice the screen, which is made of chemically-strengthened glass that didn't exist twenty years ago. Notice the case, which has rare earth metals in it that were mined on three continents. Notice the little camera, which has a chip in it that can capture more detail than your eye can see. Notice the battery, which contains lithium that was probably evaporated out of saltwater in the Atacama Desert.

Now notice that this whole object — this genuinely astonishing object, maybe the most complex thing a person is likely to own in their lifetime — exists because of a global supply chain involving

roughly two million people, seventy countries, and some of the most precisely engineered machines ever built.

You bought it for \$800, or your parents did, or someone handed it to you used. You have probably dropped it at least once. You use it mostly to watch videos and text your friends.

None of this is a criticism. I use my phone to watch videos and text my friends too. The point is just that something incredibly strange is happening: we have become, in the space of a couple of generations, casual owners of technology so complex that no single person on Earth fully understands all of it. The chip in your phone was designed by one set of humans, manufactured by another, assembled by another, and programmed by another, and none of them know what any of the others do in detail. The whole thing only works because every link in the chain does its job and trusts the other links.

This is worth thinking about. Because the next time someone tells you, with great confidence, what the “truth” about AI or data centers or chips is — including me, in this book — you should remember that the thing they are confidently describing is a thing almost no one fully understands, and which no single person has seen all of, and which is growing faster than our ability to keep up with it.

Humility, in this domain, is the correct starting point.

Let’s begin.



A note on sources: the transistor count figures are from SEMI industry reports and estimates of cumulative chip production since the first commercial integrated circuit in 1961; the “more transistors than grains of sand” claim is widely repeated but the exact ratio depends on how you count sand, so I have used conservative estimates in both directions. The fab cost figures are from public disclosures by TSMC, Samsung, and Intel. The EUV lithography details are from ASML, the Dutch company that makes

the machines. The Manhattan-to-rice-grain scale comparison is my own, done the back-of-envelope way, and you should feel free to check it.



CHAPTER 2

Sand, Light, and the World's Most Precise Factories

How a Grain of Sand Becomes the Thing in Your Pocket



Pick up a single grain of sand.

Not metaphorically. If you're somewhere you can get to sand — a beach, a sandbox, a kid's toy — go get a grain. Put it on your finger. If you can't get a grain, look at a pinch of salt on your kitchen counter; it's about the same size. Zoom in on it in your head.

That grain of sand is made mostly of silicon dioxide — silicon bonded to oxygen. To turn it into a chip, you need to pull the silicon out of it, purify the silicon until it is almost unbelievably pure, grow it into a perfect crystal, slice that crystal into wafers, and then — through about 500 to 1,000 separate process steps conducted in the cleanest rooms ever built — draw billions of transistors onto it using light that doesn't exist anywhere in nature.

The whole process takes about three months. It uses machines that cost \$350 million each. It happens, for the most advanced chips, in about five buildings in the world, most of them on one island in the Pacific. And if any one of those 500 steps goes wrong — if a single speck of dust lands in the wrong place, or a single machine is off by three nanometers, or a single drop of the wrong chemical touches the wafer — the chip is ruined.

This chapter is going to walk you through the whole thing, from the grain of sand to the finished chip in your phone. It's the most interesting chapter in the book, in my opinion, because the process is so physically astonishing that it sounds made up. It is not made up. It is just very strange.



Step 1: Making the Silicon

Regular sand is maybe 50–80% silicon dioxide, mixed with a lot of other stuff. That's not pure enough. For chips, you need something close to pure silicon.

The first step is to heat silicon-rich sand in a furnace with carbon, which pulls the oxygen away and leaves “metallurgical grade” silicon — about 98% pure. That's pure enough for solar panels and some industrial uses. For chips, it is laughably impure.

So the metallurgical-grade silicon gets converted into a gas called trichlorosilane, which is then distilled — the same way you would distill vodka, but at much higher stakes — and the distillation is repeated over and over until the silicon is **99.999999% pure**. Nine nines. This is called “9N” purity in the industry. It means there is less than one non-silicon atom per billion silicon atoms.

To give you a sense of what this level of purity means: if you took a swimming pool full of pure water and dropped a single grain of salt in it, the salt would be about a thousand times more contaminant than the impurities in electronic-grade silicon. This is the cleanest substance humans routinely manufacture.

The pure silicon comes out of the distillation process as a powder. Now you need to turn it into a single, enormous, flawless crystal.



Step 2: Growing the Ingot

Here is where the process first starts to feel genuinely magical.

The silicon powder gets melted in a crucible at about 1,400°C. A tiny “seed crystal” of pure silicon — a fingertip-sized piece with the crystal structure already established — is lowered on a rod until it just touches the surface of the molten silicon. Then the rod is very slowly pulled upward, and rotated, while the crucible is also rotated, in the opposite direction.

As the seed crystal rises, silicon atoms from the melt latch onto it one by one, in the orientation dictated by the seed’s crystal structure. A cylinder of perfect, single-crystal silicon grows downward from the seed. This process is called the **Czochralski method**, after Jan Czochralski, who invented it in 1915 after — the story goes — accidentally dipping his pen into molten tin instead of his inkwell.

The finished cylinder — called an **ingot** — is about 300 millimeters across (the diameter of a large pizza), up to 2 meters long, and weighs several hundred kilograms. Inside, the atoms are arranged in a single continuous crystal lattice. Not one crystal per region, connected to other crystals at grain boundaries like most metals. *One* crystal. The entire ingot is, in a real sense, a single molecule. This is maybe the largest perfect thing humans ever make on purpose.

Pulling the ingot takes about two days. If you do it wrong — rotate it too fast, pull it too quickly, let a vibration shake the crucible — the crystal structure gets disrupted and you have to melt it down and start over.



Step 3: Slicing the Wafers

The ingot now gets sliced into thin disks, the way you would slice a salami.

The slicing is done with diamond-coated wires — essentially microscopic cheese wires studded with industrial diamond dust — that cut through the silicon at carefully controlled speeds. Each slice is about 0.7 millimeters thick. A single 2-meter ingot yields a few thousand wafers.

Then the wafers get polished. And then polished some more. And then polished even more. The finished wafer is a mirror so flat that if you expanded it to the size of North America, the tallest “mountain” on its surface would be about the height of a paperback book. The flatness is measured in atoms.

One polished 300mm wafer — the size they use for most modern chips — sells for maybe \$200 before any processing. By the time the chips on it are finished, that same wafer is worth somewhere between \$15,000 and \$20,000, depending on what’s on it. Essentially all of that value is added by what happens next.



Step 4: The Cleanroom

Before we talk about what happens next, we have to talk about where it happens, because the environment is part of the story.

A modern chip fab is built around one big room — or, in the largest fabs, a few big rooms — called the **cleanroom**. The cleanroom has to be cleaner than anywhere else on Earth, including operating rooms, including labs handling dangerous pathogens, including the clean rooms NASA uses to build spacecraft. The reason is that a chip’s features are so small — a few nanometers — that a single dust particle landing on the wafer at the wrong moment covers thousands of transistors and ruins the chip.

The cleanroom is rated by particle count. A normal room has somewhere between 100,000 and 10 million particles per cubic foot of air (particles larger than 0.5 microns). A hospital operating room might be 10,000. A chip fab’s most sensitive areas — called Class 1

or ISO 3 cleanrooms — have **fewer than 12 particles per cubic foot**. Often fewer than 1.

To achieve this, the air in the cleanroom is completely replaced about 600 times per hour, flowing downward from the ceiling through HEPA filters, past the wafers, and out through gratings in the floor. The floor is raised and the ceiling is mostly filters; air comes down from above and is sucked out from below, in a continuous vertical river that keeps particles from getting anywhere near the wafers.

The humans in the cleanroom — and there are humans, though fewer every year — wear full-body “bunny suits” with hoods, face masks, gloves, and booties. The suits are not for the humans’ protection. They are for the chips’ protection. A person is, from the chip’s perspective, a giant walking contamination machine that sheds thousands of skin cells, hair fragments, and fibers per minute. The suit is there to contain all of that.

Humans in the cleanroom move slowly, to avoid disturbing the airflow. They don’t sneeze (or they go immediately to a decontamination area if they do). They can’t wear makeup or perfume. Their movements are tracked, and the entry airlocks involve an elaborate sequence of gowning in a specific order — booties first, then the suit, then the hood, then the mask, then the gloves — that takes about ten minutes to do correctly.

All of this is because, again: a single speck of dust can ruin a chip worth hundreds of dollars, and the chip is one of hundreds on a wafer worth \$20,000.



Step 5: The Recipe

Inside the cleanroom, the wafer will go through somewhere between 500 and 1,000 process steps before it becomes a finished chip. That’s not a typo. Five hundred to a thousand.

Each step does one small thing. Deposit a thin layer of material. Etch a pattern into that layer. Implant some atoms. Clean the surface. Inspect. Deposit the next layer. Etch the next pattern. And so on, for months.

The whole sequence is called a “process flow” or “recipe,” and it is the single most valuable piece of intellectual property at a chip company. The recipe for a cutting-edge chip is the collective work of thousands of engineers over years and costs hundreds of millions of dollars to develop. A rival company that stole the recipe could save itself a decade of work.

I am going to describe the three most important kinds of steps, because once you have them, you have the basic building blocks of everything a fab does.



Step 6: Photolithography (The One That Matters)

If you are going to understand only one thing about chip making, make it this.

Almost every step in the recipe depends on *patterning* — on making a specific, precise shape on the wafer. The transistors are shapes. The wires connecting them are shapes. The insulation between them is shaped. Every layer of the chip is a shape, and the shapes have to line up, layer after layer, to within a few nanometers of accuracy.

The way you make these shapes is called **photolithography**, and it works like this.

You start with a wafer coated in a thin layer of a chemical called **photoresist**. Photoresist is goo that changes its chemistry when light hits it. You’ve used something a bit like it if you’ve ever been to a summer camp that made sun-print T-shirts.

You take a mask — a quartz plate with an incredibly precise pattern etched into it in chrome — and you shine a specific kind of light through the mask onto the photoresist-coated wafer. The light

passes through the clear parts of the mask and is blocked by the chrome parts. Where the light hits the photoresist, it changes the resist's chemistry. Then you rinse the wafer with a developer solution, which washes away either the exposed parts or the unexposed parts (depending on which kind of resist you used), leaving behind a precise pattern.

That pattern is now the template for whatever you do next. If you etch the wafer, the etching happens only where the resist is gone. If you deposit metal, the metal sticks to the exposed silicon and sits on top of the resist. And so on.

This process — expose, develop, etch/deposit — gets repeated *dozens of times* per chip, with different masks for each layer. One modern chip might go through 80 different lithography steps, each with its own mask, each of which has to line up with the ones before it to nanometer precision.

Here is the catch. The smallest feature you can make with photolithography is roughly the wavelength of the light you're using. Visible light has wavelengths between 400 and 700 nanometers, which means you can't pattern features smaller than a few hundred nanometers with visible light. The chip industry passed "a few hundred nanometers" in the 1990s, which means for the last 30 years the industry has been desperately trying to use shorter and shorter wavelengths of light to make smaller and smaller features.

Now we need to talk about the hero of this whole book.



Step 7: EUV — The Most Absurd Machine on Earth

The current state of the art is **extreme ultraviolet lithography**, or EUV. The light used has a wavelength of 13.5 nanometers. That is about 30 times shorter than the shortest visible light. It is, technically, a form of soft X-ray.

Here is how you generate EUV light. You get a tiny drop of molten tin, about 30 micrometers across. You drop it through a vacuum

chamber. As it falls, you hit it with a high-power carbon dioxide laser — first a small pulse to flatten it into a pancake, then a much bigger pulse to vaporize it into a plasma hotter than the surface of the Sun. The plasma, briefly, emits EUV light. You catch some of that light with a curved mirror and bounce it through a series of other mirrors — ordinary lenses don't work for EUV because EUV is absorbed by basically everything, including air — and eventually onto the wafer.

You do this 50,000 times per second. One tin droplet, two laser pulses, one flash of EUV, fifty thousand times a second, for the entire time the machine is running.

The mirrors that carry the EUV light are themselves miracles. EUV is absorbed by glass, by quartz, by almost any material. The only thing that reflects EUV efficiently is a multilayer stack of about 100 alternating layers of molybdenum and silicon, each about two nanometers thick, deposited with atomic precision. Even these mirrors only reflect about 70% of the light that hits them. After bouncing off a dozen mirrors in the machine, only a few percent of the original light reaches the wafer.

The machine that does all of this — called an EUV scanner — is built by exactly one company in the world, a Dutch company called ASML. Each scanner:

- Contains about 100,000 precisely-engineered parts.
- Costs about \$350 million.
- Weighs 180 metric tons.
- Ships from ASML's factory in the Netherlands to the fab in 40 shipping containers, typically via a dedicated fleet of Boeing 747 cargo planes (because shipping by sea takes too long and doesn't handle vibration well enough).
- Takes several months to install and calibrate.
- Is so sensitive that the gravitational pull of the Moon causes small but measurable shifts in its alignment, which have to be corrected for.

ASML has a backlog of orders that stretches years into the future. Every leading-edge chip fab in the world has to buy their EUV machines from this one factory in one Dutch city. If anything happened to ASML, the chip industry as we know it would freeze.

This situation — the entire advanced computing industry depending on a single Dutch company, which in turn depends on a single Belgian company (Zeiss SMT) for the mirrors — is one of the strangest facts about the modern world. Most people don't know it. Now you do.



Step 8: Etching, Deposition, and Doping

Lithography just draws the pattern. The other steps actually build the chip.

Deposition is how you add material. Various kinds of deposition machines — chemical vapor deposition, atomic layer deposition, sputtering — add thin films of metal, insulator, or other materials to the wafer, sometimes one atomic layer at a time. A modern chip is made of 50 to 100 of these layers, stacked precisely on top of each other.

Etching is how you remove material. Where the lithography left exposed silicon or metal, etching tools use plasmas or chemicals to dissolve the exposed material away, leaving only the pattern you wanted. Modern etching is so precise it can remove material one atomic layer at a time, without touching the adjacent layer at all. This capability didn't exist 20 years ago. Now it's routine.

Doping is how you make the silicon actually behave as a transistor. Pure silicon, interestingly, is a lousy conductor. You make it useful by injecting small amounts of other elements — boron, phosphorus, arsenic — into specific regions of the silicon. These “dopants” change how electrons move through those regions, and different arrangements of dopants create transistors that either block current or let it flow, depending on the voltage applied. The whole

idea of a transistor is, at bottom, a choreography of doped silicon regions. Modern fabs do this with ion implantation machines that accelerate the dopant atoms to high speeds and shoot them into the silicon surface like a very, very small gun.

These three operations — deposit a layer, pattern it with lithography, etch it into shape, occasionally dope it — get repeated, in different combinations, for every one of the hundred-ish layers of a modern chip. This is why the process has 500–1,000 steps.



Step 9: Testing, Dicing, Packaging

After about three months in the fab, the wafer is covered in hundreds of chips, each a few square millimeters across. The chips are still all attached to each other on the wafer, and none of them have been tested yet.

Now comes **wafer probe** — a machine that lands tiny needles on each chip, one at a time, and runs a battery of electrical tests. Chips that pass get marked “good”; chips that fail get marked “bad.” The fraction that pass is called the **yield**, and it is one of the most important numbers in the industry. A fab running at 90% yield is doing great. A fab at 50% yield is losing a lot of money. When a new process is introduced, yield can start as low as 10–20% and gradually climb as engineers debug the process over months and years.

After probing, the wafer gets **diced** — cut up into individual chips with a precise saw. Each chip is now a tiny, fragile piece of silicon. These are called **die** (plural: also die; never “dice” in the industry, for reasons).

Each good die gets **packaged**: bonded to a substrate, connected to external pins or balls, and encapsulated in the black plastic or ceramic case that’s what you see when you look at a chip in a circuit. Packaging is a whole additional manufacturing process, often done in a different factory in a different country than the fab.

Then the packaged chips get tested *again*, more thoroughly, and the ones that pass go into boxes and ship to customers.

The whole thing, from silicon powder to boxed chip, takes roughly three to four months. The chip in your phone was being made, somewhere, while you were doing something in the spring, and it reached your phone by Christmas.



Where All This Happens

Here's a fact that matters more than it should: *the number of places in the world that can make the most advanced chips is very small.*

TSMC (Taiwan Semiconductor Manufacturing Company), on the island of Taiwan, makes about 90% of the world's most advanced chips. Their most advanced fabs — the ones making 3-nanometer chips in 2026 — are mostly clustered in the Hsinchu Science Park in northern Taiwan. One island, a couple of industrial parks, a large fraction of the world's leading-edge computing.

Samsung, in South Korea, is the main competitor to TSMC at the leading edge and makes a large share of the world's memory chips. Their fabs are in Pyeongtaek and a few other sites in South Korea.

Intel makes its own chips in fabs in Arizona, Oregon, New Mexico, Ireland, and Israel. Intel used to be the world leader in manufacturing technology; they have fallen behind TSMC and Samsung over the last decade and are trying to catch back up.

SMIC (Semiconductor Manufacturing International Corporation), in China, makes less-advanced chips in large volume, and is slowly pushing into more advanced nodes despite being cut off from buying EUV machines by US export controls.

ASML, in the Netherlands, makes the lithography machines that all of the above depend on.

Zeiss SMT, in Germany, makes the optics for ASML's machines.

If you drew a map of "places a modern chip's existence depends on," it would be a handful of dots: Taiwan, South Korea, the Nether-

lands, Germany, the US West Coast and Israel. That's it. Everything else — the software, the chemicals, the inspection equipment — is distributed. But the chokepoints, the places where a fire or an earthquake or a war could take down the whole industry, are in those few dots.

This is why you have probably heard, in the news, a lot of talk about Taiwan, about US chip subsidies (the CHIPS Act in 2022), about export controls on EUV machines to China, about new fabs being built in Arizona and Ohio. The geopolitics of chips is not a weird niche topic. It is one of the central issues of our time, and most people's picture of it is vague. The concentration of manufacturing in a few places that are, in various ways, fragile — Taiwan's proximity to China; Korea's proximity to North Korea; the Netherlands' dependence on a single company — is a story that is going to keep being in the news for your entire life.



What This Chapter Is For

Here is why I wanted you to know all of this.

When you hear someone talk about “chips” — the chip shortage, AI chips, data center chips, the cost of chips — it is easy to picture a chip as a commodity, something that gets made the way t-shirts or toasters get made. Lots of factories around the world, competing on price, cranking out the product.

It's not like that. The manufacture of modern chips is almost nothing like the manufacture of almost anything else. It happens in five or six buildings, in four or five countries, with machines built by two or three companies, using a process so delicate that a power outage can ruin a week's production. It is more like a cathedral than a t-shirt factory. The whole modern world — every phone, every data center, every car with airbags, every hospital monitor, every AI model — sits on top of this strange, concentrated, fragile pyramid of manufacturing.

When people talk about the environmental cost of AI and ignore the fact that a fab uses more water than a city, that is missing something real. But when people talk about the environmental cost of AI and forget that the fab also makes the chips in the medical equipment that kept their grandmother alive, that is also missing something real. Everything is connected to everything. You can't tell a sensible story about one part of it without at least knowing the other parts exist.

The next chapter is about what the chips in those packages actually *do* — how a CPU is different from a GPU is different from the chip in an airbag sensor, and why the differences matter. After that, we go to the data center, which is where most of the chips that run AI end up.

But for now, close your eyes for a second and picture it: somewhere on Earth, right now, a tin droplet is falling through a vacuum chamber. A laser is pulsing. A flash of ultraviolet light that doesn't exist in nature is bouncing off a stack of molybdenum mirrors. It hits a silicon wafer in a room cleaner than surgery. An atom-thin layer of material is patterned by that light. And the pattern — one of the hundred that will be made on that wafer that day — becomes, eventually, the thing you use to text your friends about how your day is going.

It's a weird world. You are living in it.



A note on sources: the ASML machine specifications and the details of EUV lithography are drawn from ASML's own technical publications, from Chris Miller's Chip War (which I recommend if this chapter was interesting to you), and from academic coverage of semiconductor manufacturing. The cleanroom particulate specifications are ISO 14644-1 standards. The 9N purity figure is standard industry practice for electronic-grade silicon. The percentages of global leading-edge chip production are from industry

reports; they shift year to year but the concentration in Taiwan has been roughly stable at 85–92% for about a decade.



CHAPTER 3

What Chips Actually Do

Why "A Chip" Is Not a Thing



Here is a small quiz.

How many chips are in the phone in your pocket?

Don't answer yet. Really think. Picture the phone, and picture what's inside it. What are the chips for?

If you said "one" — the main processor — you're off by about thirty. If you said "a few," you're still off by about twenty. The answer, for a typical 2026 smartphone, is somewhere between 30 and 60 separate chips, depending on how you count.

There's the main processor, obviously. But there's also a separate chip for the cellular modem, one for Wi-Fi, one (or sometimes the same one) for Bluetooth, one for each of the three or four cameras, one for the fingerprint sensor, one for the touchscreen, one for the accelerometer, one for the gyroscope, one for the microphone array, one for the speaker amplifier, one for managing the battery, one for wireless charging, one for NFC (the tap-to-pay chip), one for GPS, several for different kinds of memory, and a bunch of tiny specialized ones for everything from haptic feedback to ambient light sensing.

All of these are chips. All of them are made in fabs like the ones in the last chapter. And all of them do radically different things, in radically different ways, at radically different scales.

When someone says “AI chips” or “the chips in data centers” or “chip manufacturing,” they are talking about one small corner of a very large, very varied universe of chips. This chapter is going to map out that universe, so you know what you’re talking about when you talk about it.

The short version: **“chip” is a shape, not a job.** Like “machine” or “vehicle.” A cargo ship and a dirt bike are both vehicles, and knowing that tells you nearly nothing about either one.



The Big Four Categories

Most chips fall into one of four broad categories: **CPUs**, **GPUs**, **memory**, and **everything else**. Let me take them in order.

CPUs: The Thinkers

CPU stands for **Central Processing Unit**. A CPU is what most people picture when they picture a computer chip. It’s the “brain” of a general-purpose computer — your laptop, the main processor in your phone, the thing running the operating system.

CPUs are good at doing complicated things one at a time, very fast. “One at a time” is a slight lie — modern CPUs have between 4 and 64 “cores” that can each do something in parallel — but compared to the other things on this list, they are specialists in sequential, branching, decision-heavy work. If a task requires a long series of decisions, each of which depends on the previous one, a CPU is what you want.

Running your operating system is this kind of task. So is most software you interact with directly: web browsers, word processors, spreadsheets, games (for the non-graphics parts), compilers, databases. The logic of these programs is full of “if this, then that” steps, where the next move depends on what just happened. CPUs are optimized to race through that kind of work.

The CPU in your phone is roughly as powerful as the fastest supercomputer on Earth was in 1998. The CPU in a modern laptop is roughly as powerful as the fastest supercomputer of 2003. Both of these things are astonishing, and neither one is made by particularly famous companies — Apple makes theirs, Qualcomm makes theirs, AMD and Intel make most of the laptop ones. The CPU in your phone has, probably, eight cores and runs at around 3 GHz, which means each core is doing roughly three billion operations per second.

GPUs: The Parallel Processors That Accidentally Became the AI Chips

GPU stands for **Graphics Processing Unit**. GPUs were invented in the late 1990s to draw graphics in video games.

Drawing 3D graphics in real time is a weird computational problem. You don't have to make a lot of sequential decisions. But you do have to perform roughly the same simple calculation — figure out what color this pixel should be, given where the light is coming from and what's in front of it — *millions of times at once*, for every pixel on the screen, every time a new frame is drawn. 60 times a second, 120 times a second, higher.

CPUs are bad at this. A CPU with 8 powerful cores can only do 8 things at once, and for graphics you want to do a million things at once. So GPU designers built chips the opposite way: instead of a few powerful cores, they made *thousands* of small, simple cores that could all work in parallel on the same kind of task. A modern GPU has between 5,000 and 20,000 cores. Each individual core is much less capable than a CPU core. But all of them working together can do some specific kinds of calculations vastly faster than a CPU can.

For about twenty years, this was a fine but specialized niche in the chip industry. GPUs made video games look good and were mostly bought by gamers. A company called **Nvidia** was the market leader, selling a few billion dollars' worth of GPUs a year, mostly to people who wanted to play Call of Duty in higher resolution.

Then something unexpected happened.

Around 2012, researchers figured out that the kind of math GPUs were good at — the same simple operation performed on millions of numbers in parallel — was also exactly the kind of math you needed to train **neural networks**. The whole mathematical guts of a neural network is, mostly, one operation: multiply a big matrix of numbers by another big matrix of numbers. GPUs could do that operation hundreds of times faster than CPUs.

Almost overnight, AI researchers started buying GPUs instead of CPUs. Nvidia's business grew. Then, starting around 2020, AI got *really* big, and Nvidia's business grew ridiculously. As of 2026, Nvidia is one of the most valuable companies in the world, primarily because they make the GPUs that are used to train and run large AI models.

The chips in big AI data centers are mostly GPUs. The chip shortages you hear about are mostly about GPUs. The CHIPS Act, the export controls, the rivalry with China over AI — all of it, at the object level, is mostly about GPUs. The current most advanced GPU costs about \$40,000, and the companies making it can't make them fast enough to meet demand.

It is, genuinely, one of the stranger pivots in industrial history: the chips designed to render dragons in video games turned out to be the same chips that run ChatGPT, and the company that made them was sitting there, quietly growing for twenty years, until suddenly everyone on Earth needed them.

Memory: The Rememberers

Computers need to remember things. There are basically two kinds of memory chips, and they do it in two different ways.

DRAM (Dynamic Random Access Memory) is fast, volatile memory. It holds whatever the computer is currently working on. When you open an app, it gets loaded from storage into DRAM, and the CPU reads and writes DRAM as it runs the app. DRAM is very fast — it can be read or written in tens of nanoseconds — but the moment the power goes off, everything in DRAM vanishes. The

chip physically cannot hold its contents without a steady supply of electricity refreshing the values thousands of times a second. DRAM is how the computer thinks in the moment.

Your phone probably has 8 or 12 gigabytes of DRAM. Your laptop probably has 16 or 32. A server in a data center might have 1,000. The largest DRAM chips hold around 16 gigabytes each, and a device might combine dozens of them.

Flash memory (and its big cousin, the SSD, or Solid-State Drive) is slower but persistent. When the power goes off, flash remembers. Every photo on your phone, every app on your laptop, every file on a cloud server — almost all of it lives on flash memory when it's not actively being used. Reading flash is about a hundred times slower than reading DRAM, and writing is several thousand times slower, but flash can be much denser and much cheaper per gigabyte.

Memory chips are their own industry. **Samsung** and **SK Hynix** in South Korea dominate DRAM. Samsung and **Kioxia** (which used to be the memory arm of Toshiba) dominate flash. Memory is one of the few parts of the chip industry where Korea, not Taiwan, is the center of gravity. It's also one of the most cyclical — memory prices rise and crash in multi-year cycles that regularly bankrupt entire companies.

You will mostly not think about memory chips. But every phone, every laptop, every server, and every data center contains vastly more memory chips than logic chips, and the total value of the memory industry is about the same as the total value of the CPU industry.

Microcontrollers and Specialized Chips: Everything Else

Now we get to the part of the chip universe you almost never hear about, which is also most of it.

A **microcontroller** is a small, cheap, specialized chip that does one specific job. Not “runs a general-purpose computer” — one job. The chip in your microwave that reads the buttons and controls the timer is a microcontroller. The chip in your car that reads the pedal

position and tells the engine what to do is a microcontroller. The chip in a traffic light, a thermostat, a pacemaker, an electric toothbrush, a smoke detector, a dishwasher — all microcontrollers. They cost pennies to dollars each, and there are about 30 billion of them sold a year.

Microcontrollers are made on older, cheaper process technologies — often the “mature” nodes that big fabs moved on from 15 or 20 years ago. You don’t need EUV lithography to make a chip that reads a thermostat. The economics are completely different from cutting-edge processors. So are the suppliers: a lot of microcontrollers come from companies you’ve never heard of, in fabs in places like Germany, France, and mainland China, not from TSMC at 3 nanometers.

Sensor chips are a category of specialized chip that converts some physical thing — light, sound, pressure, acceleration, magnetic field — into an electrical signal the computer can read. Every camera has an image sensor chip (mostly made by **Sony**, which has about half the market). Every phone has accelerometer and gyroscope chips so it can tell when you rotate it. Every car has dozens of sensor chips, each watching some specific thing. These chips are genuinely clever: the image sensor in your phone has about 50 million light-sensitive pixels, each one a tiny electronic eye, all on a piece of silicon the size of your pinky fingernail.

Radio chips handle wireless communication — cellular, Wi-Fi, Bluetooth, GPS. Each one is a small masterpiece of analog engineering, because radio frequencies behave in ways that are much harder to control than digital logic. **Qualcomm**, **MediaTek**, and **Broadcom** dominate these markets.

Application-specific integrated circuits, or ASICs, are chips designed to do one specialized thing extremely well. Bitcoin mining chips are ASICs. The chips in a credit card terminal are ASICs. Google’s custom AI chips, called TPUs, are ASICs — they’re chips designed by Google to be better than general-purpose GPUs for Google’s specific AI workloads. Tesla’s self-driving chip is an ASIC.

When a company's computational needs are narrow enough and large enough, it can be worth hundreds of millions of dollars to design a custom chip that does that one thing better than a general-purpose chip can.

The ASIC trend is accelerating in AI. Every big tech company — Google, Amazon, Meta, Microsoft, OpenAI — is now designing custom chips for their AI workloads. Some of these are already in production. Others are still in development. One of the interesting things to watch over the next few years is whether Nvidia's dominance in AI chips survives a world where every major AI customer is trying to design their own alternative.



A Tour Through Your Day

Let's do the same thing we did with energy in Chapter 5, but with chips. Let me walk through an ordinary day and point at every chip you interact with, roughly in order.

You wake up. Your phone alarm wakes you. The chip that decided to play the alarm is a microcontroller inside the phone, running a scheduled task on battery-saver settings. It signals the main processor to wake up, which signals the audio chip to play a sound through the speaker amplifier chip.

You look at your phone. Your glance is picked up by a proximity sensor chip that tells the display to turn on. The display is driven by a display controller chip. Your thumbprint is read by a fingerprint sensor chip, which has its own small processor to match the print. The result is verified by a security chip (Apple calls theirs the "Secure Enclave") that is deliberately isolated from the main processor so that even if the main processor is hacked, the fingerprint data is safe.

You check the weather. The weather app runs on the main CPU. The forecast data comes over the cellular network, handled by the modem chip. The modem talks to a cell tower, which has its own

chips handling thousands of connections simultaneously. The weather data comes from a server, which is a big CPU in a data center somewhere.

You drive to school. Your car has, on average, about 1,500 chips in it. The engine is managed by an engine control chip. The transmission has its own chip. Every airbag has a chip watching an accelerometer for sudden deceleration. The anti-lock braking system has a chip. The backup camera has an image sensor and a processing chip. The dashboard is driven by a dashboard computer. The infotainment system has its own chips. The stereo has chips. If the car is electric, the battery management has dozens of chips. If the car is self-driving or partly self-driving, it has specialized AI chips running inference on the sensor data many times per second.

You arrive at school. You tap an RFID card to enter. The card has a chip. The reader has a chip. The door controller has a chip. The security system talking to all the readers has chips.

You open your laptop. The laptop has a CPU (maybe 20 billion transistors), a GPU (maybe another 10 billion), memory chips (dozens of billion more), a chipset coordinating everything (another billion), a Wi-Fi chip, a Bluetooth chip, a camera chip, a microphone chip, a battery management chip, and a trackpad controller, among others. You open Google Docs. The document lives on a server in a Google data center, which has tens of thousands of servers with hundreds of thousands of chips.

You pay for lunch with your phone. Your phone's NFC chip transmits an encrypted token to the payment terminal's NFC chip. The terminal sends it to the payment processor, which is running on servers that use specialized security chips for encryption. The bank that issued your card has its own servers. The merchant's bank has its own servers. The chips involved in processing this one \$7 sandwich payment probably number in the millions, scattered across four continents. All of this happens in about a second.

You go home. You turn on Netflix. The TV has chips. The streaming is served by yet another data center. The video is compressed by

specialized video chips and decompressed by chips in your TV. The recommendation algorithm that picked the thumbnail you saw is running on AI chips somewhere.

You go to bed. Your phone charges on a wireless charging pad, which has its own chips controlling the power transfer. Your smart thermostat adjusts the temperature based on a chip that's been watching your habits for weeks. The smoke detector on the ceiling has a chip checking for particles every few seconds.

All day, chips. Roughly 100,000 to 10 million individual chips participated in your ordinary, unremarkable day, depending on how you count. Most of them you never thought about and could not have pointed to if someone asked. All of them, together, made your day work.



Why This Matters for Everything Else

Here's why I wanted to make the point that chips are heterogeneous.

When someone says "chip manufacturing is environmentally bad," they might mean: "the leading-edge GPU fabs use a lot of water and power." Which is true. Or they might mean: "the global chip industry is bad for the planet." Which is a much bigger, vaguer, and largely wrong claim, because the vast majority of chips — the microcontrollers, the sensor chips, the older-process chips in your microwave — have a modest environmental footprint and enable things like medical devices and energy efficiency that have big environmental *benefits*.

When someone says "AI is using all the chips," they might mean: "the most advanced GPUs are in short supply and expensive." Which is true. Or they might mean: "AI is consuming the world's chip production." Which is false — AI uses a small and fast-growing fraction of chip production, overwhelmingly concentrated in one specific kind of chip (leading-edge GPUs) from one specific

manufacturer (TSMC) made for a handful of specific buyers (Nvidia and a few custom-chip companies). The microcontrollers in your dishwasher are not in shortage because of AI.

When someone says “we need to reduce chip dependence,” they might mean: “we need a backup source for leading-edge logic chips in case of a Taiwan crisis.” Which is a sensible strategic concern. Or they might mean: “we should make fewer chips.” Which would mean: fewer medical devices, fewer safety sensors in cars, fewer grid sensors managing renewable energy, fewer satellites, fewer of a lot of things that most people wouldn’t trade away even if they thought they would.

Most discourse about chips is vague in ways that conceal these distinctions. Keeping them straight is, again, most of what good thinking about this stuff consists of.



What Ends Up Where

One last picture to leave you with before the next chapter.

The chips in your phone and laptop? They live with you, in your hand or on your desk. If you get a new phone, the chips in the old one will eventually end up in recycling or landfill (see Chapter 7).

The chips in your car, your microwave, your thermostat, your medical devices? They live in those devices, sometimes for 20 years.

The CPUs, GPUs, and memory chips that handle the *cloud* side of things — the part of the internet that isn’t in your hand — live somewhere else entirely. They live in enormous buildings, mostly in a few dozen places in the world, running 24 hours a day, every day of the year. The websites you visit, the games you play online, the AI models you use, the streams you watch — all of them are served from these buildings.

Those buildings are what the next chapter is about.

They’re strange places, and almost no one has seen one.



A note on sources: the phone chip counts are from public teardown analyses by iFixit, TechInsights, and System Plus. The Nvidia/AI GPU history draws on Chris Miller's Chip War and on Nvidia's own public filings. The microcontroller and specialized-chip figures are from the World Semiconductor Trade Statistics (WSTS), the industry's standard aggregator. When I've given specific chip counts for specific devices, I've taken them from publicly-available teardowns; they will drift over time as designs evolve.



CHAPTER 4

Data Centers: The Cathedrals Nobody Visits

A Tour of the Place Your Internet Actually Lives



When you open Instagram, the photos you see do not come from your phone. They come from a building.

You probably know this in some abstract sense. “The cloud” is a metaphor; “the internet” is a metaphor; everyone has heard that this stuff is stored *somewhere*. But almost no one has a concrete image of where. The cloud is nowhere in particular, in the same way that money is nowhere in particular. It’s just... there. Out there. In the internet.

The internet, in fact, is a very specific set of buildings. They have addresses. They have parking lots. They have security guards who have badges and bad coffee. If you drove to one, you would see a beige rectangle the size of a big-box store, with a fenced perimeter and a sign at the gate that said something like “Data Campus 4B” and gave you no information about what was inside. You could not go in. There would not be a visitors’ center. There would not even be a receptionist — the building is mostly not staffed.

Inside that building, in rows, are the machines that serve you Instagram. Also Google, YouTube, Netflix, Gmail, Spotify, every video game with online multiplayer, every corporate database in the world, every hospital electronic health record, every airline reserva-

tion system, every bank's back-end, and — the part of the story that has gotten big in the last few years — all of the AI models that the world now runs on.

This chapter is a tour of one of those buildings. Since you can't actually go inside, I'm going to try to take you there in prose. It's worth knowing what this place is like, because it's where most of the things the rest of the book argues about actually live.



Where They Are

Data centers are not random. They cluster.

The biggest cluster in the world is in **Northern Virginia**, specifically Loudoun County, which sits just west of Washington, D.C. About 70% of the world's internet traffic passes through Loudoun County on its way to somewhere else. If you have used the internet today — any service, anywhere — there is a good chance your data went through Loudoun County at some point. The county has something like 200 data centers in a roughly 30-square-mile area. The combined electrical load is larger than the cities of Seattle and San Francisco combined. Driving through the area, you see mostly: beige rectangles, fences, transformers, and fiber-optic utility markings along every road.

Other major clusters, in rough order of size: Dallas, Phoenix, the Pacific Northwest (Oregon and Washington), Santa Clara (Silicon Valley proper), Chicago, Atlanta, Dublin (in Ireland), London, Frankfurt, Amsterdam, Singapore, Tokyo, Sydney. Iceland has a small but growing cluster because it has cheap geothermal power and is cold. Norway has one for the same reasons. There is a growing cluster in Abu Dhabi that's being built for Middle Eastern AI services. There's talk of one in Greenland.

Data centers cluster for reasons that are mostly not obvious. You might think “they should spread out, because bandwidth.” In fact they cluster, because:

1. **Fiber optic cables are not uniform.** Some places have vastly more fiber connectivity than others. Loudoun County has more fiber than many small countries, because it was an early hub and every subsequent network built to connect to the existing networks. Once fiber is there, more fiber comes.
2. **Power is geographically uneven.** Data centers need enormous amounts of reliable electricity. Places with cheap power and stable grids (the Pacific Northwest, with its hydroelectric dams; Virginia, with its nuclear plants; Ireland, with its wind) draw more data centers. Places with expensive or unreliable power (California, most of the Northeast) draw fewer.
3. **Cool climates are cheaper to run.** Cooling is a huge fraction of a data center’s operating cost. A data center in Ireland or Finland can use outside air for most of the year and barely needs active cooling; a data center in Phoenix has to actively refrigerate 24/7.
4. **Proximity matters for some workloads.** Financial trading companies pay enormous amounts to put their servers physically close to stock exchange servers, because even a few milliseconds of network delay can cost them money. This is why lower Manhattan and parts of New Jersey have specialized financial data centers.
5. **Tax incentives.** State and local governments offer data center operators huge tax breaks to locate in their jurisdiction. Virginia’s data center tax treatment is famously generous; this is part of why Loudoun County has what it has.
6. **Political stability and rule of law.** Data centers hold enormously valuable data. Companies are not going to put

them in places where the government might seize them, or where they might be destroyed in a war.

Add all these forces together, and you get: a handful of places with most of the world's data centers, and a long tail of smaller clusters everywhere else.



The Building

Let's go inside one. A medium-large modern data center — about 50,000 square feet, or roughly the size of a big-box hardware store.

From the outside, there is nothing to see. No logos, usually. No windows. The walls are concrete, reinforced. There are loading docks and transformers and sometimes giant diesel tanks on the side for backup generators. A chain-link fence with barbed wire. A single entrance with a guard. Security cameras everywhere. Some facilities have vehicle barriers designed to stop a ramming truck.

Inside the entrance, you go through a series of airlocks — security checkpoints where badge access is tested, sometimes with biometrics, sometimes with mantrap doors that only let one person through at a time. You sign NDAs. You are escorted.

Then you come to the main floor.

The first thing you notice is the **noise**. Data centers are loud. Not crashing, alarming loud — a continuous, deep, enormous hum, plus a higher whine from thousands of fans, plus the lower rumble of air conditioning. In the hot aisles (more on those in a moment) you actually need hearing protection; the noise level reaches 90 dB, which is roughly a lawnmower that never stops. If you stand still for five minutes, your ears start to ache.

The second thing you notice is the **cold**. The data center floor is kept at around 18–22°C (65–72°F), sometimes colder. There is a breeze — a real breeze, constant — from the cooling system. If you wear a short-sleeved shirt, you will be uncomfortable within twenty minutes.

The third thing you notice is how **empty** it is of people. A modern data center that occupies a football-field-sized room might have zero humans in it most of the time. Technicians come in for specific tasks — replacing a failed drive, installing new equipment, fixing a broken fan — and then leave. The servers run themselves. If you walked in unannounced on a Tuesday at 2 pm, you would very likely be the only human being in the room.



The Racks

The main floor is full of **racks**. A rack is a metal frame, about the size of a tall, thin refrigerator, designed to hold standardized computer equipment.

Racks are arranged in long rows, often 50 or 100 feet long, sometimes longer. The rows are oriented so that the *fronts* of the racks face each other across a “cold aisle” — the aisle from which cold air is supplied — and the *backs* of the racks face each other across a “hot aisle” — the aisle into which the hot exhaust is dumped. Air flows from cold aisle, through the servers, out into the hot aisle, and then back up to the cooling system, where it is re-cooled and recirculated. This “hot aisle / cold aisle” architecture is one of those small but important innovations that revolutionized data center efficiency in the 2000s. Before it, cooling was a mess of mixed air flows. Now it’s a clean loop.

Each rack holds somewhere between 20 and 50 **servers**. A server is a specialized computer — no monitor, no keyboard, designed to be stacked in a rack and remotely managed. Each server is about the thickness of a pizza box (and they’re measured in “U,” where 1U is about 1.75 inches; a 2U server is the thickness of two pizza boxes). A typical server has some combination of CPUs, GPUs, memory chips, storage, and network interfaces. It has a lot of fans, because it throws off a lot of heat.

A rack might have: - 20 servers, each with 2 CPUs, 16 memory sticks, 4 network cards, and a bunch of storage. - Or, for AI workloads, 8 servers, each with 8 of the most powerful GPUs plus a couple of CPUs and a lot of high-speed memory. An 8-GPU AI server can cost \$200,000 to \$400,000 on its own. - Or, for memory-intensive workloads, a handful of “fat” servers with 4 terabytes of RAM each.

The rack itself has power distribution, networking switches, and sometimes special cooling plumbing at the top or bottom. The front door of the rack has a cable-management system that looks like an elegant spiderweb if the data center is well-run and like a hairball if it isn't.

A typical medium-large data center has somewhere between 1,000 and 5,000 racks. That's 20,000 to 250,000 servers, depending on density. The very largest hyperscale facilities — the Google and Meta and Microsoft campuses that are essentially dedicated to one company — have over a million servers across multiple buildings on one site.



The Power

Data centers use an enormous amount of electricity, and how that power gets delivered is one of the most interesting (and environmentally consequential) parts of the story.

A medium-large data center draws 10 to 50 megawatts. A big one draws 100 megawatts. A giant “hyperscale” campus might draw 500 megawatts or more. For comparison, a nuclear reactor produces about 1,000 megawatts; a typical American house uses about 1.5 kilowatts on average. A single big data center campus, in other words, uses as much electricity as several hundred thousand houses.

This electricity does not arrive by magic. A data center has its own substation, often on-site, where high-voltage power from the

grid is stepped down to something the servers can use. From the substation, power flows into the building through enormously thick cables, gets distributed to racks, and ends up running chips.

About half of the power goes to the chips themselves. The other half goes to cooling (fans, pumps, chillers, air handlers), networking equipment, lighting, and conversion losses. The industry measures this efficiency with a metric called **PUE**, Power Usage Effectiveness, which is the ratio of total facility power to IT power. A PUE of 1.0 would mean every watt goes to the servers and none to overhead. The best modern data centers achieve 1.1 or better. Older ones might be 1.6 or worse — 60% of their power going to cooling and overhead.

Because data centers cannot, absolutely cannot, tolerate power outages, every data center has layers of backup.

First, **uninterruptible power supplies**, or UPS: giant banks of batteries that can run the whole facility for a few minutes. The moment the grid power hiccups, the UPS takes over, seamlessly. Modern UPS batteries are often lithium-ion, sometimes flywheels, sometimes huge banks of lead-acid batteries the size of rooms.

Second, **backup generators**. Most data centers have enough diesel generators on-site to run the whole facility for several days. When a grid outage lasts longer than the UPS can handle, the generators start up — a noisy affair — and take over. Data centers have on-site fuel tanks holding tens or hundreds of thousands of gallons of diesel. The generators get tested regularly, which is when you hear them in operation if you're nearby.

Third, **redundant grid connections**. A serious data center has power coming from two or more separate substations on two or more separate grid paths, so that a failure of one substation doesn't take the building down.

The layers of redundancy mean a well-run data center is genuinely reliable. The industry measures uptime in "nines": 99.99% uptime ("four nines") means less than an hour of downtime per year. 99.999% ("five nines") means about five minutes. The best data

centers achieve six or seven nines — a few seconds of downtime per year, across everything.



The Cooling

Because chips waste enormous amounts of their input energy as heat, cooling is most of what a data center *does*.

The basic method is air cooling. Cool air is supplied from under a raised floor or from overhead ducts, flows through the servers, picks up heat, and is exhausted into the hot aisle. From the hot aisle, it returns to a cooling unit — essentially a huge air conditioner — which removes the heat and sends cool air back.

For big data centers, the heat removed from the air has to go somewhere. This is where the environmental questions from Chapters 5 and 6 come back in. The heat can be dumped into the outside air through a cooling tower, which uses evaporating water to carry the heat away (efficient, but uses water). Or it can be dumped into a closed loop of water that's cooled by mechanical chillers (uses less water, but uses more electricity). Or, in cold climates, it can just be blown out into the cold outside air for most of the year (“free cooling”), which uses little of either.

Newer data centers — especially ones built for AI workloads, which produce much more heat per rack than traditional workloads — increasingly use **liquid cooling**. Water or other coolant is piped directly to the chips, either to cold plates that sit on top of the chips or, in the most extreme version, by **immersion cooling**: the servers are literally submerged in a specialized fluid that conducts heat but not electricity. Immersion cooling looks exactly as science-fictionary as it sounds: racks of server parts sitting in what looks like aquariums full of mineral oil. It is becoming more common, especially for AI.

AI workloads, in general, are pushing data center design hard. A traditional rack of servers drew maybe 5 to 15 kilowatts. An AI rack can draw 80 to 100 kilowatts — an order of magnitude more, in the

same physical space. The cooling systems are having to be re-designed around this, which is part of why new AI-oriented data centers look different from older general-purpose ones.



Who Owns Them

The global data center industry is dominated, at the top end, by a handful of companies often called the **hyperscalers**:

- **Amazon Web Services (AWS)**: the largest cloud provider in the world. Runs data centers in most of the major regions listed earlier.
- **Microsoft Azure**: the second largest. Also has significant exclusivity deals with OpenAI for AI infrastructure.
- **Google Cloud (GCP)**: third. Runs on infrastructure Google originally built for its own services.
- **Meta**: builds its own data centers for Facebook, Instagram, WhatsApp, and its AI research.
- **Apple**: builds data centers for iCloud and related services, though smaller than the others.
- **Oracle Cloud**: a newer but rapidly-growing player, particularly in AI infrastructure.
- **Alibaba Cloud** and **Tencent Cloud**: dominant in China, significant in Southeast Asia.

Below the hyperscalers are a layer of **colocation providers** — **Equinix, Digital Realty, CoreSite, NTT**, and others — that build data center buildings and rent space in them to companies that need data center capacity but don't want to own the buildings themselves. Colocation is where most smaller companies run their infrastructure: banks, retailers, media companies, universities, hospitals.

Below colocation are enterprise data centers: individual large companies running their own data centers. This model is declining; most companies have moved to the cloud over the last 15 years.

And there is a small but growing category of **specialized AI infrastructure providers** — **CoreWeave**, **Lambda Labs**, and others — that build data centers specifically for AI training and inference, often renting GPUs by the hour to AI companies. As of 2026, some of these companies are valued in the tens of billions of dollars, despite being almost unknown outside the industry a few years ago.

Finally, there are **neo-cloud and sovereign data center projects** — countries and regions building their own data centers so that they don't depend on American (or Chinese) companies for AI infrastructure. The EU has a major initiative on this. The UAE. Saudi Arabia. India. This is a politically growing space, partly driven by the observation that whoever owns the data centers owns significant leverage over whoever uses them.



The Two Kinds of AI Work

For reasons that matter for the next chapters, it's worth separating out two very different kinds of work that happen in AI-oriented data centers.

Training is what happens when a new AI model is being built. A huge cluster of thousands of GPUs runs continuously for weeks or months, processing enormous amounts of data, adjusting billions of numerical weights in the model. Training is done once per model version. It is extremely expensive — the training of a frontier AI model in 2026 costs, roughly, between 100 million and a few billion dollars in compute. It uses an enormous amount of power in one place, briefly. It is the thing that grabs headlines.

Inference is what happens when you actually talk to the AI. You send a query. A much smaller cluster of GPUs — often just one or two, sometimes a whole rack — runs the trained model against your specific query and returns an answer. Inference is cheap per query, but it happens billions of times a day, spread across hundreds of data centers worldwide. The total compute spent on inference,

across all users of all AI services, is now larger than the compute spent on training, though training gets most of the attention.

The environmental story differs between these two. Training is a big one-time spike. Inference is a continuous, distributed load that is probably going to grow faster than training in the coming years. The energy and water figures from Chapter 5 combined both into averages; more careful analyses separate them.



What They Enable

Before we move on to the chapters about costs, it is worth saying plainly: these buildings run the modern world.

The list of things that would immediately stop working if the data centers stopped working, for even an hour, is essentially a list of the things that make 2026 distinguishable from 1970. Banking. Airlines. Hospitals. Supply chains. Phones. Entertainment. News. Commerce. Scientific research. Communication. Navigation. Weather prediction. Drug discovery. Law enforcement. Energy grid operation. Stock markets. Social media, for better and worse. AI, for better and worse. Most of the things you did today.

This is not a reason to ignore their costs. It is a reason to take their costs seriously, because they are not going away. The buildings are the foundation of the world you live in, and they are getting bigger, and they are getting more concentrated, and they are using more power every year.

The next three chapters are about what those costs actually are, in numbers.

Now that you've been in one of the buildings — in your head, anyway — the numbers should land differently than they would have a few chapters ago.



A note on sources: the Loudoun County figures are from the Loudoun County Department of Economic Development and the Virginia Data Center Coalition. The PUE figures and industry metrics are from the Uptime Institute, which is the industry's main authority on these things. The hyperscaler rankings and market shares are from Synergy Research Group's quarterly reports. The AI training cost estimates are from Epoch AI, whose public database is a good resource if you want to look up the numbers yourself. The descriptions of data center interiors are from personal accounts, industry trade publications, and the small number of facilities that have been opened to journalists or researchers.



CHAPTER 5

The Energy Question, Honestly

*What a Kilowatt-Hour Actually Is, and Why Nobody Will Tell
You*



Most of what you have heard about the energy cost of the internet, of AI, of data centers, of chips, is either exaggerated or too vague to be useful. This is not because people are lying to you. It is because energy is invisible, and when things are invisible, humans get the numbers wrong.

This chapter is going to give you the numbers. Some of them will be smaller than you expect. Some of them will be bigger. A few of them we don't actually know, because the companies that would know aren't telling anyone, and that itself is part of the story.

By the end of the chapter you should be able to do something that, strangely, almost no adult can do: look at a claim about the energy cost of some technology — a ChatGPT query, a streaming show, a Bitcoin transaction, a flight — and have a rough sense of whether the number is plausible, and what to compare it to.



The Unit

Before anything else, you need a unit. Otherwise the numbers are just shapes.

The unit is the **watt-hour**, abbreviated **Wh**. It is the amount of energy it takes to run a one-watt device for one hour.

A watt is small. An LED bulb is about 10 watts. A laptop at full tilt is about 50 watts. A microwave is about 1,000 watts, which is why we usually talk about microwaves in **kilowatt-hours** — thousands of watt-hours, abbreviated **kWh**. One kWh is one kilowatt running for one hour. It's the unit your electric bill uses.

Here are some things it is useful to memorize, because once you have them in your head, every other number becomes legible.

Thing	Energy
Phone charge (full)	15 Wh
Laptop for one hour	50 Wh
LED bulb for one hour	10 Wh
Incandescent bulb for one hour	60 Wh
Refrigerator for one day	1,500 Wh (1.5 kWh)
One hour of Netflix on a TV	100 Wh
One mile in an electric car	300 Wh
One mile in a gas car	1,100 Wh equivalent
One load of laundry	2,000 Wh (2 kWh)
One hour of air conditioning	3,500 Wh (3.5 kWh)
One cheeseburger (to produce)	3,000 Wh (3 kWh)
One transatlantic flight, per passenger	900,000 Wh (900 kWh)

Take a minute with that table. The flight is worth staring at. One round trip from New York to London uses, per passenger, roughly as much energy as running a refrigerator continuously for a year and a half, or driving an electric car 3,000 miles, or leaving every LED bulb in a medium-sized house on for four years straight.

This is the context you need for every other number in this chapter.



A Day in Your Energy Life

Let's walk through an ordinary day for a 14-year-old in a typical American household, and count the Wh.

You wake up. The alarm on your phone wakes you up. Your phone charged overnight — 15 Wh. The house was air-conditioned while you slept — call it eight hours of AC, roughly 20,000 Wh for the whole house, or about 7,000 Wh for your share. Already, before you are out of bed, your share of the day's energy use is 7,015 Wh.

You take a shower. Hot water heater, ten minutes, maybe 3,000 Wh. You get dressed in clothes that went through a washer and dryer — amortized over the life of the clothes, about 100 Wh for today's outfit. Breakfast: toaster, 200 Wh; refrigerator's share of today, 500 Wh; the embodied energy in the food itself, which is huge but hard to count, let's skip it for now.

You look at your phone for twenty minutes while eating. The phone itself: maybe 3 Wh for that session. But — and this is the part nobody talks about — every video you watched, every TikTok you scrolled, every image that loaded, was served to you from a data center somewhere. The data-center energy per minute of video streaming is small, maybe 1 Wh per minute. Twenty minutes of scrolling: maybe 20 Wh of data-center energy on your behalf, most of it not for you specifically but spread across all users of that service.

You go to school. If you take a bus, your share of the diesel is about 2,000 Wh for the round trip. If a parent drives you in an SUV, it's more like 8,000 Wh. If you walk, zero.

At school, you use a laptop for four hours. 200 Wh. The school's lights, heating, and AC on your behalf: roughly 15,000 Wh for the day.

After school, you come home and do homework. You ask ChatGPT three questions to help you understand something about the French Revolution. Each query uses somewhere between 1 and 10 Wh of data-center energy, depending on the model and how long

the answer is. Call it 15 Wh total for the session. Your laptop during the homework: another 200 Wh.

Then you watch two hours of a show on your TV. 200 Wh for the TV, another 50 Wh for your share of the data center streaming it.

Dinner: the oven, 3,000 Wh. The dishwasher later, 1,500 Wh.

Your phone all evening: 10 Wh.

You charge your phone for tomorrow: 15 Wh.

Total, roughly, for your share of one day: about 40,000 Wh, or 40 kWh.

In the context of this day, here is what the ChatGPT session cost: **0.04%** of your day's energy use.

That is not a typo. Every AI query you made today, added together, was one twenty-five-hundredth of your day's energy footprint. The AC alone used roughly 500 times as much. The car ride used roughly 500 times as much. The cheeseburger at lunch used roughly 200 times as much. Even the dryer used more.

This is the part of the story that gets almost no airtime on TikTok.



Okay, But What About Training?

Here is where the numbers get more serious, and where I have to be honest about what we don't know.

The AI models you use — ChatGPT, Claude, Gemini, whichever — were *trained* before anyone ever talked to them. Training is the expensive part. It involves running thousands of specialized chips, nonstop, for weeks or months, processing trillions of words of text.

The companies that train these models do not, for the most part, publish their energy numbers. We have estimates.

A model the size of GPT-4, the best public estimate is that training used roughly **50 gigawatt-hours** of electricity. A gigawatt-hour is a million kilowatt-hours, so this is 50 million kWh. A big number.

Here is what 50 million kWh is, in context:

- About what 5,000 American homes use in a year.
- About what one large airport uses in a year.
- About what 60 fully-loaded transatlantic flights use, total.
- About one-seventh of the electricity used in Las Vegas in a single day.
- About 0.0002% of global annual electricity consumption.

Here is the other thing you need to know about training. It happens *once*. The model you talk to today was trained a year ago, and the energy was spent then, and it's not being spent again. Every conversation any person has with the model from now until it's retired comes out of that fixed training cost, spread across hundreds of billions of conversations.

Amortized across the conversations it will eventually have, the training cost of a single query is very small. Much smaller than the inference cost — the cost of actually generating the response you see. The headlines about AI training's energy use are real, but they are a one-time capital expense, not a recurring cost.



Data Centers, Globally

Now zoom all the way out. All the data centers in the world — every Google search, every Instagram photo, every streaming show, every AI model, every email, every spreadsheet stored in the cloud, every video call, every game, every bank transaction, every hospital record — together use about **1 to 1.5% of global electricity**. Current estimates are around 300–400 terawatt-hours a year, out of a global total of around 28,000.

For comparison:

- Residential lighting worldwide: about 4%.
- Cement production: about 2%.
- Aluminum smelting: about 1%.

- Cryptocurrency mining: about 0.5%.
- Air conditioning: about 10% and rising fast.
- Livestock agriculture (all energy inputs): about 15%.

Data centers are real. They use real energy. But they are not, in 2026, remotely the largest use of electricity on Earth, and they are not obviously the fastest-growing, and they are not going to “boil the oceans,” which is a phrase you may have encountered online and which does not correspond to any number anyone can produce.

Where the number *is* rising fast is AI specifically. The best estimates suggest that AI-related data center load could grow from maybe 10% of total data center energy today to 30% or 40% by 2030. That’s the part of the story that has the energy analysts actually worried. Whether that growth is fine or alarming depends on two things: how fast the grid decarbonizes (how much of that electricity comes from the sun, wind, and atom rather than gas and coal), and how fast AI chip efficiency improves (which, historically, has been very fast — AI chips have gotten roughly 30% more efficient per year for the last decade).

Both of those trajectories are uncertain, and honest people disagree about them. If you hear someone claim certainty in either direction, they are probably selling something.



But Globally Hides Locally

The “1 to 1.5% globally” number is true. It is also, on its own, misleading, in a way that the rest of the chapter will spend some time correcting.

Here is the problem with global percentages. The Earth is large. A small percentage of a very large total is still an enormous amount, but more importantly, the small percentage is *unevenly distributed*. A handful of regions host most of the world’s data centers, and in those regions, the share of electricity going to data centers is not

1.5%. It is closer to 20%, 30%, or 40%, depending on the specific region and how far the buildout has gotten.

This matters for two reasons. First, because the people living in those regions experience the buildout's effects regardless of what the global average is. Second, because the financial cost of expanding the grid to serve the data centers is mostly paid by the residents and businesses *in the regions where the buildout is happening*, not spread across the global population whose AI queries the data centers are serving.

Let me show you what this looks like in numbers, using the United States's largest regional electricity market.

PJM Interconnection is the grid operator covering all or part of thirteen states and the District of Columbia, from Illinois east to New Jersey, south to Virginia and North Carolina. Roughly 65 million Americans get their electricity through PJM's grid. PJM runs an annual capacity auction in which it pays generators to be available for future demand. The prices set in those auctions feed directly into residential electricity bills across the entire region.

For the 2025–2026 delivery year, PJM's capacity prices were the highest in the system's history. PJM's own Independent Market Monitor — the office whose statutory job is to evaluate the market for the grid operator — concluded that data center demand was *the primary cause* of the price increase. The Monitor's analysis attributed \$9.3 billion in additional capacity costs to data center load, an increase of 174% over what prices would have been without that load. This is not an activist claim. It is the grid operator's own analyst, applying the grid operator's own methodology, saying clearly that residential and commercial electricity customers across thirteen states are paying billions of additional dollars because of data centers.

Retail electricity prices in PJM and adjacent markets have reflected this. Between March 2021 and March 2026, average residential electricity prices rose:

- 94% in Washington, D.C.

- 74% in Maryland
- 73% in Maine
- 58% in New York
- Substantial increases in essentially every state across PJM, ISO-NE, and NYISO

Nationally, electricity prices rose nearly 40% over five years — the fastest sustained increase in American history. Utilities requested \$31 billion in rate increases in 2025 alone, double the 2024 figure. About half of the 2025 requests had not yet been approved by early 2026, which means the largest wave of rate increases is still working its way through the system.

Not all of this is data centers. Some is general inflation. Some is the cost of grid maintenance that was deferred for decades. Some is the transition away from older generation. But a substantial portion — by the grid operator’s own analyst, *the primary portion* of the recent capacity-cost increase — is data center load.

The structural feature of this that I most want you to understand is the *who pays* part. The grid is a shared system. When a new very large customer connects to it — say, a 500-megawatt data center — the cost of the new substation, the new transmission lines, the new generation capacity all get distributed across every customer in the region, according to formulas set by utility regulators. The data center pays its share. Its share is calculated as if it were a customer rather than as the *cause* of the new infrastructure. The rest of the customers — meaning ordinary households and small businesses, mostly — pay the difference.

The principle that “the customer causing the cost should pay for the cost” is well established in utility economics. It is called *cost causation*. Cost causation, properly applied, would put most of the grid expansion cost on the data centers requiring the expansion, rather than on residential ratepayers. In practice, this has mostly not happened, because the data centers and their parent companies

have substantial legal and political resources to participate in rate-case proceedings, while residential ratepayers do not.

Some states are starting to push back. Virginia's Senate Bill 253, introduced in 2026, would direct the state's utility commission to shift more capacity costs onto data centers specifically. The analysis projected residential rates would drop about 3.4% and data center rates would rise about 15.8%. By mid-2026, at least eighteen states had introduced bills creating special rate classes for very large energy users, with the explicit goal of preventing the cost shifts described above. The fact that these bills are being introduced *at all* tells you that the current rate structure is widely understood to be unfair. The fact that they have mostly not passed tells you who has the political resources to fight them.

So: globally, data centers are 1–1.5% of electricity. Regionally, in the areas hosting the buildout, the share is much higher, the prices are much higher, and the costs are falling on people who mostly do not use what the data centers produce. The global number is real. The local experience is also real. Both are true. The local experience is the one most people in PJM are actually living with.



Where the Real Problems Actually Are

So far this chapter might read as if I'm saying "the energy cost of AI is small, stop worrying." That is not what I'm saying. There are real problems, and I want to name them clearly.

Local grid strain. Discussed at length in the previous section. The short version is that the global "1–1.5% of electricity" figure hides regional concentrations that strain specific grids, and the cost of expanding those grids falls disproportionately on residents and small businesses who do not use the data centers driving the expansion. This is not a future problem. It is a present problem, visible in PJM's capacity auction results and in residential electricity bills across thirteen states.

Water. Chip fabrication uses enormous amounts of ultrapure water — a big fab can use 10 million gallons a day. Data center cooling also uses water, though less than fabs, and mostly in regions where water is actually scarce. Some new data centers being built in Arizona and Nevada are using techniques that use almost no water (air cooling, closed loops), but older ones in drought-affected areas are a genuine problem. You are not wrong to worry about this; you should just know that the industry is aware and the best new designs are much better than the old ones.

Power source matters more than power amount. A data center running on solar-plus-battery in Arizona has a very different environmental impact than a data center running on coal in West Virginia. Two data centers doing the exact same work, using the exact same amount of electricity, can have carbon footprints that differ by a factor of twenty depending on where their power comes from. When people argue about the environmental cost of AI, they are often unknowingly arguing about different data centers in different places, using different power, and talking past each other.

Embodied energy in the chips themselves. Every chip in a data center took energy to make — mining the materials, refining the silicon, running the fab, shipping the chip. For a server that will run for five years, the embodied energy is typically 10–20% of its lifetime total. This is not huge, but it is not nothing, and it is often left out of the accounting.

What we don't know, which matters. The big AI companies do not publish detailed energy figures for their training runs or their inference infrastructure. Estimates come from leaks, indirect calculations, and informed guesswork. The numbers in this chapter are the best public estimates, but they could be off by a factor of two in either direction, and the lack of transparency is itself a reason to be skeptical of any confident claim — including, to be fair, the confident claims in this chapter.



How Much Of The Growth Is Real?

There is one more thing about the energy projections that the rest of the book will spend more time on, and that I want to flag here because it changes how you should read the “rising fast” language.

The energy growth projections for data centers — the ones that have utility commissions and grid operators worried, the ones that justify the capacity buildouts described above, the ones that show data center electricity going from 1.5% to 4% or more of total U.S. electricity by 2030 — all rest on a specific assumption. They assume that the *current pattern* of AI usage continues and grows. Specifically, they assume that the substantial majority of AI inference (the moment-by-moment use of trained AI models to answer queries, generate text, produce images, and so on) continues to flow through hyperscale data centers, billed through frontier-model APIs, at something like the current pricing.

This assumption is not guaranteed to hold, and a later chapter takes apart the reasons it may not. The short version: AI capability is migrating to edge devices faster than the projections assume. Phones and laptops in 2026 run models that three years ago would have needed hyperscale clusters, and a 4-billion-parameter model on a phone handles the substantial majority of what ordinary people actually use AI for.

The cost gap is about three orders of magnitude — local inference runs at roughly one-thousandth the cost of a frontier API, for tasks the local model handles adequately. As users and enterprises wake up to that gap, much of the inference now running in hyperscale data centers migrates to the edge, where the energy is amortized over devices people already own and drawn from a wall outlet at retail rates.

This matters because much of the buildout is justified by demand projections that the people funding it have a financial interest in keeping high. The hyperscalers want high projections to justify their capacity and their valuations; the utilities want them to justify

rate-case requests and recoverable infrastructure spending; the financial system around the buildout wants them to support its equity valuations and bond issuances. None of these parties has any incentive to forecast that demand will soften, and the projections that reach news coverage almost all come from them or their consultants. The result is a systematic upward bias in the numbers that drive energy planning.

This does not make the projections worthless. Some growth is real, especially for training and specialized applications. But the honest version of “AI energy use is rising fast” is narrower: it is rising fast *at the current architecture*, and a substantial share of the rise comes from inference patterns that may not persist as small models improve and enterprises rationalize their cloud spending. That is the sentence the projections should be using and mostly are not — and the residential ratepayers funding the resulting grid expansions are being asked to bet that the softer reading is wrong.



Architecture, Not Amount

Connected to the above point but worth its own treatment: the energy cost of AI depends much more on *where* the inference runs than on *how much* of it there is.

Consider a specific task — say, summarizing an email — and trace the energy cost under different architectures.

If the summary runs through a frontier-model API: your phone sends the email text to a hyperscale data center, which loads the email into a model running on \$40,000 GPUs cooled by industrial-scale chillers, in a building drawing 100 megawatts of power, on a grid that has been expanded to serve that data center at residential ratepayer expense. The inference itself takes perhaps half a second of GPU time, consuming a few watt-hours. The total energy cost of the inference, properly accounted, includes the share of the data center’s idle-but-running infrastructure, the share of the cooling

load, the share of the grid expansion. The end-to-end energy might be ten to twenty watt-hours, depending on how you count.

If the summary runs on a 4-billion-parameter model on your phone: your phone never sends the email anywhere. The model, already loaded into the phone's neural processing unit, runs the inference in a few hundred milliseconds, drawing perhaps 0.1 watt-hours from the phone's battery — energy that came from the wall outlet you plugged the phone into last night, at residential rates, with no marginal grid expansion required. The end-to-end energy cost is one one-hundredth of the cloud version, and you didn't ask any utility or any data center to add capacity to serve you.

Same task. Two architectures. Two orders of magnitude difference in energy cost. The difference is not in how good AI is or how much people use it. It is in *where the inference runs*.

The chapter on self-driving — which comes a few chapters later in the book — describes the same architecture argument in more detail. Self-driving was forced onto an edge-inference architecture by the physical constraints of real-time driving. The result is that the world's fleet of partially and fully autonomous vehicles, running billions of inference operations per day, has an energy footprint that mostly does not show up in hyperscale data center demand. Each car's compute runs on the car's battery, paid for at the rates the car owner pays for charging, with no centralized grid expansion required.

Route general inference the way self-driving routes it, and the demand-growth projections shrink dramatically. The “rising fast” trajectory is partly real growth and partly an artifact of sending inference to hyperscale data centers that did not need to handle it — a business-model preference on the providers' part, not a technical necessity, whose energy cost is paid by ratepayers and host communities.



What This Actually Means

The honest version of the energy story, after everything in this chapter, is more complicated than the version most people will tell you.

The global aggregate is small. Data centers worldwide use 1 to 1.5% of global electricity. AI is a fraction of that. Your individual AI queries cost a few watt-hours apiece, less than almost anything else you do. The “AI will boil the oceans” framing is not supported by any number anyone can produce. The people telling you that AI energy use is catastrophic at a global level are mostly wrong, in the literal sense that their numbers do not add up.

The regional reality is large. In the specific regions hosting the build-out — Northern Virginia, parts of Ohio, the Memphis metropolitan area, parts of Arizona, Texas, Oregon, Iowa, and several other concentrated regions — data centers are a substantial share of the local electricity load, and the cost of expanding the grid to serve them is falling, through capacity-market mechanisms and rate-case proceedings, on residential and small-business customers across thirteen states in PJM alone. The 94% residential electricity price increase in Washington, D.C. is not abstract. It is paid, every month, by ordinary households who do not use the data centers driving the cost. The cost-shifting is real, large, and unfair.

The architecture is wrong. Much of the inference currently flowing through hyperscale data centers does not need to be there. The same tasks could be done on edge devices, at one-hundredth the energy cost, with no marginal grid expansion required. The reason most inference runs in the cloud is not technical necessity; it is business-model preference on the part of the providers. The energy implications of those preferences are being paid by people who do not benefit from them.

The growth projections are biased upward. The “AI energy demand will quadruple by 2030” projections all assume that the current routing of inference through hyperscale data centers continues. The

chapters that follow this one will explain in detail why that assumption is likely to be partly wrong. To the extent it is wrong, the grid expansions being built right now are being sized for a demand pattern that may not hold, and the residential ratepayers being asked to fund those expansions are being asked to fund infrastructure that may end up underutilized.

The grid decarbonization argument is still real. Whatever the AI energy load turns out to be, its carbon footprint depends on what powers it. A buildout that runs on solar, wind, and nuclear has a small carbon footprint. A buildout that runs on natural gas — including the on-site gas turbines now being installed by AI infrastructure operators in residential neighborhoods, often without permits — has a large one. The grid leverage point remains the most important environmental leverage point, and people working on grid decarbonization are working on the right thing.

None of this is a reason to abandon AI as a category. The technology is real, the productivity gains are real, and some of the applications (self-driving, scientific computing, drug discovery, genuine education tools) are worth the energy they consume. The argument is about *how* to build the infrastructure, *where*, and *who pays for it*, not about whether AI should exist.

What follows is that “the energy cost of AI” is not a single number to be for or against. It is a set of specific costs, falling on specific people, produced by specific architectural and financial choices that could mostly be made differently.

Which is what separates the vague version from the actionable one. “AI uses too much electricity” is not a policy. “Data centers should pay for the grid capacity they require through their own rate class, rather than spreading the cost across residential ratepayers” is a policy. So is “AI inference should run on edge devices by default, with cloud routing reserved for tasks that genuinely require frontier-scale capability.” So is “on-site combustion power in residential areas should require full permitting and full emissions accounting.”

Each of those is specific enough to argue about, fund, or fight. The vague feeling is not.



The Skeptic's Checklist

When someone tells you the energy cost of a technology, here are the questions to ask, in order:

1. **What's the unit?** If they don't tell you Wh, kWh, MWh, or GWh, they probably don't know.
2. **Per what?** Per query? Per user? Per year? Per training run? A number without a "per" is meaningless.
3. **Compared to what?** Five kWh is a lot compared to a phone charge and nothing compared to a flight. Every claim needs a baseline.
4. **Which data center?** Powered by what? In what climate? Using what cooling?
5. **Is this the total, or the marginal?** The total cost of all data centers is a big number; the marginal cost of your specific use is a small one. Don't confuse them.
6. **Who's the source, and what do they want you to conclude?** An environmental group estimating high, an AI company estimating low, and a utility grid operator estimating carefully have very different incentives. Read accordingly.

If a claim survives all six questions, it is probably worth taking seriously. Most claims, unfortunately, don't survive the first two.



What Comes Next

The next chapter is about water; the one after that, mining and e-waste. Each has the same structure: real numbers, honest uncertainty, and a careful separation of the parts that are a problem from the parts that only feel like one.

Several chapters later, the book reaches the parts this one has been gesturing at — the specific costs imposed on the specific people who live near the infrastructure, the architecture that would do less damage if the industry chose to use it, the financial structure of the buildout and who is funding it, and the question of whether the current execution is producing outcomes worth the cost. Those chapters are darker than this one, because the more you learn about the specifics, the harder it becomes to read the global aggregates as the whole story.

One more thing worth flagging, which the rest of the book will come back to: what if the scarce resource isn't the electricity at all? What if it's the political legitimacy that gets spent every time a community is forced to host infrastructure it did not consent to? What if the durable cost is not measured in watt-hours but in the corrosion of the systems we rely on to make collective decisions? Hold the question. The book will come back to it.



A note on sources: the per-query AI energy figures are based on published estimates from the International Energy Agency, academic studies (particularly those by Alex de Vries and the Hugging Face team), and leaked or disclosed figures from the major AI companies when available. The 50 GWh training figure for GPT-4-class models is a midpoint of public estimates; the real number could be half or double that. The comparison figures (flights, burgers, AC) are from standard life-cycle assessments. When in doubt, I've used the higher end of plausible estimates for AI and the middle of the range for everything else, so that the comparison is conservative in AI's disfavor. Even with that bias, AI's per-query energy use is small compared to almost anything else a person does in a day.



CHAPTER 6

The Water Question, Honestly

Why the Answer Depends Almost Entirely on Where You're Standing



If you spend any time on the internet, you have probably seen some version of this claim: *training GPT-4 used enough water to fill a swimming pool. Every ChatGPT query you send drinks a bottle of water. Data centers are draining the planet.*

Some of that is roughly right. Some of it is exaggerated by a factor of ten. Some of it is technically true in ways that turn out, when you look closely, to mean almost nothing. And some of it is genuinely concerning, but for reasons that are very different from what most people mean when they repeat it.

This chapter is going to walk through the actual water story — where water is used, why, how much, and in which places it actually matters. The honest summary, which I'll expand on below, is this: **the water problem with chips and data centers is real, but it is overwhelmingly a local problem, not a global one, and the local problems are concentrated in a small number of places that most of the scary statistics don't distinguish.**



How Water Gets Used, and Where

There are two completely different ways chips and data centers use water, and it's important to keep them straight because the stories are very different.

The fab uses water to make the chip. A chip fabrication plant — the factory from Chapter 3 — uses enormous amounts of ultrapure water to rinse wafers between processing steps. Ultrapure water is regular water that has been filtered to the point where it is actually slightly toxic to drink (it leaches minerals out of your body). Fabs use this water because chips are so small that even trace contaminants in ordinary water would ruin them.

A single large fab uses somewhere between 5 million and 20 million gallons of water per day. TSMC, which makes most of the world's most advanced chips, uses about 150,000 metric tons of water per day across its facilities in Taiwan. That is a real and large number.

The data center uses water to stay cool. A big data center generates enormous amounts of heat — tens of megawatts of it, which is roughly the heat output of a small town. That heat has to go somewhere, and the cheapest, most efficient way to get rid of it is evaporative cooling: spray water on a surface, let the water evaporate, and the evaporation carries away the heat.

A big data center using evaporative cooling can use 1 to 5 million gallons of water per day. The amount varies enormously depending on how the data center is built.

These two uses — fabs and data centers — add up to, globally, something like 0.1–0.2% of total human water use. That's a real number. It's not zero. But it's also much, much less than people think.

For comparison:

- Agriculture: about 70% of all human water use, globally.

- Livestock (just the animals, not including feed): about 30% of agricultural water, or 20% of total.
- One pound of beef: about 1,800 gallons of embedded water.
- One cotton T-shirt: about 700 gallons.
- One pair of jeans: about 2,000 gallons.
- One almond: about 1.1 gallons.
- One American golf course: about 500,000 gallons of water a day.
- One American family, residential use: about 300 gallons a day.

Put another way: the entire global chip and data center industry uses less water per year than American residential lawn watering. Not lawn care in total — just the watering of grass.

This is the kind of context that almost never makes it into the viral posts, and it matters.



The “Water Per Query” Thing

There was a study in 2023 — a good one, by researchers at UC Riverside — that estimated the water cost of an AI query. The number they came up with was that a ChatGPT conversation of 20–50 responses used about 500 milliliters of water, or roughly one plastic water bottle.

This number went everywhere. It is the single most quoted statistic about AI and the environment. It is also more misleading than it appears, for reasons worth understanding.

First: that 500 ml includes *both* the data center’s direct cooling water *and* the water used to generate the electricity the data center consumes, upstream at the power plant. Power plants — especially coal and gas plants — use a lot of water for their own cooling. If the data center is running on coal, the water footprint is big. If it’s running on solar or wind, the water footprint is almost zero. The 500

ml figure assumes an average American grid mix circa 2022, which is mostly coal and gas. A data center running on solar could have a water footprint 50 times smaller.

Second: the 500 ml is for a 20–50 response conversation, not a single query. A single query is more like 10–25 ml — one or two tablespoons. The number got repeated as “a bottle of water per query,” which is, at a minimum, off by a factor of 20.

Third: even 500 ml of water per 50-response conversation is a small number in context. One hamburger contains about 1,700 gallons of embedded water — enough water for 13,000 ChatGPT conversations at the high-end estimate. The water you used to shower this morning was enough for several thousand of those conversations.

Fourth — and this is the part that matters most — **the water cost of a query depends almost entirely on where the data center is and how its electricity is made.** A query answered by a data center in Iowa running on wind-and-solar with closed-loop cooling uses essentially zero fresh water. The same query, answered by a data center in a desert running on gas with evaporative cooling, uses quite a lot. Averaging these together produces a number that describes no actual data center anywhere.

None of this means the 500 ml number is “wrong.” It’s a real estimate of a real, if complicated, quantity. It means that using that number as a cudgel to argue that AI is destroying water supplies is a misuse of the science. The researchers themselves have said as much, in interviews that got much less traction than the original number.



Where It Actually Matters

Now the other side of the honesty.

In some specific places, at some specific times, the water use of chip fabs and data centers is a real, serious, local problem. I want to

name these because a book that only defended the industry would be as dishonest as one that only attacked it.

Taiwan. TSMC's fabs sit in a region that has experienced serious droughts in recent years. In 2021, Taiwan went through its worst drought in 50 years, and the government had to prioritize water for fabs over agriculture, because the fabs are more economically valuable per gallon. This is a real tradeoff, and Taiwanese farmers were, legitimately, upset. TSMC has since built water recycling facilities that dramatically reduce their fresh water draw, but the fundamental vulnerability — the world's most economically important fabs sitting in a drought-prone island — remains real.

Arizona. The data center boom of the 2020s has put a lot of new capacity in the Phoenix area, which is in the middle of a 20-year megadrought in the Colorado River basin. Some of the new data centers use closed-loop cooling that consumes almost no water. Some use evaporative cooling that consumes a lot. Which kind of data center is built where has become a live political question in Arizona, and reasonably so.

Parts of Spain. Similar story: new European data center capacity being built in water-stressed regions, with local communities justifiably wanting to know which kind of cooling is going in.

Google's and Microsoft's disclosures. Both companies publish annual environmental reports, and both have reported rising water use year-over-year as they've built out AI infrastructure. Google's water use grew 20% in 2022 over 2021, to about 5.5 billion gallons across all operations. That is a real increase, concentrated in real places, and critics who point it out are not making it up. The companies have committed to "water positive" operations by 2030, meaning they will return more water to local watersheds than they consume, but those promises are not yet fulfilled, and skepticism is warranted until they are.

The fabs that are coming. The US government is spending tens of billions of dollars to subsidize new chip fabs being built in Arizona, Texas, Ohio, and New York. All of these will use millions

of gallons a day. In some of the locations, water availability is already a stress point. The policy debate about chip reshoring has not, so far, grappled seriously with the water side of the question, and it probably should.



Fresh Water vs. Withdrawn vs. Consumed

One more technical distinction, and then we'll be done with the unpleasant part.

When people talk about “water use,” they often conflate three different things:

Withdrawn water is water taken from a river, lake, or aquifer. Most of it is returned to the source (sometimes warmer, sometimes dirtier, but returned). This is the biggest number when you see industrial water figures.

Consumed water is water that is actually used up — evaporated, absorbed into products, or otherwise removed from the local water system. This is typically 10–20% of withdrawn water for industrial uses, and it's the number that actually matters for whether the local water table is being drawn down.

Potable fresh water is water that was suitable for drinking before it was used. A data center that uses drinking water is competing with human and agricultural needs. A data center that uses recycled wastewater or seawater or reclaimed greywater is not.

When a data center reports “5 million gallons of water used per day,” it might mean 5 million gallons withdrawn (most of which goes back to the river), or 5 million gallons consumed (which doesn't), or 5 million gallons of drinking water (which competes with human use) or 5 million gallons of reclaimed water (which doesn't). These are all very different, and the headlines almost always fail to distinguish them.

When you read a scary number about data center water use, the question to ask is: *withdrawn or consumed? Fresh or reclaimed? In what*

watershed? If the article doesn't answer those questions, the article doesn't actually know what it's claiming.



The Technologies That Change the Answer

The water story is not static. Several things are changing fast enough that the numbers in this chapter might be outdated by the time you read this book, and mostly in the direction of lower water use.

Closed-loop cooling. Newer data centers increasingly use systems where the cooling water circulates continuously through a closed loop and is cooled by contact with outside air, rather than evaporating into the atmosphere. These systems use a tiny fraction of the water of evaporative systems — sometimes less than 1% — but they use more electricity, so they're most efficient in cool climates. Microsoft's data centers in Ireland and Sweden use essentially no water for cooling. The tradeoff is real (more electricity vs. less water), and the right choice depends on the local grid and the local water situation.

Immersion cooling. A newer approach where servers are literally submerged in a liquid that isn't water — a dielectric fluid that conducts heat but not electricity. Zero water consumption. Higher capital cost, but catching on fast, especially for AI workloads that run very hot.

Reclaimed and non-potable water. Several data center operators now use recycled wastewater or seawater for cooling, which means they're not competing with drinking water supplies at all. Google's data center in Douglas County, Georgia, runs on treated sewage. This is weirder to think about than it sounds, and it is also obviously the right answer in a lot of places.

Water recycling in fabs. TSMC now recycles over 85% of the water it uses, up from around 30% a decade ago. Newer fabs are being built with even higher recycling rates. The chip industry's per-chip

water use has dropped significantly over the last 15 years, even as total chip production has gone up.

None of this erases the concerns, but it does mean that the *direction of travel* is better than it used to be, and the “water use will spiral out of control” scenarios assume a continuation of old practices rather than current best ones.



Putting It Together

The honest summary of the water question looks like this.

Globally, chip fabs and data centers use a small fraction of human water — somewhere around 0.1–0.2% of the total. This is real but small compared to almost anything else (agriculture, lawns, golf courses, the cooling needs of power plants themselves).

Locally, in specific drought-affected regions — Taiwan, Arizona, parts of Spain, parts of Texas — the water use is a real problem that is worth paying attention to and regulating carefully. The question is less “should data centers exist” and more “what kind of cooling should they use, and in which watersheds.”

The technology is moving in the right direction (closed-loop, immersion, recycled water), but not fast enough that you should ignore the problem. It’s also not the case that every new data center is making things worse; some of them are genuinely well-designed and use essentially no fresh water.

The “bottle of water per query” meme is based on a real study, but that study’s numbers are mostly *upstream power plant water*, which drops to near-zero when the grid is clean. As grids decarbonize, the water footprint of AI drops with it — the same coupling we saw with energy in Chapter 5.

The companies’ disclosures are getting better but still aren’t good enough. “Water positive by 2030” is a promise, not a result, and promises are worth exactly what past promises from the same companies have been worth, which is a mixed record.



The Checklist, Again

When someone tells you the water cost of a technology, the questions to ask, in order:

1. **What's the unit?** Gallons? Liters? Per day? Per year? Per query?
2. **Withdrawn, consumed, or potable?** These are different by a lot.
3. **Where?** A data center in Arizona is not a data center in Finland.
4. **What kind of cooling?** Evaporative, closed-loop, immersion?
5. **Is the upstream power plant water included?** Because that's often the biggest piece, and it depends on the grid.
6. **Compared to what?** 500 gallons a day is almost nothing for a data center and almost everything for a house.

A number without answers to these questions is not an argument. It is a vibe. Vibes are not what you use to decide things.



A Personal Observation

The water chapter is, honestly, the chapter I find hardest to write, because the emotional valence of water is so strong.

Energy is abstract. Kilowatt-hours don't evoke much. But *water* — water is what you drink, it's what you shower in, it's what keeps your body alive, it's what goes into your dog's bowl. When someone says a data center “drank” water, you viscerally feel like it took something from you. The metaphor is doing a lot of the work.

The reality is more mundane. Most data center water is returned to the local watershed, often cleaner than it came in (because it had

to be filtered to ultrapure standards before use). The fraction that's actually consumed — evaporated away — is small. And almost all of it is in specific places that are specific problems, not a diffuse global crisis.

I want you to be able to tell the difference between a *visceral* claim and a *quantitative* one. Visceral claims are easy to make, hard to argue with, and frequently wrong. Quantitative claims take more work to state, more work to understand, and more work to criticize — and they are what you need if you are going to make actual decisions about actual policy.

The data center in Arizona that is drawing down the aquifer is a real problem, and you should care about it. The ChatGPT query you sent this morning is not. These are different facts, and keeping them separate is most of what good thinking about this stuff consists of.



A note on sources: the TSMC water figures come from TSMC's own sustainability reports, which are unusually detailed. The UC Riverside study is Li et al. (2023), "Making AI Less 'Thirsty.'" Data center industry water figures come from the Lawrence Berkeley National Laboratory's data center reports and from the hyperscalers' own annual sustainability reports. The agricultural water comparisons are from the Water Footprint Network, which is the standard reference. When figures conflicted — and they often did — I used the midpoint of credible estimates and flagged the uncertainty.



CHAPTER 7

The Mining and E-Waste Question, Honestly

The Chapter Where the Problems Are Actually as Bad as You've Heard



I owe you this chapter. After two chapters where the answer was “the problem is smaller and more local than you’ve been told,” it might be tempting to assume this whole book is going to be a long defense of the industry. It isn’t, and this chapter is where that becomes clear.

The energy and water stories, honestly assessed, are mostly stories of real but manageable problems, improving technology, and misleading headlines. The mining and e-waste story is different. In places, at the extremes, it is as bad as the worst things you have heard. Some of it is worse than you have heard, because the worst parts don’t get much airtime in the West.

This chapter is going to be uncomfortable. I think that’s appropriate.



Two Categories, Kept Separate

First, a distinction that most coverage smooshes together.

When people talk about “mining for tech,” they are usually talking about *two different supply chains with two different problems*, and it helps a lot to keep them straight.

The chip supply chain. Silicon (for the chips themselves), copper (for wiring), a small amount of gold and other metals for contacts, some rare earth elements for specific applications. Globally, chip manufacturing is a small consumer of raw materials. The silicon is basically purified sand. The quantities of metals involved per chip are tiny. The environmental footprint of the raw materials going *into* chips is relatively modest.

The battery and device supply chain. Lithium, cobalt, nickel, rare earth magnets, all the metals in a modern rechargeable battery. Globally, this is enormous and getting bigger — because electric cars, grid storage, and all those phones and laptops have huge battery demand. The chip inside your phone has a small material footprint. The battery next to it has a large one.

Almost everything people worry about — child labor, environmental devastation, human rights violations — is in the *second* supply chain, not the first. The phone in your pocket is, materially speaking, mostly a battery with a chip stuck to it, and the moral problems are mostly in the battery.

This matters because arguments about “the ethics of AI chips” sometimes borrow moral weight from mining practices that have approximately nothing to do with AI chips. Cobalt mining in the Democratic Republic of the Congo is genuinely appalling. It is also almost entirely about batteries, not compute. A data center full of GPUs involves very little cobalt. A Tesla involves a lot.

There is, I should say up front, a *third* lens that this book in particular needs, beyond these two supply chains: the hardware churn of the data centers themselves — what the buildout buys, runs hot, and retires. That is the part of the mining-and-waste story that is genuinely about AI rather than about phones, and I’ll come to it near the end of the chapter. But the human horrors are mostly in the device chain, so that is where I’ll start.

I am going to cover all of this, because your phone has both things in it, the buildout has a hardware life of its own, and you deserve the full picture. But please notice the distinctions, because they will come up again.



The Cobalt Story

Let me start with the worst one, because if you've heard one mining horror story, it is probably this one.

About 70% of the world's cobalt comes from the Democratic Republic of the Congo — specifically from Katanga Province in the southern part of the country. Cobalt is used in lithium-ion battery cathodes. It makes the batteries more stable and longer-lived. Every smartphone, every laptop, and most electric cars on Earth today contain cobalt that was, with high probability, mined in the DRC.

Most DRC cobalt — maybe 80% — comes from industrial mines, which are bad in the ordinary ways big mines are bad (environmental destruction, corruption, poor labor conditions), but roughly comparable to mines elsewhere in the world.

The other 20% comes from what are called “artisanal” mines. This is a polite word. In practice, artisanal mining means people — including, yes, children — digging cobalt ore out of the ground with hand tools, sometimes in shafts that have no structural support, sometimes with no protective equipment of any kind. Workers get paid a few dollars a day. Cave-ins happen regularly. Long-term exposure to cobalt dust causes respiratory diseases and cancer. Children working in these mines — estimated at 40,000 in the DRC, though real numbers are impossible to verify — are sometimes as young as seven.

The material from artisanal mines flows into the same supply chain as the material from industrial mines, gets refined at the same facilities (mostly in China), and ends up in batteries indistinguishable from the “clean” kind. Unless the battery manufacturer has

invested heavily in supply chain traceability — and most haven't, until recently, under pressure from regulators and reporters — there is no way to know whether the cobalt in a specific device came from an industrial mine or a collapsing pit with a 12-year-old in it.

This is an unambiguously bad situation. I am not going to soft-pedal it. You have likely benefited from it, as have I. The smartphone economy is partially built on it. Pretending otherwise is dishonest, and pretending it will fix itself is naive.

What *is* happening, slowly: major companies (Apple, Tesla, the big automakers) are spending serious money on traceability. Some are reducing cobalt content in their batteries, because it's valuable in its own right to depend less on DRC supply. Battery chemistries that use less cobalt or no cobalt (lithium iron phosphate, for instance) are growing fast — China's new EVs are mostly cobalt-free. This is progress. It is also slow, and the industry has known about artisanal mining for at least 15 years, so the progress should have been faster.

If you want to do something about this specific issue, the most effective thing is probably to pay attention to which companies are being transparent about their supply chains and which aren't, and to support regulations (like the EU's Battery Passport initiative) that force everyone to disclose. Individual consumer choice here has limited leverage — all smartphones source from roughly the same refineries — but collective pressure has moved the industry, slowly, in the right direction.



The Lithium Story

Lithium is the other battery material worth talking about, and its story is different from cobalt's.

Lithium mostly comes from two places: hard rock mines in Australia, and brine pools in the “lithium triangle” — the salt flats of Chile, Bolivia, and Argentina. Australia and Chile together produce about 75% of the world's lithium.

The Australian mines are ordinary open-pit mining operations, environmentally unremarkable by the standards of mining generally. The labor conditions are fine (Australia has strong labor laws). The environmental impact is real but contained.

The South American brine extraction is more interesting and more controversial. Lithium-rich salty water is pumped from underground aquifers into huge evaporation ponds. Over 18 months, the water evaporates, and lithium-rich concentrate remains, which is then processed into battery-grade lithium carbonate.

The issue is that those aquifers are shared with local communities — indigenous Atacameño people in Chile, among others — and with the local ecosystem, including flamingo populations that depend on the saline wetlands. How much the lithium extraction is actually drawing down the local water, and whether it is harming the ecosystem, is genuinely disputed in the scientific literature. Some studies show major impacts. Others show minimal ones. Reasonable scientists disagree.

What is *not* disputed is that the decisions are being made mostly by distant companies and governments, and the people who live there are not getting much of the money or much of a say. Whether or not the environmental impact turns out to be severe, the *procedural* injustice — local communities being asked to bear whatever the impact is, without meaningful consent or compensation — is real.

This is a different kind of problem than the cobalt one. It's less about graphic suffering and more about fairness, consent, and scientific uncertainty. But it's a real problem, and the "transition to clean energy" that electric vehicles and grid batteries are part of is, to a non-trivial degree, being built on top of it.



The Rare Earths Story

"Rare earth elements" are a group of 17 metals — lanthanum, neodymium, dysprosium, and so on — that are used in small quantities

in lots of high-tech things, including the powerful magnets in electric motors, certain lasers, some chip manufacturing processes, and wind turbine generators.

They are not actually rare. The name is historical. What is rare is the economically viable way to extract them, because the ores they come from are almost always mixed with thorium and other mildly radioactive elements, which makes processing them messy.

About 60% of the world's rare earth production is in China, mostly in Inner Mongolia and Sichuan. The main processing site, near Baotou, has a tailings lake that is one of the most polluted places on Earth — a black, muddy, mildly radioactive 10-square-kilometer sludge pit that local villages have had to relocate away from. You can find photographs of it. They are as bad as they sound.

This is sometimes held up as a reason to be worried about green technology, and the worry is not crazy: if wind turbines and EVs depend on neodymium magnets, and neodymium comes from Baotou, then the “clean” technology has a genuinely dirty upstream.

A few things to add to that picture, though. First, *per unit of rare earth*, Chinese processing has gotten much cleaner over the last 15 years under domestic pressure. Second, new rare earth production outside of China is coming online — Lynas in Australia, Mountain Pass in California — with much better environmental practices. Third, the total quantity of rare earths involved in a chip or an EV is very small — a few grams per car for the motor magnets — which means the problem scales less than you might think. Fourth, a lot of high-end applications are moving away from rare-earth magnets toward alternatives, partly for supply chain reasons, partly for cost.

Still: a lot of the magical thinking about “green tech” involves pretending that the upstream doesn't exist. It exists. Someone lives downstream of it.



The E-Waste Story

Now we come to the back end of the supply chain, which is what happens to your phone and your laptop when you're done with them.

Globally, humans generate about 60 million metric tons of electronic waste per year. This is growing at about 3–4% per year. Only about 20% of it is formally recycled. The rest goes into landfills, gets stockpiled, or — most concerning — ends up in informal processing operations in low-income countries.

The informal processing is the part you should know about. There are specific places — Agbogbloshie in Ghana, Guiyu in China (now largely cleaned up, but for twenty years the worst in the world), parts of India and Pakistan — where large quantities of rich-world e-waste get shipped, sometimes legally and sometimes not, to be disassembled for the metals inside.

“Disassembled” in this context means: people, often young people, often without protective gear, burning plastic off copper wires to recover the copper, soaking circuit boards in acid to extract the gold, smashing CRT monitors to get at the components inside. The air in these places is full of lead, mercury, and cadmium. Groundwater is contaminated. Soil is contaminated. Local cancer rates are elevated. Infant mortality is elevated.

This is happening. It is happening now. It has been happening for thirty years. It is improving in some places — Guiyu has been partially shut down; Ghana's Agbogbloshie was formally closed in 2021, though some processing has moved to other sites — but the fundamental dynamic remains: rich countries produce e-waste, poor countries process it, and the people doing the processing get sick.

The device in your pocket will eventually become e-waste. What happens to it when you're done with it depends heavily on where you are and what you do. In the EU and increasingly in some US states, there are laws requiring electronics retailers to take back

devices for proper recycling. In most of the world, most devices end up in the regular waste stream, and a significant fraction end up, eventually, in one of these informal processing sites.

The best thing an individual person can do about this is the most boring thing: keep your devices longer, and when you do get rid of them, take them to a certified e-waste recycler (R2 or e-Stewards certified in the US). This actually matters. Not because your individual phone is a big deal, but because the choice of where you send your old phone flows into the broader system in a way that individual choices don't flow into most environmental problems.



The Silicon and Chip Footprint, For Completeness

Given that this book is nominally about chips, let me actually cover the chip-specific material story, which is — genuinely, thankfully — not terrible.

Silicon. The silicon in chips is refined from quartz sand, which is one of the most abundant materials on Earth. The refining is energy-intensive, but not particularly environmentally damaging in any other way. Sand mining, in general, has environmental issues — coastal erosion, illegal sand extraction for concrete — but the quantities used for chips are vanishingly small compared to those used for construction. A year of global chip production uses maybe 25,000 tons of silicon. A year of global concrete production uses 30 billion tons of sand. These are different orders of magnitude.

Chemicals and gases. Chip fabrication uses an enormous variety of highly toxic and/or environmentally persistent chemicals — per-fluorinated compounds, arsine, phosphine, hydrogen fluoride, nitrogen trifluoride, and so on. Fabs are tightly regulated in the US, EU, Japan, Korea, and Taiwan, and emissions are low. Fabs in less-regulated places are more variable. Some of the gases — particularly nitrogen trifluoride, used in cleaning — are very potent greenhouse

gases, and industry emissions are non-trivial but well under 0.1% of the global total.

The fab's total footprint. Depends on the fab and the assessment, but the *material* footprint of a chip is small compared to almost anything else in the device it ends up in. Most of the environmental cost of making a phone is in the mining and assembly, not in the chip itself.

This doesn't mean chip manufacturing is zero-impact. It means that, of the things to worry about in the device stack, the chip itself is probably not the biggest one.



The Data Center's Own Hardware

So far this chapter has mostly been about the device in your pocket — its battery, its eventual disposal. That is where the worst of it is, in human terms. But this book is about data centers specifically, and a data center has a hardware life of its own, which is worth looking at directly, because it is large, it is growing fast, and it is the part of the mining-and-waste story that is actually about AI rather than about phones.

Start with how fast the hardware turns over. A traditional server was expected to last five to seven years. AI servers have compressed that to something between eighteen months and three years. Part of the reason is that each new generation of GPU is dramatically faster, which makes the previous generation economically inefficient to keep running at scale. But part of it is physical. The training GPUs run at sixty to seventy percent utilization, around the clock, each one drawing more than seven hundred watts and turning nearly all of it into heat. Run that hot, that constantly, and the hardware wears out. An anonymous senior engineer at one of the major labs put the real working life of a heavily loaded training GPU at one to three years.

There is an accounting wrinkle here worth knowing, because it connects to the financial story later in the book. On their books, the hyperscalers depreciate this hardware over *six* years — a schedule they all moved to in 2023 and 2024, and one that conveniently raises their reported operating income by billions of dollars, because the cost of the hardware is spread across more years. Several careful observers — the investor Michael Burry most loudly — have argued that this overstates how long the hardware actually lasts, and that the real figure is closer to two or three years. If they are right, the buildout is not only consuming more capital than it appears to; it is also generating retired hardware faster than the official schedules imply.

Either way, the volume is large and rising. A single facility like Colossus holds more than two hundred thousand GPUs; cycle that every few years and you have an enormous, recurring stream of retired hardware — not just the GPUs but the CPUs, the memory, the circuit boards, the networking gear, and the backup batteries. A study in *Nature Computational Science* estimated that, under aggressive adoption, large language models alone could generate something like two and a half million tonnes of electronic waste a year by 2030 — up from almost nothing in 2023. That is a new waste stream, created in roughly a decade, specifically by the AI buildout.

Here, though, is the honest other half, and it is genuinely better news than the phone story. Data center hardware is, so far, handled far better than the phone-in-a-drawer-to-Agbogbloshie pipeline. There is a real and active secondary market: a retired training GPU is not garbage, it is a slightly older but still valuable inference chip, and it gets resold rather than smashed. The “value cascade” in the industry runs a GPU from front-line training in years one and two, to inference in years three and four, to cheap batch work after that — one company, CoreWeave, has rebooked H100s coming off expired contracts at ninety-five percent of their original price. The large operators run formal asset-disposition programs with documented chain of custody; Microsoft reported reusing or recycling

more than ninety percent of its decommissioned server components in 2024. So the data center's hardware footprint, unlike the consumer one, is for now mostly captured rather than dumped.

The open question is whether that holds. The secondary market and the recycling programs work well for the big, reputation-conscious, regulator-watched operators. They work less well, or not at all, for the long tail of smaller data centers and the newer “neo-cloud” companies racing to deploy hardware as fast as they can, who have weaker incentives and thinner processes. And the volumes are about to get much larger. A waste stream that is well-managed at two and a half million tonnes by the disciplined players is not necessarily well-managed once the undisciplined ones are added in. The data center hardware story is, for now, a relatively good one. Whether it stays good is a question of whether the management scales as fast as the buildout does.

One last thing, to close the loop with the cobalt and lithium stories above. I said earlier that a data center full of GPUs involves very little cobalt, and that is true of the GPUs. It is not quite true of the building. Data centers need backup power — banks of batteries that hold the servers up for the seconds it takes the diesel generators to start during an outage — and those battery banks, increasingly lithium-based, are a genuine, data-center-specific piece of the battery supply chain. It is far smaller than the electric-vehicle fleet or the phone fleet, and it doesn't change the basic picture that the worst battery horrors are about cars and phones rather than compute. But it is not zero, and a book being honest about the buildout should say so.



What Actually Helps

After all that, what actually makes things better?

Keep your devices longer. The most effective personal action in this whole chapter, by a wide margin, is using your phone and

laptop for four years instead of two. Doubling device lifespan roughly halves your lifetime contribution to both upstream mining demand and downstream e-waste. This is not even close. It beats everything else by an order of magnitude. It is also boring and unsatisfying and does not go viral on TikTok.

Proper recycling when you do dispose. Certified recyclers actually recover the metals in your device, in controlled conditions. Non-certified recycling often means “export to a place with no regulations.” In the US, look for R2 or e-Stewards certifications. In the EU, the WEEE directive covers it.

Supply chain transparency. Support regulation that requires companies to disclose where their materials come from. This is actually happening — the EU’s Battery Passport, the US’s conflict minerals rules, California’s Transparency in Supply Chains Act. These laws do change company behavior, slowly. They work better when they have teeth.

Don’t buy the “eco-phone” thing uncritically. Companies market devices as sustainable in ways that mostly don’t hold up to scrutiny. An “eco-friendly” phone made with 10% recycled aluminum is, in the scheme of things, about 1% better than a regular one. Keeping a regular phone for four years is about 50% better. The marketing tells you the first thing; the math tells you the second.

The systemic stuff, if you care and are old enough to care about it. Regulation of battery supply chains, mandatory take-back laws for electronics, international enforcement against e-waste dumping, investment in cobalt-free battery chemistries, domestic mining and processing under environmental controls rather than offshored to places without them — and, for the data center hardware specifically, requirements that operators document where their retired GPUs and batteries actually end up, rather than taking “we recycle” on faith. These are policy questions, not personal ones. But policy is how things like this actually change, and the people voting on the policy in 2035 are the people reading this book in 2026.



The Summary, Honestly

The mining and e-waste problems with modern technology are real, they are worse than the energy and water problems, and they are in a lot of cases as bad as the most critical coverage suggests.

The specific problems — child cobalt mining in the DRC, informal e-waste processing in West Africa and South Asia, the environmental costs of rare earth processing — are things you should know about, and you should not let yourself be talked out of caring about them by a smooth recitation of progress statistics. The progress is real, but it has been slow, it has lagged the industry's knowledge of the problems, and in some cases it hasn't happened at all.

But notice where the worst of it lives, because the location matters for a book about AI. The graphic human horrors — the children in the pits, the people burning circuit boards over open fires — are mostly in the *device and battery* supply chain: the phones, the laptops, the electric cars. The chip itself, the fingernail-sized piece of silicon, has a modest material footprint. A data center full of GPUs is not where the cobalt and the acid baths are.

What the data center *does* have is a hardware life of its own — and this is the part of the story that is genuinely about the AI buildout. It runs an enormous quantity of expensive hardware extremely hot, retires it in a few years, and generates, as a result, a large and fast-growing stream of electronic waste that barely existed a decade ago. The relatively good news is that this stream is, for now, handled far better than consumer e-waste: the hardware is valuable, there is a real resale market for it, and the big operators recover most of it. The open question is whether that discipline survives as the volumes explode and the careless players take a larger share.

So there are two honest conclusions, and you should carry both. First: the worst suffering in the supply chain is much more directly

about the phone in your hand than about the ChatGPT query you sent with it. Second: the buildout still creates a real, specific, growing materials-and-waste footprint of its own, separate from the energy and water costs of the last two chapters, and a book that is honest about the costs has to count it.

You can hold both truths at the same time. You should, in fact. Either one alone is incomplete.



A note on sources: the cobalt figures are from Amnesty International's reports on the DRC, from the OECD's due diligence guidance, and from academic studies of artisanal mining. The lithium figures are from the USGS and from academic hydrogeology studies of the Atacama. The rare earth figures are from Chinese government disclosures (such as they are) and from Adamas Intelligence. The consumer e-waste figures are from the UN Global E-waste Monitor, the most reliable public source on the topic. The data center hardware figures draw on the UN Global E-waste Monitor 2024, the Nature Computational Science study projecting AI-driven e-waste growth, reporting on hyperscaler GPU depreciation schedules and the debate over their realism, and disclosed corporate IT-asset-disposition figures. Where the numbers are uncertain — particularly on artisanal mining, on informal e-waste, and on the real working life of data center GPUs — I have said so.



CHAPTER 8

The Highway in the Bronx

What It's Like to Live Next Door



In 1948, East Tremont was one of the nicer neighborhoods in the Bronx — a stable, mostly working-class area, Jewish and Italian and Black and Puerto Rican, with tree-lined streets, sturdy apartment buildings, small businesses on the avenues, a synagogue, a hospital, decent schools, low crime, and rising property values. People bought apartments there expecting to live in them for thirty years and pass them to their children.

In 1955, Robert Moses decided that the Cross-Bronx Expressway needed to cut through it. In *The Power Broker*, his 1974 biography of Moses, Robert Caro documented that alternative routes were available — including one two blocks north that would have used an existing rail right-of-way and displaced almost no one — and that the route through East Tremont was chosen instead. Caro's account is that Moses did not visit the neighborhood or meet the residents. He looked at a map, drew a line, and the line went through their lives.

About 60,000 people lost their homes to make way for the trench. Eviction notices arrived with ninety days' warning and compensation set by a city formula that, Caro documented, systematically undervalued what families were losing. The construction took most of a decade, and during it, and for years afterward, the neighbor-

hood began to come apart in ways that were not immediately obvious as collapse.

Property values for the buildings *next to* the trench dropped first, because nobody wanted to live next to a freeway. Falling values meant owners couldn't refinance to maintain the buildings, so maintenance was deferred, which made the buildings worse, which dropped values further. Owners sold at a loss. New owners, paying less, had less to spend on upkeep, and some sold to absentee landlords with no personal stake in the neighborhood.

By 1965, the businesses on the commercial streets began to close. The synagogue closed in 1968. By 1975 came the fires that became the iconic image of the South Bronx — buildings worth more burned than standing. By 1980, East Tremont was one of the most devastated places in America.

Hold the timeline in your head, because it is the one this chapter is about: 1948, *stable working-class neighborhood*. 1955, *decision made by someone who did not live there*. 1958, *evictions*. 1965, *decline visible*. 1972, *irreversibility*. 1980, *ghetto*. Twenty-seven years from the last good year to total collapse. By the halfway point, anyone paying attention could see the trajectory. By the three-quarter point it was irreversible — the capital had left, the institutions had closed, the people with options had moved out.

Caro's was not a unique story. The same pattern recurred where Interstate 375 cut through Black Bottom and Paradise Valley in Detroit, where I-95 took out Overtown in Miami, where I-10 went through Tremé in New Orleans, where I-94 destroyed Rondo in St. Paul — a neighborhood that, historians have documented, the city's planners chose partly because they expected the residents would have difficulty fighting back — and where I-40 took the Jefferson Street corridor in Nashville.

The Federal-Aid Highway Act of 1956 authorized 41,000 miles of interstate, and the urban portions destroyed an estimated *one million* American homes between 1956 and 1972, disproportionately in Black and working-class neighborhoods. None of those neighbor-

hoods looked like ghettos when the highway decisions were made. Most became ghettos within twenty-five years. None came back.

I open the chapter this way because the pattern I want to describe — what is happening now, in 2026, as AI data centers are built near residential areas — rhymes with the highway pattern closely enough that the comparison is worth taking seriously. It is being executed with newer legal tools, by a different industry, in some of the same kinds of neighborhoods, and the same twenty-five-year trajectory is available to anyone willing to look ahead.

Whether a given neighborhood follows that trajectory is not fixed — sometimes communities organize and stop a development, sometimes a facility is sited or designed so the cascade never starts — but absent any of that, the trajectory is the default. This chapter is about what it is like to live next to one of these facilities. The two that follow are about how the developments get approved, and about the mechanics of the cascade itself.



What Living Next To One Is Like

Brittany Heights is a neighborhood in Chandler, Arizona. Until 2014 it was a quiet residential development — backyards, barbecues, the normal sounds of suburbia, rising property values, kids on bikes. In late 2014, a data center came in next door. The construction took most of a year, and residents tolerated it, because construction ends. Then the data center came online, and they discovered that what comes after construction is not quiet.

It is the continuous low-frequency hum of the cooling system, twenty-four hours a day, every day of the year. As one resident put it, it is “like living with your refrigerator running, except the refrigerator is the size of a Costco and you can’t unplug it.”

They tried earplugs, noise-canceling headphones indoors, white noise machines, sleeping on the far side of the house, planting trees as a sound barrier. They wrote letters and came to council meetings.

The complaints went nowhere, because Chandler’s noise ordinance — like most municipal noise ordinances in America — was written for block parties and barking dogs, not 24-hour industrial cooling. Measured by the standard method, the data center’s noise came in below the legal threshold.

That word — *threshold* — is doing a lot of work, and we will come back to it. For now: the city had no legal basis to act, the operator had no incentive to change anything, and the residents had no recourse. They had complained correctly, to the correct authorities, using the correct procedures, and the system told them, in effect, that what they were experiencing was not a problem it was equipped to recognize.

Some moved away — the ones who could afford to. After eight years of organizing that produced media coverage, lawsuits, and political pressure, Chandler amended its zoning in 2022 to make new data centers harder to site, and in 2025 the council voted unanimously against a proposed one, citing Brittany Heights. The original data center is still there. The hum is still there.

That is the longest-running documented case, but it is not unique. In Bristow, Virginia, Donna Gallant installed triple-pane windows in her home next to Google’s “Mango Farm” complex; the sound still penetrates her bedroom walls, and during load testing — when cooling systems cycle to maximum at unpredictable intervals — she has described anxiety attacks and sleepless nights. “It literally rocks your core,” she told a *U.S. News* reporter in April 2026. There are thousands of people like her in Loudoun and Prince William counties, the world’s largest data center cluster — nearly 300 facilities in roughly thirty square miles. Most of them lived in their houses before the data centers existed.

In South Memphis, in the historically Black neighborhood of Box-town, residents have been breathing exhaust from gas turbines installed by Elon Musk’s xAI to power its Colossus supercomputer. Reporting and the subsequent legal filings describe turbines installed in 2024 without the air permits the Clean Air Act requires for

major emission sources; the company's position was that the turbines were "temporary" and therefore exempt, a reading the EPA has rejected.

The Southern Environmental Law Center, representing the Memphis NAACP, filed a notice of intent to sue under the Clean Air Act in 2025; xAI then removed most of the turbines and obtained permits for the remainder, and said the facility would be "the lowest emitting of its kind in the country." A subsequent federal lawsuit filed by the NAACP and Earthjustice in April 2026 concerns a further set of turbines installed across the state line in Southaven, Mississippi, to power a second facility in the predominantly Black Whitehaven neighborhood.

According to that complaint, the turbines have the potential to emit, per year, on the order of 1,700 tons of nitrogen oxides, 180 tons of fine particulate matter, 500 tons of carbon monoxide, and 19 tons of formaldehyde. The litigation is ongoing, and the company disputes aspects of the residents' characterization. What is not in dispute is the shape of the situation: emissions are being produced near homes, schools, and churches; the chatbot the facility trains is used by people who live nowhere near Memphis; and the economic benefit and the air-quality burden land in very different places.

The list goes on. In Aurora, Illinois, residents organized against a CyrusOne facility, telling ABC7 Chicago in 2026 that they were not adequately informed before approval. In Warrenton, Virginia, residents have organized against a proposed Amazon Web Services facility, having watched Amazon's existing Prince William County data centers generate thousands of noise complaints. Across essentially every state with significant data center development, at least one neighborhood is organizing against impacts that residents never agreed to host.

These are the people the next two chapters are about.



Why the Numbers Lie

Before the legal machinery, one piece of technical sleight-of-hand, because it runs under the whole story — and once you see it, you can spot it in every claim that a facility is “within regulatory limits.”

Go back to Brittany Heights for a moment. The residents complained; the city measured; the meter reported that the data center was *quieter than a normal conversation*, somewhere around 50 decibels. The reading was accurate. It was also why nothing happened. The catch is that the meter and the body are measuring two different things, for three reasons worth carrying with you.

The first is that it never stops. A conversation lasts minutes. The hum runs every minute of every hour of every year, across an operating life now expected to stretch thirty to fifty years, and the body handles those two cases differently. A noise you know will end is one your nervous system can set aside; a noise that has run for eight years and will run for fifty more is one it never stops bracing against.

The World Health Organization recommends that nighttime outdoor noise stay under 40 decibels on average, because above that, studies tie it to high blood pressure, heart disease, broken sleep, and — in children — measurable drops in reading comprehension and memory. The European Environment Agency has called environmental noise the second-largest environmental cause of ill health in Europe, after air pollution. Data center noise routinely runs past that nighttime guideline, and after dark it stands out more, not less, as the world around it goes quiet.

The second is that it's the wrong kind of sound to measure the usual way. The scale built into nearly every noise ordinance — A-weighting — was designed in the 1930s to gauge factory hearing damage, so it deliberately discounts low frequencies, which don't harm hearing the way high ones do. Sensible for the 1930s problem. But a data center's noise — thousands of cooling fans over the

deeper rumble of chillers and pumps — lives mostly between 50 and 200 hertz, exactly where the A-weighted meter looks away.

So the official number understates what people actually hear. As one noise-control engineer put it in a 2025 trade publication, a low hum “contributes little to the dBA level, but causes complaints.” The most maddening part of the sound is the part the law doesn’t count.

The third is that it goes through walls. High-frequency sound bounces off windows and insulation; low-frequency sound passes through them, the way a neighbor’s bass reaches you when the melody doesn’t. That is why Donna Gallant’s triple-pane windows didn’t save her — they block the high notes the brochure emphasizes, not the low rumble she lives with. The same is true of most of what operators install under pressure: sound walls, acoustic louvers, rows of trees, all aimed at the high frequencies. The measures that work on the low end cost more and show up less often.

Hold those three, because the same move repeats through everything that follows: technical compliance with a measurement that misses the actual harm. Air emissions get checked at the fence line, on metrics that don’t capture what accumulates over a neighborhood. Water draws are measured against permits that ignore how long an aquifer takes to refill. Rate increases are modeled in ways that never ask who caused the new cost. Light mostly isn’t measured at all.

Nobody is lying. The regulators are using the tools they have, and the operators are following the rules as written — but the rules were written before the thing they regulate existed in this form. A noise ordinance built for backyard parties doesn’t have to be anyone’s deliberate gift. It just has to sit on the books, unchanged, while the facility is sited in perfect compliance with it.



What the Air Is Like

Noise gets the news coverage. The air does more of the lasting damage, and it arrives for a reason that has nothing to do with malice and everything to do with timing.

Data centers need backup power, because their tenants can't tolerate even a few seconds of downtime. For years that meant diesel generators that ran rarely — a test now and then, the occasional outage — so the air impact was real but small. What changed is speed. The new AI campuses are too big for the existing grid to power on any normal schedule: expanding the grid for a major new load can take five to ten years, and these facilities are going up in two. The gap gets filled with on-site combustion, increasingly gas turbines, which install fast and, unlike backup generators, can run for months at a stretch.

The exhaust lands on whoever lives next door, and the pollutants are the familiar ones. Nitrogen oxides, which form smog and track with childhood asthma. Fine particulate matter — the PM2.5 linked to heart disease, stroke, lung cancer, and early death, with no level considered safe. Carbon monoxide. Formaldehyde and benzene, both carcinogens. Sulfur dioxide. Run enough turbines continuously in a residential area, and neighborhood levels can climb toward or past the limits the EPA sets, depending on weather, distance, and how many are running. Memphis is the most public case, but not the only one; across the country, new facilities are coming online with substantial on-site generation — some permitted, some running for years under “temporary” exemptions, some unpermitted and now in court.

How big does this get? A 2025 study modeled the public-health toll of U.S. data centers on their current path and projected that by 2030, the associated air pollution could be tied to roughly 600,000 asthma-symptom cases and about 1,300 premature deaths a year, with a health burden north of \$20 billion annually — landing, disproportionately, on the host neighborhoods.

Those are projections, and they could move either way: worse if more capacity comes on combustion power, better if courts and regulators tighten up. The exact figure isn't the point. The order of magnitude is — hundreds of thousands of asthma cases, thousands of early deaths, tens of billions in health costs, concentrated in specific places — and it is a burden the people bearing it never got to vote on. Data center air pollution isn't yet at the scale of power generation or traffic. It is already at a scale that wouldn't be tolerated if it were settling over different neighborhoods.



What You Pay For

The third thing you live with next to a data center doesn't show up in a photograph, because it isn't physical — and it reaches more people than the noise and the air put together. It's the electric bill.

When a data center lands in your region, the grid has to grow to feed it — new substations, new transmission, new generation, all of it enormously expensive. The only real question is who pays for it, and across most of the United States the answer has been the people who were already there, through their monthly rates. An earlier chapter walked through the mechanics: the regional capacity auctions, the grid operator's own analyst pinning billions of dollars in higher costs on data-center demand, the residential rates that have climbed faster over the past five years than at any point in American history.

The part that belongs here, in a chapter about what it is like to live next to one of these things, is the unfairness baked into the arithmetic. The cost of serving a giant new customer — a 500-mega-watt data center — is spread across everyone in proportion to the power each already uses. The data center pays its share, but its share is figured as if it were just another customer, not the reason the new infrastructure had to be built at all. So a household whose own

usage hasn't budged watches its bill climb because someone else's data center showed up.

It is the same move as the air monitor at the fence line and the water permit that ignores how long an aquifer takes to refill: a number that is technically correct and silent on the question that actually matters — who caused the cost.

No smokestack here, no hum. But every month, in tens of millions of households, the bill is higher than it would be because the grid is being built out to run compute most of those families will never knowingly touch — and since low-income households spend a bigger share of their income on electricity, an increase that merely stings a median family can be ruinous near the poverty line.



What the Lights Are Like

Light gets less attention than the noise, the air, and the bills — but it reaches more homes, over more ground, than any of them.

A data center campus is lit all night, every night, by high-intensity security and operations lighting, and the design almost always puts operational needs ahead of the neighbors. A bedroom that faces the facility gets enough light, all night, that the people sleeping in it buy blackout curtains. Yards and patios go unused after dark. The night sky washes out for miles.

The health cost of that is well documented. Light at night scrambles the body's circadian clock, suppresses melatonin, and wrecks sleep — and it tracks with higher rates of obesity, diabetes, heart disease, depression, and several cancers, in doses far below what you'd need to read by. In children, whose internal clocks are still forming, the effects on metabolism and growth appear to follow them into adulthood. It takes an ecological toll too, killing migrating birds in collisions and throwing off the behavior of nocturnal animals.

The fixes have existed for years — “dark sky” fixtures that aim light down instead of out, motion-activated lighting, warmer-colored lamps — and data centers mostly skip them, because they cost a little more. In most of the country, light pollution isn’t regulated at all, so the resident at the window has no one to call.

None of this is abstract. It is the daily life of specific people, in specific houses, who were never asked for any of it. The next chapter is about how it gets approved over their objections.



CHAPTER 9

The Legal Weapons

How These Deals Actually Get Done



On a March afternoon in 2025, in Mason County, Kentucky — a rural stretch along the Ohio River that once billed itself as the second-largest tobacco market in the world — Dr. Timothy Grosser and his son Andy sat across a table from three men who had come with an offer. Ten million dollars for the 250-acre farm the family had worked for nearly four decades. That was thirty-five times what Grosser had paid for the land in 1988, and well above anything else changing hands in the county.

There was a catch. The men wouldn't say who was buying, or what they intended to build. First, the Grossers would have to sign a non-disclosure agreement.

They said no. "I have spent half my life on this farm," Grosser told a local reporter. "I'm just not giving up my happiness and freedom for any amount of money." Months later — after a neighbor who *had* signed let something slip, and after residents went digging through public records and found a request for 2.2 gigawatts of electricity, nearly double what the local utility generated for everyone else combined — the county finally pieced it together. A hyperscale data center, backed by a Fortune 100 company that, to this day, has not put its name on the project.

That is how a great many of these deals begin: not with a public proposal a community can weigh, but with a quiet offer, a gag order, and a company that doesn't exist anywhere but on paper.

This chapter is about the rest of that playbook — the legal tools that let the buildout go forward over the objections of the people who will live beside it. I want to be clear about one thing up front: almost none of it is illegal. Most of it is ordinary commercial real-estate practice, the kind of thing that happens in every large land deal.

What's different here is the *combination* — half a dozen of these tools deployed together, on a specific kind of project that a lot of communities would turn down if you asked them plainly — and the lopsidedness of the contest. On one side, firms and law offices that have run this play a hundred times. On the other, a part-time county board reading zoning code on a weeknight. It's worth seeing the moves clearly, whichever side of one you happen to find yourself on.

It Starts in Secret

The first move is the one the Grossers ran into: the non-disclosure agreement, signed not just by sellers but by the local officials themselves, before any real negotiation begins. It bars them from telling the public who the developer is, what's being proposed, or sometimes that anything is being proposed at all.

This is not a fringe tactic. An NBC News investigation in late 2025 reviewed more than thirty data center proposals across fourteen states and found NDAs in the majority of them, with shell companies layered on top to keep the real developer's name off the public filings. The officials signing them are usually part-time, often without legal staff, and genuinely hoping to land investment for a struggling local economy. They sign documents drafted by people who do this for a living. And the practical effect is that the layer of government closest to the affected residents — the one most accountable to them — becomes legally forbidden to tell them the basic facts.

The shell company finishes the job. The actual developer never appears. A generic entity is created, usually in Delaware, and *it* signs the contracts and files for the permits. In Beaver Dam, Wisconsin, reporting by Wisconsin Watch found that Meta built toward a billion-dollar, 520-acre data center using shell companies — one of them named, with a certain whimsy, “Balloonist LLC.” The NDA referred only to “the project.” For more than a year, officials worked on it while telling residents nothing, and by the time the public learned what was coming, the land was assembled and the deal was effectively done.

So picture a resident who suspects something is happening and tries to find out what. The public records say “Balloonist LLC.” Their elected officials are under legal seal. There is, by design, almost nothing to grab onto. You cannot organize against a project you cannot name, and that is rather the point — the American Economic Liberties Project, which has tracked the pattern, puts it simply: officials are supposed to answer first to their voters, not to a corporation they’re cutting deals with behind a confidentiality clause.

It doesn’t always work. Sometimes the secrecy backfires once it breaks. In Tucson, a proposed Amazon data center known throughout the process only as “Project Blue” — a \$3.6 billion development — became public when a document reached a local reporter, and the resulting opposition was strong enough that the city council reversed itself and killed the project in 2025. But Tucson is the exception that shows what it takes: the public has to learn what’s happening early enough, and clearly enough, to push back before the dirt moves.

And They’ll Go to Court to Keep It Secret

The secrecy doesn’t always end when the building opens. Sometimes it hardens into something stranger.

In The Dalles, Oregon — a small city on the Columbia River — Google has run one of its largest data centers since 2006. As a long

drought wore on and residents watched the water drop in their wells, the local paper, *The Oregonian*, asked a simple question: how much of the city's water was Google using? The city refused to say, calling the number a trade secret. The county district attorney reviewed that claim and ordered the records released. And rather than comply, the city sued the newspaper to keep them sealed — a lawsuit that Google, it later emerged, was quietly paying for.

The fight ran thirteen months. The Reporters Committee for Freedom of the Press took the paper's side at no charge, and in the end the city gave up. When the numbers finally came out, they explained the secrecy: Google's data centers were drawing about a quarter of all the water in the city, and the figure had nearly tripled in five years.

The part that lingers is the sequence — the town council had already voted, unanimously, to approve two *more* Google data centers, and the councilors themselves hadn't seen the water projections until two weeks before that vote. The public never saw them at all. Years later the mayor admitted the episode "is coming back to haunt us," in a town where trust in the city's dealings with Google had not recovered.

It's the NDA logic carried one step further. When a public-records law starts pulling the information toward daylight, the local government can be enlisted to fight the disclosure in court — and the company can quietly fund the fight — on the theory that what a corporation draws from a community's water supply is the corporation's secret to keep.

Assembling the Land

A hyperscale campus needs a lot of contiguous land — often hundreds or thousands of acres owned by many different families. In Mason County, the unnamed company surveyed around five thousand acres. Assembling that quietly requires that no single seller realize the whole picture, because a seller who knows a Fortune 100 buyer needs *this exact parcel* will price accordingly. So the buying is

done through intermediaries, one transaction at a time, each looking like an ordinary sale until the pattern becomes undeniable.

Most of the time, the offers are generous enough that this works on its own — the Grossers' \$10 million was real money, and plenty of neighbors took similar deals. But behind the generosity sits an older tool: eminent domain, the government's power to take private land for "public use." Since the Supreme Court's 2005 *Kelo* decision, "public use" has been read broadly enough to include economic development, and the utility infrastructure a data center needs — the substations, the transmission lines — can often be condemned even when the data center itself cannot. A landowner who knows that refusal can, in the last resort, be overridden bargains from a weaker spot, whether or not anyone ever files the papers.

When the Locals Push Back

Say a community sees it coming anyway, organizes, and persuades its officials to vote the project down. Two things tend to happen next.

The first is that the fight moves to the state capitol. In 2025, West Virginia passed a law cutting local governments' power to zone and regulate data centers and their microgrids — the legislative equivalent of taking the decision out of the town's hands entirely. A similar bill in New Hampshire was defeated in May of 2026. These pre-emption bills are the industry's answer to the one thing that has reliably worked for residents: winning locally. A single company can fund lobbyists and circulate model legislation across many states at once; the residents of one town cannot. The bills are often bundled with provisions that sound good on their own — workforce training, tax-sharing — which makes the pre-emption clause harder to oppose without appearing to oppose the popular parts.

The second thing that happens is the lawsuit. Here the clearest case is Saline Township, Michigan, a community of fewer than three thousand people outside Ann Arbor. In September 2025, after impassioned public comment, the township board voted four to one to

reject rezoning 575 acres of farmland for a data center. It was exactly what the residents wanted.

Two days later, the developer sued.

The claim was “exclusionary zoning” — the argument that Saline had unreasonably barred a legitimate use, which under Michigan law a community isn’t allowed to do. It leaned on a technicality: the township had no land zoned for industrial use at all, and a data center, the suit argued, was a “necessary” use that couldn’t simply be excluded everywhere.

The township’s lawyers told the board the hard truth. A drawn-out fight would drain the town’s budget; even if they somehow won, the developer could route around local zoning entirely by partnering with the University of Michigan, which can build on a different legal footing than a private firm. And they pointed to a \$50 million judgment another Michigan community had taken in a zoning dispute as a warning of what losing could cost.

So Saline settled. The consent judgment, signed that October, won the residents some real concessions — a 55-decibel noise limit, no water-hungry evaporative cooling, no future expansion, money for farmland preservation. But the data center is being built, and it later emerged that the site would be leased largely to OpenAI and Oracle as part of the federal “Stargate” initiative. Developers filed a similar suit against Chatham County, North Carolina. The lesson other small towns are meant to draw is not subtle: opposition is expensive, and the people on the other side can afford the expense better than you can.

The Money That Comes With It

All of this arrives wrapped in the language of opportunity, and that wrapping matters, because it’s what splits a community against itself. Local development authorities describe these projects as once-in-a-generation lifelines — hundreds of jobs, millions in new tax base, a way to reverse decades of decline. In a place that has

watched its young people leave, that is a powerful pitch, and the people making it often believe it.

The arithmetic is less flattering, and one deal shows it about as starkly as a deal can. In Orangeburg, New York — twenty-five miles from JPMorgan Chase’s brand-new Manhattan headquarters — the largest bank in the country asked the local development agency for tax breaks to expand a data center, and got the exact break it requested the following month: around \$77 million, the biggest per-job data center subsidy in the country, in exchange for a single permanent job. The public hearing the law required before the vote is the part to picture. Nobody came. After twenty minutes of silence, an agency official closed the meeting. Two weeks later, the deal was approved.

Orangeburg isn’t a freak; it’s the pattern in miniature. A proposed data center in Genesee County, New York, sought \$801 million in tax breaks for 125 jobs — about \$6.4 million each. Virginia’s data center tax break began in 2008 as a modest bill meant to lure one employer to a struggling rural county; the data centers went instead to the wealthiest part of the state, and the break now runs Virginia around \$1.6 billion a year. These are highly automated buildings — a hyperscale campus might permanently employ a hundred or two — and whatever tax revenue they throw off is frequently offset, sometimes more than offset, by the incentives handed over to land them.

And the jobs that justify the breaks are rarely counted afterward. A watchdog review found that not one state tracks both the jobs a data center promises and the jobs it actually delivers, even though the promise of jobs is the whole basis for the subsidy. None of this means the deals are always bad; some net out positive, and it depends entirely on the terms. But “it’ll bring jobs and taxes” is a line to read closely rather than accept, because the specific numbers — when anyone is allowed to see them — often say something the pitch doesn’t.

The Smaller Moves

Around these big levers sits a thicker layer of smaller tactics, used selectively. A few worth knowing by name, because once you can see them they stop feeling like the natural weather of “how development works.”

The *boundary site*: a parcel placed right on the line between two jurisdictions, so the operation sits in one while the noise and traffic fall on residents of the other — residents who can’t vote out the officials who approved it. The “*temporary*” label: the move at the center of the Memphis dispute from the energy chapter, where equipment that regulations exempt as temporary stays in place far longer than “temporary” was ever meant to cover.

Manufactured urgency: a hard “approve now or we build elsewhere” deadline, usually more movable than it sounds, designed to force a decision before residents can organize. The *document dump*: hundreds of pages filed days before a hearing, so no one has time to mount a real objection. And in Hood County, Texas, when officials floated a six-month pause to study a project, a state senator urged the attorney general to step in and block the moratorium — a reminder that even the modest tool of *slowing down* can be contested.

None of these is exotic. They’re the ordinary furniture of land development, refined over decades across many industries. What makes them matter here is the scale of what’s being built and the speed at which it’s arriving.

The Real Story Is the Asymmetry

Step back from the individual moves and the actual subject of this chapter comes into view, and it isn’t any single tactic. It’s the imbalance.

Defenders of the industry make a fair point: none of these tools is unique to data centers, and singling the industry out is, in a sense, unfair. That’s true. NDAs, shell companies, exclusionary-zoning

suits — they predate this boom by decades and show up in every kind of large deal. But “common” describes how widespread a practice has become, not whether it’s fair to the people on the receiving end of it. The Grossers did not experience the gag order as standard procedure. They experienced it as being asked to sell their life’s work to someone who wouldn’t say their name.

Any one of these moves, a well-resourced community could probably counter. The trouble is that they don’t come one at a time. Combine half a dozen in a single project — the secret approach, the shell company, the quiet land assembly, the pre-emption bill, the lawsuit, the subsidy that divides the town — against a few volunteers reading municipal code after their day jobs, and the outcome is mostly settled before the public hearing ever happens. Run the same play across a whole region, by an industry that has been doing exactly this for a decade, and you get what we have now: data centers going up, over local objection, in the kinds of places least equipped to say no.

Naming the moves is most of what this chapter set out to do. Whether you end up finding them defensible or not, they are far easier to think about as a recognizable set of plays than as the vague, inevitable-seeming texture of progress arriving in your town.



CHAPTER 10

How a Neighborhood Dies

How the Damage Spreads



No one can point to the day a neighborhood died.

That is the hard part of this chapter, and it is the reason the thing it describes keeps happening. We saw in the highway chapter how East Tremont went from one of the nicer neighborhoods in the Bronx in 1948 to one of the most devastated places in America by 1980 — twenty-seven years, and not a single morning in all of them when you could have stood on the corner and said, *today it ended*. It went one softened property value, one departed family, one shuttered store at a time, each loss small enough to wave off, until the sum was a place that no longer existed.

The data center version runs slower and quieter than the highway version, because a data center doesn't tear an expressway through your block. But it runs the same way and ends somewhere similar, and the mechanism is worth watching in motion — because the whole tragedy of it is that the trajectory is visible from the start, and almost nobody acts on it, since the early frames look nothing like the last one.

So here is the film.

It opens with an announcement — or, more and more often, with the discovery that there had been something to announce all along. Construction starts: the trucks, the dust, the closed streets, the trees

coming down. People grit their teeth and wait it out, because construction ends. What they don't yet know is that what comes *after* construction is the part that doesn't.

The first people to leave are the ones who can. The young family already half-thinking about a bigger house; the couple who can work from anywhere. They read the coverage, they look at the thing rising at the end of the road, and they quietly decide to list now rather than later. Their houses sit a little longer than they would have a year ago and sell for a little less. Nobody calls that a crisis. It's just a soft patch in the market.

Then the facility switches on, and the hum begins, and it does not stop. Some people stop hearing it within a week; some never do, and stop sleeping. The complaints to the city go nowhere, because the noise ordinance was written for backyard parties, not for a cooling system the size of a shopping mall running at three in the morning. A few neighbors sue. Most of those suits settle — often with a clause barring the people who settled from ever discussing it, so that when the next family wonders whether the noise is really as bad as they fear, the people who would know aren't allowed to say. The neighborhood's own memory gets edited as it goes.

A couple of years in, the electric bills start to climb, and the utility blames vague "supply costs" rather than the enormous new customer down the road. For a household with a little slack, the extra fifty or eighty dollars a month is an irritation. For a household with none, it's a choice between the bill and something else that mattered. Some families cut the groceries. Some cut the medication. Some just go.

By the time the local school starts shedding students — the families with options have been trickling out for years now — the money follows the enrollment out the door, and the strongest teachers tend to leave first. The barber, the dry cleaner, the diner everyone used to walk to lose enough business that a few of them close, and each empty storefront makes the next one likelier. The dentist who'd been there thirty years retires, and nobody takes his place. The thing

that made it a neighborhood — that everything was close, that you knew the people — quietly thins out.

And somewhere in here it tips into something self-sustaining. Buyers who can afford to look anywhere stop looking here, so the only buyers left are investors paying low prices to rent the houses out, and the owners who want to leave find the market won't give them back what they paid. Deferred repairs make the block look tired, which makes it decline faster, because in real estate the appearance of decline is half its cause. Crime ticks up the way it does wherever fewer people who know each other are watching the street. Insurers raise premiums; banks grow wary of the mortgages; the buyer pool shrinks again. The grocery closes. The corridor fills with dollar stores and payday lenders.

Then one day someone says the neighborhood died, and they're right — and there is still no day to point to. There was just twenty years of small decisions, none of them made by the people who lived there.

Every frame of that film has run before, somewhere, near some industrial siting decision of the last hundred years — the highways, the refineries, the petrochemical stretch of Louisiana that people started calling Cancer Alley, the waste plants, the prisons. The data centers are doing it on a gentler timeline, because they break less at the moment they're built. The destination is the same.

I'm not claiming every neighborhood near a data center is bound for this. Some organize and stop the project outright. Some facilities are sited or designed so the cascade never catches. And a wealthy neighborhood with political capital can often have one as a neighbor and absorb it, because its residents have the resources to force the developer into the mitigations it would otherwise skip. The trajectory is the *default*, not the destiny. But the default is what runs when nothing interrupts it.

Here is the part worth sitting with: the early frames don't look like collapse. Three years into the East Tremont story, in 1958, the neighborhood was still mostly itself — values a little soft, a few

families gone, a couple of shops dark. It looked like a rough patch that would pass. It did not pass, and the damage done in those first years made the next years worse, and so on, until the highway had been there two decades and the place was gone.

Which means whatever leverage exists is front-loaded. By the time the bills are bad and the for-sale signs are sitting unsold, the people with options have already left and taken the capacity to do anything with them — so the only window in which the trajectory can still be bent is the early one, when the damage is least visible and therefore hardest to take seriously.

As of 2026 you can watch the same film at every frame at once: the residents of Brittany Heights in Arizona are eleven years in; Boxtown in Memphis is about two; Beaver Dam, Wisconsin, where Meta is just breaking ground, is at year zero. What's striking, looking across them, is how rarely anyone living it could read the trajectory early enough to matter — precisely because the beginning never looks like the end.

The Highway Map

Pull back and look at where all this lands geographically, and the highway parallel stops being a metaphor.

Data centers cluster where the power, fiber, and climate suit them. But *within* a cluster, the specific question — which neighborhood gets the 500,000-square-foot building, which street gets the substation — is settled by the local decisions and legal tools of the last chapter. And the pattern of where it lands is consistent enough to be unsettling.

Overlay the data center sites in Loudoun County, Virginia, on a map of household income, and they sit in the working-class subdivisions, not the wealthy parts of the county. In the Memphis area, the xAI facilities are in Boxtown and Whitehaven — historically Black, long industrially burdened — not in Germantown or Collierville. Around Phoenix, the cluster is in the southeast valley, below the metro's median income, not in north Scottsdale or Paradise Valley.

The same holds in Atlanta, in Dallas, in the Intel-anchored cluster in Licking County outside Columbus.

The places hosting the noise, the emissions, the higher bills, and the truck traffic are systematically not the places whose residents own and profit from the industry — consistently enough that you could predict, with fair accuracy, where the next big facility in a metro will go by finding the working-class edge where organized resistance is likely to be thinnest.

This is the highway map, sixty years on. The Cross-Bronx Expressway did not run through Riverdale; I-375 did not run through Grosse Pointe; I-94 did not run through Edina. The decisions about where the burden fell were made by people who would never bear it, and the legal tools made it easier to drop in some places than others. The data center decisions carry the same asymmetry, with newer instruments, landing on the same kinds of places.

A generation from now, the histories of the AI buildout will include maps — red where the facilities went, blue where the wealthy parts of those metros were — and the two colors won't overlap much, just as they didn't for the highways. The lesson, taught alongside Caro the way it's taught now, will be that the pattern was legible in advance. Whether it was legible *early enough to matter* is the part still being decided.

What the Industry Says, and Why It Doesn't Fully Survive

The industry has answers to all of this, and a fair-minded reader should hear them, because some of them have real force.

Data centers are necessary infrastructure. True. Modern life, AI included, depends on them, and no serious person wants to abolish them. The argument is about how and where they're built, not whether — and treating it as a fight over their existence misreads it.

They bring jobs and tax revenue. Sometimes. The jobs are real but few; a hyperscale building might permanently employ a hundred or two, fewer per dollar than almost any other use of the land. The tax

revenue is real but frequently offset, sometimes more than offset, by the subsidies handed over to land the project. Whether a given deal nets out positive depends entirely on its terms, and the blanket version doesn't survive a look at them.

The industry invests in mitigation. Sometimes. Some facilities are well designed; some aren't. And the investment tends to show up where there's organized opposition — reactively, after the lawsuits and the coverage and the pressure — rather than as standard practice everywhere.

They comply with the regulatory limits. Largely true, and largely beside the point, for the reason the energy chapter gave: the limits were written before this industrial form existed, calibrated against other problems, and mostly never updated. Complying with an outdated limit is not the same as doing no harm.

The people complaining are NIMBYs. This is the move that does the most work, so it's worth pausing on. *NIMBY* — “not in my back yard” — is a way to dismiss a specific objection without engaging it. The useful test, when you hear it, is whether the person using the word would take the project in *their* own neighborhood. The residents' concerns — the noise, the air, the water, the bills, the property values, the children — aren't opposition to all development. They're opposition to *this* development, in *this* place, with *these* costs, which they didn't agree to and have few good ways to refuse.

Remote siting isn't economically feasible. This is the most interesting one, and it deserves a straight answer. At the margin it's true: building out in the country costs more — fiber, workforce, logistics. What it leaves out is *who pays the cost of building close in*. The cost of the remote site is paid by the developer. The cost of the residential site is paid by the residents, in everything this chapter has described.

Those are externalities — costs pushed onto other people — and when they're large, the efficient arrangement is for the party creating them to carry them. Require developers to do that — stricter siting, real mitigation, full payment for the grid, honest regulation

— and some close-in sites stop penciling out while others survive with protections attached. The buildout costs a bit more and lands more fairly. The “infeasibility” is about margins, not physics.

The Test

Here is a test worth carrying — for data centers, and for every other industrial siting decision you’ll ever read about.

The legal limits, for noise and air and water and light, are not floors of what’s acceptable. They’re the most harm the industry could get political permission for, usually set before the harm was well understood and rarely revisited since. A noise reading under the ordinance, an emission inside the permit, a rate hike the commission signed off on — none of that is evidence the harm is acceptable. It’s evidence the harm is *legal*. Those are different things, and the gap between them is where most of this lives.

The test that cuts through it: take any siting decision and ask whether the people making it would accept the result next to their own homes.

You don’t have to pry into anyone’s private life to run it, because the Highway Map answers it in the aggregate. These facilities are not, as a rule, built in the wealthy places where the executives and officials and lawyers who shape these decisions tend to live. The endless low hum, the combustion exhaust near a school, the all-night lighting into a child’s window, the bill that climbs to pay for someone else’s infrastructure, the deal cut in secret with your own representatives gagged — these land on the working-class edge, not where the deciders chose to live.

The dark-sky fixtures that keep glare out of a bedroom exist, and get installed where the affected residents have clout. The closed-loop cooling that doesn’t hum exists. The rate structures that make a facility pay its own way exist. The technical fixes aren’t the obstacle. They get used when the people who’d benefit can insist, and skipped when those people can’t.

The test isn't complicated, and the answers it returns are uncomfortable with some consistency. What the discomfort points to is structural, not personal: the people deciding what gets built are mostly not the people who have to live with it, and the tools in the last chapter are what hold that arrangement in place. It's an observation about how the system is wired — and once you've made it, it's hard to stop seeing.

The Alternative

You can't go back. Refrigeration, air conditioning, the internet, AI — none of it is being given up, and the infrastructure underneath it will keep getting built.

But it doesn't have to be built next to people. A facility the size of a small town does not need to sit beside anyone's house. It can go twenty miles out, with its own access road and substation and water, where the land is cheaper, the legal exposure is lower, and the impact on a community is close to nothing. The workforce can commute; the fiber can be run. Nothing about a data center's actual requirements demands a residential neighbor.

The reason so many land near homes is the externality logic again: the remote site costs the developer more, while the site near an existing exurb is cheaper precisely because some of its costs fall on the residents instead. The highway parallel holds here too. The interstates did not *have* to run through East Tremont and Black Bottom and Overtown and Rondo and Tremé. Other routes existed — a little more expensive, a little more complex — and the choice to take the cheaper one at the residents' expense is a choice we now broadly agree was wrong. Those neighborhoods weren't lost to physics.

Sixty years on, that recognition is slowly producing the will to undo a little of the damage. The Embarcadero Freeway in San Francisco came down in 1991, the Park East in Milwaukee in 2003, the eastern leg of Rochester's Inner Loop was filled back in to street level beginning in 2014. Detroit now has federal money to turn I-375 — the mile of sunken freeway that finished off Black Bottom — back

into a surface boulevard, with construction staged for the late 2020s; New Orleans is pursuing the same for the Claiborne Expressway that gutted Tremé. The work is slow, expensive, and incomplete, and the neighborhoods that existed before the concrete are not coming back. But the wounds are at least being treated.

The data center industry is making the same kind of decision right now. Some of it will be undone by the lawsuits in progress and the bills being introduced; some won't; and the ones that stick will shape those places for decades. What would change the outcome is knowable and not exotic — it's the same category of change that eventually slowed urban highway construction in the 1970s.

Site the facilities away from homes. Put the grid-expansion costs on the parties that cause them. Measure low-frequency noise honestly. Enforce the air rules already on the books. Protect local zoning from pre-emption. Give the residents on the wrong side of a jurisdictional line some standing in a decision that lands on them. None of it is technically hard. All of it is politically hard, for one reason: each piece asks the buildout to absorb a cost it currently passes along, and the tools in the last chapter are exactly what have kept those costs passable.

That's the heart of it, and it's where I'd rather leave the question than tie a bow on it. The current pattern is not a law of nature. It's what a particular set of rules and incentives produces, and a different set would produce a different pattern — more of the cost carried by the people who choose the buildout, less by the people who only live near it.

Whether the rules change isn't something a book can predict, and I won't pretend otherwise. It's one of the genuinely open questions the buildout leaves on the table — and how it resolves will say something about the era: whether a society that could build a technology this powerful could also share out its costs in a way its own citizens would recognize as fair.



CHAPTER 11

No Going Back

*Why the Argument You're Having Isn't the Argument You
Think You're Having*



Here is a question worth sitting with for a minute before you read the rest of this chapter.

If someone offered you a time machine and said you could live in any year you wanted, with the average standard of living of that year, how far back would you go?

Not as a king. Not as a movie star. As an ordinary person — a kid with ordinary parents in an ordinary house.

Most people, when they actually think about this, don't go back very far. And the reasons why are the whole point of this chapter.

Three Houses

Let me show you three houses. They're all real — reconstructed from census records, household inventories, utility data, and a fair amount of historical research. They belong to families of roughly similar social standing: not rich, not poor, doing okay. The kind of family whose kids went to school and whose parents had steady work.

House One — Massachusetts, 1725

The house is about 700 square feet. That's smaller than a lot of two-car garages today. Seven people live in it: two parents, four children, and a grandmother.

There is one room that matters, and it has a fireplace in it. The fireplace is not decoration. It is the stove, the heater, the light source, and — in winter — the only place in the house where you can feel your fingers. In January, three feet from the fire, it is maybe 60°F. Ten feet away, it is 40°F. In the sleeping loft, where the children sleep under piles of wool blankets, it is whatever temperature it is outside, minus a little.

There is no running water. Water comes from a well, in buckets, carried by whoever is youngest and healthy. In winter, the top of the well freezes and someone has to break it. A bath happens once a week, in a wooden tub, with water heated on the fire one pot at a time. It takes about two hours to fill. The whole family uses the same water, youngest to oldest.

There is no refrigerator. Meat is salted or smoked. Milk lasts a day in summer, two in winter. Butter is made weekly. Vegetables come from a root cellar — potatoes, turnips, cabbages, onions, apples if you're lucky. From November to April, no one in this house eats anything green. Scurvy is a real risk if the apples run out.

Light after sunset comes from tallow candles, which smell like burning fat because that's what they are, or from the fire. Most people go to bed around 8 PM in winter because staying up is expensive and hard on the eyes. Reading at night is a luxury. The grandmother in this house is illiterate, which is not unusual.

Laundry happens once a month. It takes a full day. Water is heated in a giant kettle, clothes are scrubbed by hand on a washboard, wrung out, and hung to dry — inside in winter, which makes the house damp for days. Each person owns about three outfits. Clothes are patched, re-patched, and handed down until they disintegrate.

The children in this house have about a one-in-four chance of dying before age ten. The most common causes are diarrhea from

contaminated water, respiratory infections that today would be cured by a five-dollar antibiotic, and accidents near the fire. The mother of this family has roughly a one-in-eight chance of dying in childbirth during her lifetime. Nobody thinks this is unusual, because it isn't.

Annual energy use for this household: about 100 gigajoules total, or roughly 14 GJ per person. Almost all of it is firewood, which someone — usually the father and the older boys — has to cut, haul, split, and stack. A family this size burns about 15 cords of wood a year. That's a woodpile the size of a small truck.

This family is not poor by the standards of 1725. They are doing fine.

House Two — Ohio, 1975

The house is about 1,500 square feet. Four people live in it: two parents, two kids. Everyone has their own bedroom except the kids, who share.

There is a furnace in the basement that burns natural gas. In winter, the whole house is 68°F, day and night. In summer, there is a window air conditioner in the parents' bedroom that runs at night when it's really hot. The rest of the house has fans. On the worst August afternoons, the living room hits 85°F and everybody complains, but it is not dangerous.

Water comes out of taps. Hot water comes out of other taps, from a tank in the basement. A shower takes five minutes and uses about 15 gallons. Everyone showers every day or every other day. Baths are for kids. The idea of heating water on the stove to bathe would strike this family as something from a pioneer movie.

There is a refrigerator in the kitchen, white, about five feet tall. It has a small freezer compartment at the top that grows frost and has to be defrosted once a year, a job everyone hates. The family eats fresh milk, fresh meat, frozen vegetables in winter, and — miraculously, to a 1725 person — iceberg lettuce in February, shipped from California on refrigerated trucks. Nobody finds this remarkable.

Light comes from incandescent bulbs, one or two per room, plus a fluorescent tube in the kitchen that flickers. The kids do homework at the kitchen table after dinner. Reading at night is completely normal. The whole family can be doing different things in different rooms at 9 PM — one watching TV, one reading, one doing a puzzle — and all of it requires electricity, and none of it feels like a luxury.

Laundry happens twice a week. It takes about an hour of actual human effort, spread across a day. There is a washing machine and a dryer in the basement. Each person owns maybe 20 outfits. Clothes are replaced when they wear out, not patched.

There is one television, a color Zenith, in the living room. There are three channels, plus PBS. There is one telephone, attached to the kitchen wall by a coiled cord, and the whole family shares it. Long-distance calls are expensive and rare; you call your grandmother on Sundays because the rates are lower.

The family has one car, a station wagon. The mother drives it to the grocery store. The father takes the bus to work, or drives on days when he needs to. Nobody in this family has ever been on an airplane except the father, once, for a funeral.

The children in this house have about a one-in-fifty chance of dying before age ten. The mother's risk of dying in childbirth was about one in ten thousand. Both numbers would seem miraculous to the 1725 family.

Annual energy use for this household: about 1,200 GJ total, or roughly 300 GJ per person. That's about twenty times the 1725 figure.

This family is not rich. They are doing fine.

House Three — North Carolina, 2026

The house is about 2,400 square feet. Three people live in it: two parents, one kid.

There is a central heating and air-conditioning system that runs, in some form, about ten months of the year. In winter, the house is 70°F. In summer, it is 72°F. The family has not experienced indoor temperatures outside the 68–74°F range in over a decade. The AC is

the single largest electricity user in the house, by a wide margin, and in July it runs almost continuously.

Water comes out of taps, as in 1975, but there are more taps — three bathrooms, a kitchen, a utility sink, an outdoor spigot, a refrigerator with an ice maker. The teenager takes a 15-minute shower every morning and a shorter one after soccer practice. Nobody counts.

There are two refrigerators — one in the kitchen, one in the garage for drinks and overflow. Together they hold more food than the entire root cellar of the 1725 family stored for a winter. The family eats fresh produce from four continents, year-round, without noticing.

Light comes from LED bulbs, which use about one-tenth the electricity of the 1975 incandescents and last about 25 times longer. There are about 60 light bulbs in the house. Most of them are almost never noticed. The kid's bedroom alone has more lumens of available light than the entire 1725 house had in a year.

There are seven screens in active use: two TVs, two laptops, three phones, a tablet, and a smart display in the kitchen. There is Wi-Fi. There is broadband internet at 500 megabits per second, which is about 50,000 times faster than the fastest home connection available in 1975, which didn't really exist. The family streams about four hours of video per person per day, across all devices.

There are two cars, both SUVs. The family flies somewhere — Florida, usually, sometimes farther — about twice a year. Amazon delivers something to the house, on average, four times a week. The kid has never not had a phone.

Annual energy use for this household: about 950 GJ total, or roughly 315 GJ per person. That is almost exactly the same as the 1975 family.

Read that last sentence again.

This family is not rich. They are doing fine.

The Number That Should Surprise You

Per-person energy use in the United States has been roughly flat since about 1973. In some years it has gone down. The 2026 family uses almost exactly the same amount of energy as the 1975 family.

But look at what that energy does now, compared to 1975:

- The 2026 refrigerator is twice as large and uses about one-quarter the electricity.
- LED bulbs use about one-tenth the energy of incandescent bulbs.
- Cars go about twice as far on a gallon of gasoline.
- Houses are better insulated by a factor of two or three.
- A 2026 laptop uses a tiny fraction of the energy of a 1975 mainframe while being roughly a million times more powerful.
- A modern natural gas power plant converts fuel to electricity about twice as efficiently as a 1975 coal plant.

What has happened, quietly, over fifty years is one of the great engineering achievements in human history, and almost nobody talks about it. We are doing vastly more with the same energy. The reason our per-person energy use hasn't exploded is that every device, every building, every vehicle, and every power plant has gotten dramatically more efficient, roughly in lockstep with our appetite for more of everything.

This is a fact. It is not an opinion. You can look it up. The U.S. Energy Information Administration publishes the numbers every year. Per-capita energy use peaked around 1978 and has drifted downward since.

If you have been told — online, in the press, or by a well-meaning friend — that humans are using more and more energy per person and that the trajectory is unsustainable, the first part of that sentence is factually wrong in the rich countries, and it is worth knowing why you heard it anyway.

(The second part — about sustainability — is a real question, and we'll get to it. But it has almost nothing to do with per-person energy use in America, and almost everything to do with the fact that there are a lot more people now, and a lot of them reasonably want refrigerators.)

The Thought Experiment

Go back to the time machine question at the start of the chapter.

Would you give up refrigeration? Really think about it. No refrigerator means no dairy past a day, no meat past a day, no leftovers, no frozen anything. It means your diet in February is root vegetables, grains, and whatever meat you can salt or smoke. It means food poisoning as a regular event in your life. It means that when your two-year-old gets diarrhea, which happens more than you'd think in a world without refrigeration, there is a real chance she dies of dehydration, because that is what used to happen and that is why infant mortality was what it was.

Almost no one, confronted with this honestly, chooses to give up refrigeration.

Would you give up hot water on demand? Really think about it. A weekly bath, heated on the stove, shared by the whole family. Laundry as a full day of labor, once a month. Washing your hair as a production involving a basin and a kettle. This was the normal human condition until about 1920 in the rich countries and is still the normal condition for about a third of humanity.

Almost no one, confronted with this honestly, chooses to give it up.

Would you give up lighting after dark? Your productive day ends at sunset. In December, that's 4:30 PM. Reading by candlelight is genuinely hard — the light flickers, your eyes strain, and the candles smell. Most people in 1725 went to bed around 8 PM in winter and got up at dawn. Their lives were shaped by the sun in a way ours aren't, and almost no one, offered the trade, wants to go back to that life.

Would you give up air conditioning? This one is more interesting, because it splits by where you live. Your grandparents in Boston probably didn't have AC and were fine. Your cousins in Phoenix cannot live without it — literally cannot, because the summer temperatures there, which are rising, are now lethal to elderly and sick people within a few hours of exposure. In 1980, about 55% of American homes had AC. Today it's about 90%. This is the change you can most easily feel within a single lifetime, and it is the one most directly driven by a climate that is itself changing.

Would you give up the internet? Sit with this one. Really. Your whole social life, most of your education, most of your entertainment, most of how you find out what's happening in the world, and — if you're honest — most of the way you form your opinions about things like data centers and chips.

What This Chapter Is Not Saying

This chapter is not saying modern life is great and everyone should be grateful and stop worrying. That would be a stupid argument and you would rightly reject it.

This chapter is saying something more specific: the debate you are having is never really “modern life versus pre-modern life.” That debate is already over. It was over before you were born. No society that has acquired refrigeration, anesthesia, electric light, and the internet has ever voluntarily given any of them up, and there is no reason to think yours will be the first.

The real debate is always about specific changes at specific costs. It is always about marginal questions. Not “should we have data centers?” but “how should data centers be powered, where should they be sited, how efficient can they get, and who decides?” Not “should we have air conditioning?” but “how do we cool a warming world without making the warming worse?” Not “should we have smartphones?” but “how do we mine cobalt without exploiting children, and what do we do with the phone when you're done with it?”

These are real questions. They have real answers, some of them known and some of them still being worked out. They are the questions that will shape the next fifty years.

There is a distributional point hiding inside all that convenience, and it is worth making plainly, because the rest of this book depends on it. Almost everyone benefits from the buildout, diffusely — the cheaper service, the faster answer, the package that comes the next day. Almost no one would give those up; that is the whole argument of this chapter. But the costs do not land the way the benefits do. They concentrate, geographically and physically, on whoever happens to live next to the infrastructure that delivers them.

For the family in the third house, the data center is an abstraction somewhere over the horizon, and “no going back” is simply true. For the family whose subdivision it backs onto, it is the highway story — the noise, the bills, the slow unmaking of a neighborhood that the chapters on the people next door described in detail. Both families are real. Both stories are true at the same time. “No going back” is the honest view from one of those houses and not the other, and a book that told only the comfortable half would be lying by leaving the other half out.

But you cannot answer the real questions well if you are still secretly relitigating whether the last three hundred years of technological change were a mistake. They weren’t, and they won’t be reversed, and pretending otherwise is a way of letting yourself off the hook from the harder work of asking the actual questions.

The Quiet Part

There is one more thing worth saying, and it is the part that usually gets skipped.

You are going to live your whole life inside a civilization that runs on about 300 GJ per person per year, give or take. That number is not going to drop to 15 GJ, which is what your great-great-great-great-grandparents used. That civilization is gone. Its return would

require the voluntary death of about seven-eighths of humanity, and nobody sane is proposing that.

What that number might do, over your lifetime, is stay roughly flat while the stuff inside it changes enormously — more of it electric, more of it from the sun and the wind and the atom, more of it used by machines you can't see doing work you barely notice. The efficiency revolution that quietly happened between 1975 and 2026 will probably happen again between 2026 and 2075, and you will live through it, and most of it will be invisible to you, and that is fine.

Your job is not to reverse the last 300 years. Your job is to help decide what the next 50 look like. That is a much more interesting job, and you are, if you want to be, qualified for it.

The chips in the data centers — the ones that worried you enough to start reading this book — are part of the answer, and part of the question. They always were going to be. That is what the next chapter is about.



A note on sources: the per-capita energy figures in this chapter come from the U.S. Energy Information Administration's Annual Energy Review. The household snapshots are composites, built from census data, period household inventories, probate records, and utility studies, with the numbers rounded generously. The child mortality figures are from historical demographic studies; the exact numbers vary by region and decade but the orders of magnitude are solid. If you want to dig into any of this, Vaclav Smil's Energy and Civilization is the serious version of this chapter, and Hans Rosling's Factfulness is the readable one.



CHAPTER 12

How Much Do We Actually Need?

The Question The Industry Doesn't Want Asked



The chapters on the people next door described, at length, what is being done to people in order to build the AI data centers of the 2020s. The noise, the air, the bills, the cascade. The legal weapons and the tactical tricks. The neighborhoods entering trajectories that will, in many cases, end in collapse twenty years from now.

There is a question I have not yet asked, and it is the question the entire rest of this book is dancing around. The question is this:

How much of this do we actually need?

If the buildout described in the neighborhood chapters is paying for compute capacity that will be used to do things people genuinely want done — to cure diseases, to make scientific discoveries, to translate languages, to help children learn, to do useful work that nobody could do before — then the moral situation, while still containing real injustices that should be addressed, is at least one where there is a real benefit being created somewhere. The harm to the neighborhoods is being paid for *something*. We can argue about whether the something is worth the cost, but at least the something exists.

If, on the other hand, the buildout is paying for compute capacity that will mostly be used to produce search-engine spam, automated marketing copy, content nobody reads, ads nobody clicks on, and

AI-generated images that fill social media feeds and are scrolled past without engagement — then the moral situation is much worse. The neighborhoods are being destroyed to produce *junk*. The wealth transfer described in those chapters is happening in service of an industry whose actual output, at the scale being built, has very little value to anyone except the companies producing it.

These are different worlds, and which one we are actually in matters more than almost any other question this book can ask. The two cases look identical from outside — the same data centers, the same power draws, the same neighborhoods hosting them — right up until the demand either materializes or doesn't. And those costs are imposed before anyone knows which way it goes.

Here is the spoiler. The evidence available in mid-2026 strongly suggests we are not in the first world but something closer to the second, with some genuine demand mixed in, and the buildout is running well ahead of any rigorous demand signal. The industry knows this. Its investors are starting to. The residents of those neighborhoods mostly do not, because nobody has told them.



What We Actually Know About Utilization

The single most useful piece of information you can have when evaluating any infrastructure buildout is its utilization rate. Utilization is the fraction of installed capacity that is actually being used at any given time. If you build a power plant capable of producing 1,000 megawatts and it is producing 100 megawatts on average, your utilization is 10%. The other 90% of your capacity is sitting idle, costing you money, providing no value to anyone.

A well-run infrastructure investment, in any industry, targets a utilization rate that balances reliability against efficiency. Power plants typically target 50–90% depending on type. Highways are designed for peak rather than average traffic; their average utilization is lower than their design capacity, but the design is sized to handle

the peak. Hospitals target 80–85% occupancy. Airlines, when healthy, fly at 75–85% load factor.

When utilization rates are very low — below 20% on average — it tells you something is wrong. Either the infrastructure was over-built for the actual demand, or the operator is doing something inefficient, or both.

Here is what we now know about utilization in the AI data center buildout.

The Enterprise Numbers

VentureBeat’s Q1 2026 AI Infrastructure & Compute Market Tracker, surveying IT decision-makers across mid-sized and large enterprises, found that enterprise-deployed GPU infrastructure for AI workloads is running at *approximately 5% utilization on average*. This is not a typo. The infrastructure that companies have spent — collectively, hundreds of billions of dollars buying — is, on average, idle 95% of the time. The publication headlined the finding as “the \$401 billion AI infrastructure problem enterprises can’t keep ignoring.”

The reasons for this are not mysterious. During 2023 and 2024, enterprises bought GPUs out of fear. The narrative was that capable AI required massive compute, that compute was scarce, and that any organization not in possession of substantial GPU capacity would be left behind. Boards approved capital expenditure. CIOs placed orders. The orders, when they arrived, sat in racks waiting for the AI applications that were going to use them. Most of those applications, when they were built, turned out to need much less compute than the procurement assumed — because, as we will see below, most enterprise AI workloads do not actually require frontier-scale infrastructure. The GPUs sit. The depreciation runs. The CFOs notice.

A 5% utilization rate, in any traditional infrastructure context, would be considered catastrophic. In the AI infrastructure context, it is currently considered normal, because everyone has roughly the

same problem and nobody wants to be the first to admit it publicly. The admission is starting to leak out anyway, through earnings calls, through industry research, and through the kind of strategic shifts I will describe in the next section.

The Hyperscaler Numbers

Hyperscalers — the very large cloud providers that operate the biggest data centers — have historically had better utilization than enterprise on-premises, because they share capacity across many customers and can shift load between them. Even so, the utilization rates at hyperscale AI data centers in 2025 and 2026 are nothing like the rates you would expect from a well-designed infrastructure investment.

The most public data point comes from a place few people would have expected: xAI's Colossus 1 data center in Memphis, the one whose 35 unpermitted gas turbines we met in the neighborhood chapters. Colossus 1 is one of the largest AI training facilities ever built — more than 220,000 NVIDIA GPUs (a mix of H100, H200, and Blackwell-class), drawing roughly 300 megawatts of power, built in nineteen days from concept to construction, valued at approximately \$40 billion in capital expenditure.

The facility was built to train Grok, xAI's large language model. As of early 2026, before xAI publicly disclosed what it was about to do, the facility's *Model FLOPs Utilization* (MFU) — the standard metric for how efficiently a training cluster is using its theoretical peak compute — was reported, by xAI's own engineers in industry coverage, as approximately 11%. The phrase used in the technical press to describe this number was “embarrassingly low.”

MFU is a training metric. It does not perfectly translate to overall facility utilization. But 11% MFU on a facility built specifically for training is a damning number. It means that, on average, 89% of the theoretical computing capacity the facility was designed to deliver was *not being used*, even by the company that built it for its own training run. The facility was overbuilt for the actual workload, or

the workload was inefficiently structured, or the demand for training runs of that size was not actually present, or some combination. Whatever the explanation, the asset was sitting mostly idle, depreciating, while the gas turbines polluted Boxtown and the cooling system hummed.

In May 2026, xAI announced what it had quietly been planning for months: it leased the *entire* Colossus 1 facility to Anthropic, at \$1.25 billion per month, on a three-year contract worth more than \$40 billion. xAI characterized the deal, in its parent company SpaceX's S-1 filing with the SEC, as a way to "monetize unused compute capacity in our infrastructure." That phrase, "unused compute capacity," is doing significant work. It is a confession, written in financial-disclosure prose, that the facility had been built ahead of demand and that the company that built it was now in the business of being a landlord rather than a frontier AI lab.

Weeks later, xAI signed a second large deal: Google would pay \$920 million per month for additional Colossus capacity, through 2028, after the first \$1.25 billion was already committed to Anthropic. The combined annual revenue from these two contracts is approximately \$26 billion, against a facility build cost estimated at \$40 billion, recovering the build within roughly eighteen months.

The strategic shift is consequential. Colossus 1 was, until May 2026, xAI's identity — the proof that Elon Musk could build a frontier AI lab faster than the incumbents. After May 2026, xAI is something else: it is, in effect, a real estate trust for GPUs, leasing capacity it built for itself to competitors who actually have customer demand to fill it. The technical press that covered the deal noted, accurately, that you do not rent out the asset you used to define your strategic identity unless your strategic identity has changed.

The Colossus story tells you several things at once. Even the most aggressive builders are building ahead of their own demand — the facility ran at 11% MFU before the rental. The demand to lease that idle capacity is real and large — Anthropic is committing roughly half its annualized revenue to a single compute contract. The bottle-

neck has shifted from training to inference, with the lab that built a training facility handing it to labs that have the product demand to fill it. And the unit economics of running a frontier lab on your own hardware are bad enough that an aggressive builder will lease out the asset it used to define itself by.

There is one more thing the Colossus story tells us, which is the inversion of everything the industry has been saying publicly. The public narrative has been: *we are GPU-constrained, we cannot get enough compute, the demand vastly exceeds supply*. The Colossus story tells us: *at least one of the biggest facilities in the world was running at 11% utilization, and the company that built it leased it out because it could not fill its own capacity with its own workloads*. Both things cannot be true simultaneously. One of them is, at minimum, exaggerated.

What the Numbers Mean

Put the enterprise number (5%) next to the hyperscaler number (11% at one of the most aggressive builders, before the rental, with other facilities undisclosed but unlikely to be dramatically higher) and you get an industry that built well ahead of demand and is now scrambling to use what it has already constructed.

That is not how a demand-constrained market behaves. When American chip foundries are demand-constrained they run above 90% and customers wait months for delivery; when oil refineries are, they run above 85% and crack spreads widen; even ordinary cloud data centers, when demand-constrained, run at 60–80% and add capacity carefully, in stages, against confirmed orders.

The AI buildout behaves like a supply-side speculative one instead: capacity added on the bet that demand will appear once supply is ready. It is a familiar pattern — railroads in the 1870s, telecom in the 1990s, commercial real estate many times over. The buildouts that work still usually pass through a phase of overcapacity, write-downs, and consolidation first. The ones that don't end with the assets sold for cents on the dollar and the neighborhoods that

hosted them never recovering. We do not yet know which kind this is, and anyone who tells you with confidence is selling something.



The Stack: A Tour of What Compute Is Actually For

Before we can ask which workloads genuinely need hyperscale data centers, we have to be clear about what the different kinds of AI compute actually do — because the jargon obscures more than it illuminates. There are three fundamentally different kinds, with very different costs and infrastructure needs.

Kind One: Training

Training is the process of building a model in the first place. You take a very large amount of data — most or all of the text on the internet, or millions of images, or thousands of hours of video — and you run it through a neural network architecture many times, adjusting the network's internal parameters at each step to improve its performance at predicting the data. Training a modern frontier model (GPT-4 class, Claude-class, Gemini-class) takes weeks to months on tens of thousands of high-end GPUs, costs tens to hundreds of millions of dollars in compute alone, and consumes a comparable amount of electricity to running a small town for the duration.

Here are the actual sizes, in the unit that matters for a book about data centers, which is compute rather than money. GPT-4, trained across 2022 and 2023, took roughly 2×10^{25} floating-point operations. The 2026 frontier — the GPT-5, Claude, and Gemini-class models being trained as I write this — sits between 10^{26} and 10^{27} operations, ten to fifty times larger.

Put that in physical terms. Take a top-end GPU of the period, an H100 or one of its successors, which does on the order of 10^{15} operations a second and realizes maybe a third of that on a real training workload. A 10^{26} -operation run is then something like a hundred million GPU-hours: tens of thousands of those GPUs running

flat out for one to three months. xAI's Colossus in Memphis is exactly this kind of machine — 220,000 GPUs drawing about 300 megawatts. A single frontier training run occupies a cluster on that scale, on the order of 100 to 300 megawatts for the duration, and burns through a few hundred gigawatt-hours of electricity getting there — comparable to a small city's consumption for a year.

In dollars, for anyone keeping score, that final run is a few hundred million — closer to a billion once the failed runs and research compute are counted. But the megawatts are the number that shows up in a neighborhood.

Training is genuinely compute-intensive. It is the workload that justifies the existence of hyperscale GPU clusters in their current form. There is no practical way to train a frontier large language model on consumer hardware, on edge devices, or even on small enterprise clusters. The compute requirements are real, and the infrastructure required to meet them is real.

But training is also *rare*. A frontier model is trained once. After it is trained, it is used — for inference — for the rest of its useful life, which is typically a year or two before it is replaced by a successor. The training compute is a fixed cost, amortized over the model's deployment. The dominant compute cost over time is not training but inference.

How rare is training? In 2026, the major AI labs each train perhaps a handful of frontier models per year. There are perhaps ten labs in the world that do frontier training (OpenAI, Anthropic, Google, Meta, xAI, DeepSeek, Mistral, a few Chinese labs, a small number of others). Each of them might do one to four major training runs per year. That's perhaps fifty to a hundred major frontier training runs per year, globally, across all labs.

Each of those runs is enormously compute-intensive, but the number of runs is small. The total compute consumed by frontier training, even across all labs, is meaningful but not the bulk of total AI compute. The IEA estimates that inference accounts for 80–90% of

total AI computing in 2026, with training representing the minority share even though it gets most of the press attention.

The rarity has a physical size, too. Add up every major frontier run in a year — generously, with the big ones at the top of the range — and the world’s entire training load comes to a few gigawatts of sustained-equivalent power. That is a real number, and it is a small slice of the *tens* of gigawatts the buildout is adding: Loudoun County alone is on its way from four gigawatts to twelve. The genuinely compute-heavy part of AI is the part you do once, and globally it would fit in a corner of what is being built.

There is also a category of *fine-tuning* — taking an existing trained model and adapting it to a specific task or domain. Fine-tuning is much less compute-intensive than full training, often achievable on much smaller clusters or even on individual high-end GPUs, and is the way most organizations adapt frontier models to their specific needs. Fine-tuning is increasingly being done on the edge, or on modest cloud capacity, rather than requiring hyperscale clusters.

Kind Two: Inference

Inference is the process of using a trained model to do something useful. You give the model an input (a question, an image, a document) and it produces an output (an answer, a description, a summary). Each individual inference call is much smaller than a training run — milliseconds to seconds of compute, on one or a small number of GPUs.

But there are billions of inference calls happening every day, and the volume is growing fast. ChatGPT alone is estimated to handle hundreds of millions of queries per day, with Claude, Gemini, and other systems contributing substantial additional volume. Inference embedded in productivity software (Microsoft Copilot, Google Workspace AI features), inference embedded in consumer apps (autocomplete, photo organization, voice assistants), inference embedded in enterprise software (customer service bots, document

analysis, code completion) — all of this adds up to a vastly larger compute footprint than training.

This is where the demand picture gets interesting, and where the analysis has to get specific. Because *not all inference is the same*. The compute required for inference depends critically on two things: the size of the model you're using, and the complexity of the query you're sending it. A small model answering a simple question is many orders of magnitude cheaper than a frontier model answering a complex one. The same task, depending on how it is architected, can require anywhere from a fraction of a milliwatt to several watts of compute on the inference side.

The industry, for reasons we will examine, has substantial incentives to route inference toward the frontier-model end of this spectrum. Each frontier-model inference call generates more revenue for the company providing it. Each frontier-model inference call also consumes more compute, justifies more infrastructure buildout, and supports the narrative that compute demand is exploding. The same task done on a small local model would generate no revenue for the cloud provider and would not justify any infrastructure buildout. The route the task actually takes is, in many cases, not the route that makes economic or technical sense for the task; it is the route that makes economic sense for the *provider*. We'll come back to this.

Kind Three: Specialized Compute

There is a third category that is sometimes lumped with general AI compute but is genuinely different. *Specialized compute* covers things like protein structure prediction (AlphaFold and successors), climate modeling, materials science simulations, particle physics analysis, drug discovery pipelines, and a handful of other scientific and industrial applications that require very large compute clusters but are doing something other than training or running general-purpose language models.

This category has real, durable demand. The compute needs are large. The applications are valuable. The infrastructure required is

genuinely similar in form to AI data centers — large numbers of GPUs, high-bandwidth interconnects, substantial power and cooling. The buildout for *this* category is justified by the science it enables, and the science is enabling real things: better drugs, better climate predictions, better understanding of how proteins fold, better materials.

But this category is a small fraction of total AI infrastructure spending — a few percent, in dollar terms. AlphaFold’s footprint, dramatic as it is for what it accomplishes, is a rounding error against any one major consumer AI product. When the industry says compute demand is exploding, it is not talking about protein folding or climate models. It is talking about training and inference for general-purpose language and multimodal models, deployed against consumer and enterprise applications and monetized through subscriptions and API calls. The specialized scientific compute is real, valuable, and small. The general-purpose compute is large, mostly inference, and is what the buildout is actually for.



Where The Inference Should Run

Now to the part of the analysis that most directly addresses the inflation hypothesis. *Inference* — 80 to 90% of all AI compute — can run in many places, and the choice of *where* has dramatic implications for how much infrastructure is actually required.

There are roughly three places inference can run:

On the edge — meaning on the device the user is already holding. A phone, a laptop, a tablet, a smart speaker, a car, a robot. The device runs the model locally. No data leaves the device. No cloud round-trip happens. The latency is near zero. The cost to the user is whatever battery and compute the device uses to run the model. The cost to the provider is zero, because the provider is not providing any compute.

On nearby compute — meaning a local server, an edge data center near the user, or a small regional cloud presence. Latency is low. Data may or may not leave the user’s immediate environment. The compute is paid for by whoever runs the local server, which may be the user, the user’s employer, or a regional cloud provider.

On hyperscale cloud — meaning a frontier-model API call to OpenAI, Anthropic, Google, or one of their hyperscale competitors. The data leaves the device, is sent to a hyperscale data center somewhere (possibly thousands of miles away), processed by a very large model running on very expensive infrastructure, and the result is sent back. Latency is hundreds of milliseconds to seconds. The cost per query is meaningful — modern frontier APIs charge on the order of \$1–15 per million tokens, depending on the model and the task.

The choice among these three depends on the task. Some tasks genuinely require hyperscale cloud — frontier-level reasoning, very large context windows, multimodal generation, real-time access to up-to-date information. Some tasks can be perfectly well served on the edge — autocomplete, simple classification, transcription, basic chat, photo organization. Some tasks fall in between and benefit from hybrid architectures where simple parts run locally and complex parts route to the cloud.

The technical question is which tasks belong where. The economic question is which routing decisions get made, and why.

What Edge Compute Can Actually Do In 2026

The capabilities of edge devices have improved dramatically in the past three years, and most people — including most people in the AI industry — have not internalized how much. Some specific 2026 numbers.

Phones. A flagship 2026 phone — iPhone 17 Pro, Pixel 10, Samsung Galaxy S26 — has a neural processing unit (NPU) capable of 40–80 trillion operations per second (TOPS). This is enough to run language models with 1–3 billion parameters at reading speed, com-

fortably. A model like Microsoft’s Phi-4-mini (3.8 billion parameters) or Meta’s Llama 3.2 (3 billion parameters) or Google’s Gemini Nano runs locally on these phones with no cloud connection required.

What can a 3-billion-parameter model do? In 2023, the answer was “not much.” In 2026, it can do most of what 95% of users actually use AI for: rephrasing emails, summarizing documents, answering basic factual questions, performing simple translations, writing short text, doing basic reasoning, generating code snippets, providing tutoring on standard educational topics. The 2026 small models, trained on large amounts of high-quality data, are dramatically more capable than the 2023 small models of the same size. They are not as good as frontier models for the hardest tasks. They are entirely adequate for the median task.

Laptops. A 2026 laptop with modern silicon — Apple M4 or M5, Qualcomm Snapdragon X2, Intel Lunar Lake, AMD Ryzen AI 300 — can run models of 7–14 billion parameters comfortably, and models up to 70 billion parameters with substantial system memory. An Apple M4 Max with 128 gigabytes of unified memory can run a 70-billion-parameter model at INT4 quantization at usable speeds, locally, with no cloud connection. The cost of inference on this hardware, amortized over the laptop’s lifetime, is approximately \$0.014 per million tokens. For comparison, a frontier API call to Claude or GPT-4 class models in 2026 costs \$3–15 per million tokens depending on the model and the input/output ratio.

The cost differential is *three orders of magnitude*. Running inference locally on a laptop the user already owns costs approximately one one-thousandth of what running the same inference through a frontier API costs.

Mid-range hardware. A modest desktop with a consumer GPU (RTX 4060, 4070, 4080, equivalent AMD) can run 7–30 billion parameter models comfortably. A workstation with multiple GPUs or a Mac Studio with substantial memory can run frontier-adjacent models locally. None of this is exotic hardware. It is hardware that

millions of small businesses, professional users, and serious hobbyists already own for other reasons.

Quality of small models in 2026. Here is the part most people miss. The small models in 2026 are not the small models of 2023. They have been trained on much larger amounts of data, with much better training recipes, with much better fine-tuning, and with techniques like distillation that compress the capabilities of larger models into smaller ones. A 4-billion-parameter model in 2026 substantially outperforms a 70-billion-parameter model from 2023 on most benchmarks. The state of the art for small models has moved much faster than most people realize.

The practical implication is this: *for the substantial majority of AI use cases an ordinary person or ordinary business has, edge compute in 2026 is already adequate.* Not someday adequate. Not adequate-with-some-caveats. Adequate. Right now. The model running locally on your phone is good enough to do what you actually want done, in most cases.

What Hyperscale Cloud Is Genuinely Required For

This does not mean hyperscale cloud is unnecessary. There are categories of work where it is genuinely required, and being honest about which categories those are is part of evaluating whether the buildout is justified.

Frontier model training. Discussed above. Real, large, but rare.

Very long context reasoning. Some tasks require processing hundreds of thousands or millions of tokens of context — analyzing an entire codebase, summarizing a year of email, reasoning over a large legal document corpus. Models that can handle these context lengths efficiently currently require hyperscale-class hardware. Edge devices, in 2026, can handle contexts up to perhaps tens of thousands of tokens efficiently. Beyond that, hyperscale is currently required.

State-of-the-art reasoning on hard problems. For the hardest problems — competition mathematics, novel scientific reasoning,

complex multi-step planning — frontier models that are substantially larger than what fits on edge devices remain measurably better. The gap is closing, but it is real, and for users whose work depends on the absolute frontier (researchers, advanced developers, certain professional applications), the gap matters.

Multimodal generation at high quality. Generating images, video, or audio at production quality currently requires substantial compute. Edge devices can do real-time image generation at modest quality and small resolutions; producing the kind of output that matches dedicated graphics pipelines still requires cloud infrastructure.

Real-time access to up-to-date information. Edge models have a knowledge cutoff and cannot search the web. Tasks that require current information — news, prices, weather, recent events — require some form of connectivity, though the connectivity could in principle route to a small retrieval system rather than a frontier model.

Shared services with low individual usage. Some tasks are used rarely by any individual but used often across the population. Translation between obscure language pairs. Specialized medical diagnostic tools. Niche professional applications. Hosting these centrally, rather than installing them on every device, is more efficient.

Genuine multi-user coordination. Real-time multiplayer game AI, collaborative document editing, large-group conversation summarization — anything where the compute genuinely needs to coordinate across multiple users.

Privacy-preserving but compute-intensive workflows. Some workflows benefit from centralization for non-obvious reasons — federated learning, where models are improved using data that never leaves user devices, still requires central coordination of the aggregation step.

This is the list of things hyperscale cloud is genuinely required for. It is not nothing. It is also, in aggregate, *much smaller than the cur-*

rent buildout assumes. The list above does not justify three hundred new hyperscale data centers per year. It justifies maybe a third of one year's worth of current buildout, sustained at current capacity.

What Hyperscale Cloud Is Being Used For That Doesn't Belong There

Now consider the inverse. Look at the actual inference traffic currently routed to hyperscale cloud APIs, and ask which fraction of it genuinely requires hyperscale-class infrastructure.

The honest answer is: a substantial fraction does not. A substantial fraction of the queries currently routed to frontier models are queries that a 4-billion-parameter local model would handle adequately. Email rephrasing. Document summarization. Basic question-answering. Most coding assistance. Most customer service. Most translation. Most autocomplete. Most photo and image processing. Most personal assistant tasks. None of these benefit meaningfully from frontier-model capability over what a competent small model can do, and most of them would have lower latency, better privacy, and dramatically lower cost on the edge.

The reason they currently route to frontier models is partly inertia (the integrations were built that way), partly user-experience uniformity (the frontier model handles every case adequately even when most cases don't require it), and partly economic incentive on the part of the providers (each frontier-model call generates revenue; each local-model call does not). The economic incentive is the largest single factor. If frontier-model API providers had to compete head-to-head with local-model inference on cost and latency for the median task, they would lose, decisively, and the cloud-side infrastructure required would collapse.

This is the structural feature that makes the inflation hypothesis plausible. The current architecture, where most AI inference routes through frontier-model APIs running on hyperscale infrastructure, is not a technical optimum. It is a *commercial optimum* for the providers. The shift toward edge inference — which would dramatically reduce hyperscale infrastructure requirements — is being

slowed not by technical limits but by business model considerations. The providers do not want it to happen, and they have the resources to slow it down.

The shift is happening anyway, slowly, despite the providers. 42% of developers were running LLMs locally as of early 2026, up from a small minority two years earlier. The tooling is improving. The models are getting better. The hardware is getting more capable. The economic pressure from frontier API pricing is pushing users to find alternatives. The trajectory is clearly toward more inference on the edge and less in the cloud. The trajectory implies that the current hyperscale infrastructure is being built for a demand pattern that is going to decay before the infrastructure is paid off.

This is not a fringe view. The VentureBeat survey of IT decision-makers found that “cost per inference / total cost of ownership” jumped from 34% to 41% as a top priority *in a single quarter* in early 2026. Enterprises are starting to migrate workloads off frontier APIs and toward smaller, cheaper, often local alternatives, because the unit economics of running production systems on frontier APIs do not work. The infrastructure being built today is not being sized for the demand pattern of 2027 and 2028, when the migration will be further along. It is being sized for the demand pattern of 2024 and 2025, projected forward as if it would continue indefinitely.

There is a deeper version of this point, and it is the one that most directly inflates the numbers. The argument so far has been about model *size* — that a small model would handle the median task as well as a frontier one. But a meaningful slice of what gets routed to a frontier API is not a language task at all. It is deterministic work with a single correct answer: looking a value up in a database, converting a file from one format to another, classifying a record, routing a request, pulling a field out of a form.

A frontier model is a probabilistic machine — it returns the *likely* continuation of its input, not a guaranteed-correct one — and sending deterministic work to it is like convening a jury to decide

whether two plus two is four. Enormous overhead, real per-call cost, and a small but nonzero chance the jury returns the wrong number. A plain database query does it faster, cheaper, and correctly every single time.

This work migrates downward fastest of all, because it never needed a model to begin with — only the early-adoption reflex of reaching for the most capable tool by default, before the bill arrived. As it moves to small models, to distilled models, and to ordinary deterministic code, the demand it represented does not so much shrink as disappear: it was never durable demand for frontier inference in the first place.

This is what speculative buildouts look like in their late stages: capacity additions made on the assumption that current usage patterns will persist, while the underlying conditions that created those usage patterns are shifting.



A Worked Example: The Compute Cost Of Building The Software

The abstract claim of the last section — that inference is cheap, and that the expensive routing decisions are commercial rather than technical — gets concrete fast if you apply it, in compute, to the single most economically valuable thing AI does, which is write working software.

Take a large enterprise system: an Oracle-style ERP suite, an Amazon-style retail and logistics platform, the kind of system a big company runs its entire operation on. A substantial one is several million lines of code; the very largest are tens of millions. Built the traditional way, at an industry productivity of ten to fifty lines of *finished, tested, reviewed* code per engineer per day, a three-million-line system is five hundred to two thousand engineer-years of work, spread over the better part of a decade. This is not a hypothetical. It

is roughly the history of every large enterprise software product in existence.

Now price the compute for an AI agent that does the whole loop — plans the architecture, generates the code, writes the tests, runs them, finds the failures, regenerates until they pass.

Start with tokens and convert to operations. The finished three million lines, at about ten tokens a line, is thirty million tokens of output. But the agent doesn't write it once and stop; it drafts, tests, debugs, rewrites. Be generous and assume it generates thirty to a hundred times the final codebase getting there — every discarded draft, every test, every debugging trace. That is one to three *billion* tokens. Each generated token costs roughly two operations per active parameter, so for a frontier model with around 10^{11} active parameters, about 2×10^{11} operations a token. Multiply through: the entire build is on the order of 10^{20} to 10^{21} operations.

Sit that next to a training run. A frontier run is 10^{26} to 10^{27} operations. Building the whole enterprise system is therefore something like a *millionth* of a single training run — between a hundred thousand and ten million times less compute. On the same H100-class GPU as before, 10^{21} operations is roughly a hundred to a thousand GPU-hours: a handful of GPUs for a few days, or one server rack for a week. The electricity is a megawatt-hour or two — about what a couple of houses use in a month, and almost certainly less than the laptops and the office of the engineering team that would have built it the old way burned over the decade.

Here is the same fact from the data center's point of view, which is the view this book keeps. Colossus has 220,000 GPUs. Running for a single day, that one facility does enough inference to generate the working code for *thousands* of complete enterprise systems — each of which used to take hundreds of engineers a decade. One large data center, one day, thousands of systems. The compute was never the constraint.

I want to be careful here, because this is exactly the kind of number that gets misused. Those operations buy the *generation and*

testing of the code. They do not buy the things that actually make an enterprise system work: knowing precisely what to build, wiring it into the messy real-world data and the legacy systems it has to coexist with, verifying that it does the right thing under conditions no test suite fully anticipates, and owning the result when it doesn't.

Those are slow, they are hard, and — this is the part that matters for a book about data centers — they are *human*. They consume no GPUs, and they do not get faster when you add a gigawatt of them. The agent hands you the code for a megawatt-hour. The code was never the expensive part.

That is the whole point, and it closes the loop with the training numbers. The one genuinely hyperscale-scale workload — training — is a few gigawatts, globally, run intermittently. The most economically valuable use of the resulting models — producing software that used to take armies of engineers a decade — is a hundred-odd GPU-hours and a couple of megawatt-hours apiece: a rounding error against a single training run, and an imperceptible sliver of one data center.

The expensive workload is rare. The valuable workload is physically negligible. Neither shape is a demand curve that needs three hundred new hyperscale data centers a year. And it shuts the last escape hatch: if the coming flood of AI-written software is real — and it is — it arrives on almost no compute. The binding constraints on turning compute into value were never silicon, and tens of gigawatts built as though they were is capacity sized for the wrong problem.



The Industry Tour: Who Actually Needs What

A tour through the major industries cited as drivers of AI compute demand, in roughly decreasing order of how often each is invoked to justify the buildout — and what each one actually needs.

Consumer Internet and Social Media

This is the largest current driver of inference compute by volume, and it is the part of the picture most distorted by the inflation incentives I described above. The major consumer AI products — ChatGPT, Claude, Gemini, Grok, Meta AI, Perplexity, and their many imitators — are running enormous inference volumes against frontier models, paid for partly by subscription revenue (Claude Pro, ChatGPT Plus, etc.) and partly by free-tier usage that is subsidized by investor capital while the providers try to figure out how to monetize it.

What is the usage actually for? Surveys and product analytics, where they have been published, suggest that the substantial majority of consumer AI queries are mundane: rephrasing emails, generating short writing, asking factual questions, getting recipe ideas, debugging code, explaining concepts, drafting messages, summarizing articles. The hardest queries — novel reasoning, complex multi-step planning, frontier-grade creative work — are a small minority of total usage.

A 4-billion-parameter model running on the user's own device could handle the substantial majority of this load. The user-experience differential between local and cloud for most of these queries is small and often runs in the opposite direction from what the providers claim — local inference has lower latency, works offline, preserves privacy, and costs the user nothing per query. The provider's profit motive is the dominant reason the queries route to the cloud.

The IEA projects that consumer AI inference will grow substantially over the next five years. In user-engagement terms that's probably right — more people using AI more often. But what it means for hyperscale demand depends entirely on where the inference runs. If the growth happens on the edge, hyperscale demand comes in dramatically below current projections; if it happens in the cloud, the projections are roughly right.

That choice will be made by users, developers, and platform designers, not by abstract demand — and users have not, on net, expressed a strong preference yet. They will, as the latency, privacy, and cost advantages of edge inference become more visible.

Enterprise SaaS

The second-largest current driver, by spending if not by volume, is enterprise software-as-a-service. Microsoft, Google, Salesforce, Adobe, Oracle, SAP, and essentially every other major enterprise software company has added AI features to their products and is charging premium rates for them. The marketing is that these features will transform productivity. The reality, in early 2026, is more uneven.

Some enterprise AI features have demonstrated real value. Code assistance (Copilot, Claude Code, Cursor) has measurably improved developer productivity for many users. Document analysis and summarization for legal, financial, and research workflows has saved meaningful time. Customer service automation has reduced costs in many call centers, though the quality is uneven.

Many enterprise AI features have not demonstrated value. The “AI summary” at the top of every Slack channel and Microsoft Teams meeting and Salesforce dashboard, for the most part, has not been shown to materially improve outcomes; it just exists, consuming compute, because the product manager added it as a feature. The “AI insight” sidebar on every analytics product. The “ask AI” button on every interface. Most of these add marginal value at best, and they consume a substantial amount of inference per impression.

The honest unit economics: *some* of these features justify their compute cost, *some* are net negative once API costs are subtracted from the productivity gains, and *most* sit in between. As enterprises learn to evaluate them individually instead of buying them in bundles, the spending will rationalize toward the ones that work — and the ones that work mostly run on smaller models. Code assistance is fine on 7–14B parameter models; so is document summariza-

tion; so is routine customer service. The cases that genuinely require frontier-scale models are narrower than the marketing suggests, and the compute footprint of the valuable features, properly sized, is much smaller than the current allocation.

Robotics

Robotics is the category most commonly invoked in pitches for sustained, large-scale AI compute demand, and it is also the category where the realistic compute picture is most different from the marketing.

Robotics has two compute needs that are quite different. *Training* of robot control models — using simulation, real-world data, or reinforcement learning to build the models that decide how a robot should move — is genuinely compute-intensive and benefits from hyperscale infrastructure. The major robotics labs (Boston Dynamics, Tesla, Figure, 1X, Apptронik, the academic robotics programs at Stanford and CMU and Berkeley, the major Chinese robotics efforts) all use substantial GPU clusters for training.

Inference — the actual on-robot computation that runs while a robot is operating — is overwhelmingly an *edge* problem. A robot operating in the real world cannot tolerate the latency of a cloud round-trip for every motor decision. The control loops run at hundreds to thousands of cycles per second. The compute has to be on the robot, with low latency to the sensors and actuators. Modern robot platforms increasingly include substantial on-board compute (Nvidia Jetson, AMD edge platforms, custom silicon from the robot vendors themselves) precisely because cloud-based inference is unworkable for real-time control.

So the robotics story, for the buildout, is: *training is real and contributes to hyperscale demand; inference is mostly on the edge*. The training contribution is genuinely large but represents a small number of major training runs per year per company, with the total compute footprint comparable to general-purpose AI training rather than larger.

The robotics fleet scale is also smaller than the marketing implies. The total number of advanced robots deployed in the world, as of early 2026, is in the low hundreds of thousands and growing fast but from a small base. Even if every robot in the world were doing some cloud-side inference (which they mostly are not), the per-robot compute would be modest, and the total fleet compute would be substantially smaller than current consumer inference loads. Robotics as a sustained driver of hyperscale demand is a story that may be true a decade from now and is not yet true today.

The World Model Question

There is a substantive debate happening, right now, about whether the next phase of AI even runs on the architecture the current buildout is being constructed for. It is worth describing carefully, because the debate has real implications for whether the hyperscale infrastructure being built today is the right infrastructure for what's coming, or whether it is being built for a technological moment that may already be partly past.

The dominant approach since 2022 has been *large language models* — systems trained on vast quantities of text (and increasingly images, audio, and video) to produce capable general-purpose behavior. GPT-4, Claude, Gemini, Grok, DeepSeek, and the other frontier systems are all variants of this approach. The bet behind the current buildout is, in significant part, that scaling these systems further will continue to produce capability gains across most domains, including domains that require reasoning about the physical world, embodied action, and long-horizon planning.

A substantial and growing camp of senior AI researchers disagrees. Yann LeCun, who ran Meta's AI research for a decade and is one of the three Turing Award winners credited with founding the deep learning era, left Meta in late 2025 to launch AMI Labs (Advanced Machine Intelligence Labs), explicitly to pursue *world models* rather than language models. LeCun has argued, publicly and at length, that the LLM paradigm has a ceiling — that language alone

cannot give a system the physics, causation, persistent state, and embodied common sense required for systems that operate competently in the physical world. His bet, and the bet of his new lab, is that fundamentally different architectures, trained on different kinds of data, are required for the next phase of AI capability.

Fei-Fei Li — the Stanford professor known as “the godmother of AI” for her role in creating the ImageNet dataset that ignited the deep learning era — has taken a similar position with her startup World Labs. As of February 2026, World Labs had raised \$1.23 billion at a \$5 billion valuation, with investors including NVIDIA, AMD, Andreessen Horowitz, Autodesk (\$200 million, with an advisory relationship), Emerson Collective, Fidelity, and Sea. The company’s product, Marble, generates persistent navigable 3D environments from text, images, or video. Li’s framing of the work is that *spatial intelligence* — the ability to reason about the three-dimensional world rather than about flat text and images — is a distinct technological category that the language-centric platforms cannot fully address.

Jeff Bezos, in his first hands-on operational role since stepping back from Amazon in 2021, returned in November 2025 to co-lead Project Prometheus alongside Vik Bajaj, a chemist and physicist formerly of Google X. Prometheus initially raised \$6.2 billion, then closed a \$12 billion Series B in June 2026 at a \$41 billion valuation — total funding of approximately \$18.2 billion in about seven months, making it one of the best-capitalized AI startups in history. Prometheus’s focus is *physical AI* — applying machine learning to engineering and manufacturing of complex physical systems including semiconductors, automobiles, spacecraft, jet engines, and pharmaceutical compounds. Bezos has described the long-term goal as building “an artificial general engineer.”

Google DeepMind has been working on world models in parallel, with the Genie family of models capable of generating and simulating navigable 3D environments. NVIDIA’s Cosmos platform is an explicit infrastructure bet on world foundation models, positioning

the company to sell into whichever camp wins. Physical Intelligence (a Bezos portfolio company since 2024), Figure AI, 1X, and several other embodied-AI startups have positioned themselves along the spectrum between language-model fine-tuning and pure-world-model training.

The substantive technical question between the two camps is whether the language-centered approach can extend to embodied and physical-world reasoning by being scaled and given access to multimodal data, or whether the language approach has a real architectural ceiling that requires a different paradigm to surpass. The honest answer, in mid-2026, is that we do not yet know. Both camps are building, both are making progress, and the decisive empirical evidence will not be in for several more years.

What this means for the buildout question, though, is significant. *The current hyperscale infrastructure is designed around language-model training and inference.* The very large GPU clusters, the specific networking topologies, the optimizations for token-by-token generation, the deployment patterns for serving frontier language APIs — all of these are infrastructure choices that match the LLM bet. If the LLM bet wins, or even if it remains dominant, the infrastructure is well-matched to the demand. If the world-model camp is right that fundamentally different architectures are needed, then a meaningful fraction of the current infrastructure is being built for a paradigm that is at or near its capability ceiling, while the infrastructure that will actually be needed for the next phase is different and is not, yet, being built at scale.

The world-model camp also has an honest version of its own demand picture. The training compute for world models is different from the training compute for language models — it requires more multimodal data, more simulation, more physics, more 3D structure. Some of this compute can run on existing hyperscale infrastructure; some of it requires different architectures. The world-model camp's inference is mostly *edge* in the same way self-driving inference is edge — physical-world AI runs on the device that oper-

ates in the world, not in a remote data center. The total hyperscale demand from world-model AI, if that camp is right, is probably smaller than the demand the LLM-everything thesis projects.

There are, then, three coexisting bets. The *LLM-extension bet* — that scaling language models, with multimodal extensions, will be enough — is what most of the current hyperscale buildout is sized for, and the major language labs (OpenAI, Anthropic, the language sides of Google and Meta) are operating on it. The *world-model bet* — that the next phase needs different architectures, data, and infrastructure — is World Labs, AMI Labs, Project Prometheus, NVIDIA’s Cosmos, the embodied-AI startups, and much of Google DeepMind. The *hybrid bet* — that the eventual buildout is a mixture sized for both — is where the large integrated players (Google, Meta, NVIDIA, Microsoft) are hedging.

Honest assessment, in mid-2026: language-centered systems have, for most of the practical purposes ordinary users and most enterprises actually want, reached a state of capability that is hard to materially improve. Language tasks — writing, summarization, translation, most coding, most reasoning at human-tutorable difficulty levels — work well at the current frontier and are unlikely to dramatically improve through pure scaling. The remaining hard problems in language-centered AI are mostly *agentic* — getting models to take multi-step actions in complex environments — and *factual* — keeping the models honest about what they know versus what they’re confabulating. Substantial engineering effort is going into both, and progress is real but incremental.

The harder, less-solved problems are in the world-model domain. Embodied common sense, robust physical reasoning, sample-efficient learning from real-world interaction, durable transfer between tasks in unstructured environments — these are domains where current systems still perform substantially worse than humans, and where the world-model camp’s argument that fundamentally different approaches are needed has the most traction.

For the buildout, the implication is the one computing history keeps repeating: each era builds infrastructure for the workloads of its moment — mainframe, minicomputer, PC, internet, cloud — and the next era is rarely the one the last anticipated. The current hyperscale buildout is well-matched to a language-paradigm phase that, by most accounts, is approaching its useful ceiling. None of it is wasted; the language work will remain useful for much of what's being built. But the demand projections justifying it assume the continued centrality of a paradigm that some of the field's most credible people are betting tens of billions against — a bet far less settled than the buildout's defenders imply.

Scientific Computing

Scientific applications are the most defensible category of hyperscale AI compute demand, and they are also the category that is least overhyped. The work being done is real, the compute requirements are genuine, and the value delivered is substantial.

Protein structure prediction (AlphaFold and successors) requires substantial GPU compute, has produced demonstrably valuable scientific results, and continues to scale as it is applied to broader biological problems. Climate modeling, particularly AI-enhanced climate modeling that combines traditional physical simulation with learned components, is a genuine and growing user of compute. Materials science simulations using AI for property prediction and inverse design have shown real capability in discovering new compounds, alloys, and structures. Particle physics analysis at CERN and elsewhere uses AI to filter and interpret experimental data. Genomics, transcriptomics, and broader bioinformatics workloads use AI for pattern recognition, function prediction, and disease association.

The aggregate footprint, however, is small — perhaps 1–3% of the compute deployed for general-purpose AI. AlphaFold's costs, dramatic for the research groups running them, are dwarfed by the inference traffic of any one consumer AI product; the entire Nation-

al Science Foundation AI-relevant computing budget is a fraction of what a hyperscaler spends on expansion in a single quarter. “Scientific applications” gets deployed rhetorically to justify the whole buildout, but it justifies a small slice and would be served just as well if the rest never happened.

Drug Discovery and Biotech

Closely related but worth treating separately. Drug discovery uses AI for a growing set of tasks: target identification, molecule generation, ADMET (absorption, distribution, metabolism, excretion, toxicity) prediction, protein-ligand binding affinity estimation, clinical trial optimization, and others. Major pharmaceutical companies (Pfizer, Merck, Novartis, AstraZeneca) and AI-first biotech companies (Recursion, Insitro, Generate, Insilico Medicine) have built substantial AI capabilities.

The footprint is bounded, though: a major pharma’s AI spend runs in the tens to low hundreds of millions a year, and global pharmaceutical R&D compute totals a few billion at most — against the hundreds of billions being deployed for general-purpose infrastructure, and increasingly run on providers’ specialized scientific offerings rather than consumer-facing AI hardware. It is a genuine positive, a small piece of the total, and routinely invoked in ways that overstate its share of the buildout.

Healthcare

Beyond drug discovery, healthcare AI covers diagnostic imaging analysis, electronic health record summarization, clinical decision support, patient communication automation, billing and coding automation, and a growing set of patient-facing applications.

Three properties push most of it away from general-purpose hyperscale, though. Patient data under HIPAA is legally awkward to route through general cloud APIs; diagnostic support in clinical settings often can’t tolerate a cloud round-trip; and the expensive frontier-API model is mostly unaffordable for routine use. So healthcare AI, where it works, runs mostly on hospital data centers, regu-

lated healthcare-specific clouds, or edge devices. The growth is real and substantial; the share that flows into the hyperscale buildout is small, even though healthcare is frequently invoked in the marketing.

Education

Personal tutoring, content generation, assessment automation, accessibility tools — education has been one of the most-discussed applications of AI, with major investments by Khan Academy (Khanmigo), Duolingo, Pearson, and many others.

The footprint, done well, is modest. A tutoring app on a phone, laptop, or school-issued Chromebook can run a 4–8 billion-parameter model and deliver genuinely good outcomes; the compute per student is small. Even at the limit — every student in America tutored by AI daily — the total would be a small fraction of what’s currently being built. The promise is real, the scale required to deliver it is small, and the marketing that invokes it to justify large buildouts is overselling.

Finance

Financial services use AI for many things: fraud detection, risk modeling, trading signal generation, customer service, document processing, regulatory compliance, anti-money-laundering monitoring. The total compute spend is meaningful — financial services is a major customer of cloud AI services — but the bulk of it is batch processing and modest-scale inference rather than frontier-scale workloads.

Algorithmic trading, which gets associated with AI in public coverage, is mostly *not* AI in the modern frontier-model sense. HFT infrastructure is dedicated and architecturally distinct — FPGAs, custom ASICs, low-latency networking, not the GPU clusters being built for AI. The trading firms (Citadel, Two Sigma, Renaissance, Jane Street, DE Shaw) run substantial AI research, but their revenue-generating infrastructure is not what the buildout is for. Finance is a moderate contributor to general-purpose demand, not a driver of it.

Gaming and Entertainment

Gaming AI has historically been local — NPC behavior, pathfinding, decision-making — and there is no strong technical reason for that to change. Generative AI for game content (procedurally generated levels, AI-generated character dialogue, dynamic narrative) is a real area but is mostly being implemented with local inference, because the latency and cost economics of cloud-based generative AI inside a fast-paced game do not work.

Streaming entertainment — Netflix, Spotify, YouTube — uses AI extensively for recommendations and content moderation, but these workloads are batch and modest in compute. They are not driving the buildout.

The most compute-intensive gaming and entertainment workload is *generative video and image creation*, used by content creators rather than embedded in games. Tools like Sora, Runway, Pika, Kling, Veo, and their successors use very substantial inference compute per minute of generated content. The user base for these tools is currently small but growing fast. If generative video becomes a major content category, the inference footprint could become significant. As of mid-2026, it is one of the legitimate sources of compute demand growth, though its scale is small relative to text inference.

Government and Defense

Government and defense AI workloads cover surveillance, intelligence analysis, autonomous systems, cybersecurity, logistics, citizen services, and policy analysis. The compute footprint is substantial, and much of it is run on dedicated government and defense facilities rather than commercial hyperscale infrastructure, for security reasons. The classified portion is large but largely opaque. The unclassified portion is meaningful but not dominant.

The U.S. government's spending on commercial AI cloud services is large in absolute terms but small relative to the hyperscale buildout. The Department of Defense's AI compute spending, in-

cluding its arrangements with the major hyperscalers (Microsoft, Google, Amazon, Oracle), is meaningful. None of this comes close to justifying the buildout being described as “necessary for AI.”

Media and Content Creation

I have saved this for last, because it is the category where the inflation hypothesis is most directly visible and where the question of *what we are actually getting for this compute* becomes hardest to answer in the affirmative.

The major identifiable use case for AI in media and content creation, by inference volume, is *the generation of content for distribution online*. Articles for SEO ranking. Marketing copy. Social media posts. Email newsletters. Product descriptions. Reviews. Listicles. Blog posts. Images for thumbnails, ads, and stock photography. Videos for short-form social media. Voice narration for low-budget audio content. The vast majority of this content is generated in volumes far exceeding what humans would have produced, distributed through algorithmic platforms whose engagement metrics increasingly reward volume over quality, and consumed — to the extent it is consumed at all — by audiences whose engagement with it is shallow, brief, and often involuntary.



The Ceiling the Models Leave Out

Every argument so far has been about misallocation — compute pointed at tasks that don’t require it. There is a simpler argument underneath all of them, and it sets a hard ceiling the demand projections rarely include: the people the intelligence is for.

Intelligence can be made essentially infinite and essentially free. The capacity to absorb it cannot. A person can metabolize only so much real content in a day — on the order of twenty to thirty thousand words of genuinely new, substantive material, the figure the chapter on attention develops at length — and that number is set by

human biology, not by how many GPUs are running. You can double the supply of generated intelligence every year for a decade; you cannot double how much of it a human being takes in. Past a point, additional output isn't consumed. It is produced and skimmed, or produced and discarded, or produced and never opened at all.

The honest counterargument is that not all of it is for humans. A growing share of AI output feeds other software — agents calling agents, pipelines processing documents, systems talking to systems — and that machinery has no attention limit. True, and it matters. But a large fraction of what the projections count is still human-facing, and that fraction runs straight into the wall. A demand model that assumes consumption scales with capacity is modeling a consumer who does not exist. And what gets produced past the ceiling does not simply vanish. It becomes the subject of the next section.

The Slop Economy

There is a word that has entered the technical and public vocabulary in the past two years for the category of AI-generated content I just described. The word is *slop*. It is used as a noun. The standard definition, which has stabilized across recent industry coverage, is *low-quality content generated at scale by AI systems for the primary purpose of capturing algorithmic distribution and monetization, with minimal regard for whether the content is useful, true, or wanted by anyone*.

Slop has economic logic. The marginal cost of producing AI content approaches zero; the marginal revenue from ads or affiliate links on a piece that gets even minimal engagement is small but positive. When cost is near zero and revenue per piece is small but positive, the rational play is volume: produce as much as possible, accept low engagement on each, and aggregate the thin positive margin across an enormous number of pieces. That is the strategy, and it is being run at vast scale by individuals, agencies, and platforms across the global content economy.

The compute consumed by slop production is substantial. Each AI-generated article is a few thousand tokens of inference. Each AI-generated image is a model invocation. Each AI-generated video is a much larger model invocation. Each is, individually, small. Across billions of pieces of content per day, the compute load is enormous. The hyperscale infrastructure being built is, in significant part, supporting this load.

The output of this compute, however, is consumed at miserable engagement rates. A Bynder study found that 62% of consumers are *less* likely to engage with content they believe is AI-generated. A SmythOS finding showed that 52% reduce engagement with content they identify as AI-generated. The Capgemini and Baringa surveys found that 77% want to know when content is AI-generated, and that a substantial fraction of those informed users then disengage.

The readership metrics for AI-generated SEO articles, where they've been studied, show short dwell times, high bounce rates, and minimal social sharing. The content is *technically* consumed — briefly loaded and scrolled past — but not *engaged with* in any meaningful sense.

The economic model of slop only works because of an arbitrage between the content's actual value (near zero) and the algorithmic systems that distribute and monetize it. Search engines, recommendation algorithms, and advertising networks pay out based on technical metrics — clicks, impressions, time-on-page — that do not measure whether the user got anything valuable from the interaction. AI-generated content can score well on these technical metrics through volume and optimization, even when the actual value to the human users is negative.

Google's March 2026 Core Update was an attempt to penalize this pattern. Google explicitly stated that the update would "reward genuinely helpful content" and would target sites publishing AI content "primarily to manipulate rankings." The update has had some effect, with several high-profile content farms losing substantial traffic. But the underlying economics still favor slop production.

The penalty for being caught is occasional ranking demotion; the reward for not being caught is steady ad revenue. The producers run the cost-benefit calculation and continue producing, with marginal adjustments to evade detection.

The compute footprint of slop production is the most morally indefensible component of the AI infrastructure buildout. It is compute being expended to produce content that, by every available measure of engagement, the human population does not want, consumed by recommendation algorithms that distribute it primarily because the production is cheaper than the alternatives, supporting an advertising economy that monetizes attention without measuring whether attention was actually paid.

When the residents of Boxtown breathe formaldehyde from gas turbines, some of the electricity those turbines produce is being used to train models that will be used to produce slop that nobody will read, monetized through advertising networks that pay out based on metrics that do not measure value. This is the moral situation the chapter title is reaching toward. The harm in the neighborhood chapters is being imposed in service of an output that, in significant part, *is itself harmful to the people consuming it*, by polluting the information environment, by drowning out genuine human-created content, and by training the algorithmic systems we use to find information to surface slop ahead of more valuable alternatives.

I do not know what fraction of the AI infrastructure buildout is supporting slop production specifically. The accounting is not public, and the providers have no incentive to break it out. A reasonable estimate, based on the engagement data and the visible content economics, is that *something like a third* of consumer-facing AI inference is supporting content production that, by reasonable measures of human value, is net negative. Another third is supporting content that has modest positive value. The final third is supporting content and applications that have substantial positive value — the genuine productivity gains, the real assistive applications, the educational and informational uses that work.

These fractions are necessarily approximate. The exact numbers depend on how you weight different kinds of value, and reasonable people will disagree. The point is not the precise number. The point is that, on any reasonable accounting, *a substantial fraction of the AI compute being built is being used to produce things that the human population does not net value*. The buildout is not, in the simple narrative, producing useful capability that justifies its cost. It is producing a mix of useful capability, useless capability, and actively harmful capability, with the harmful component large enough that it deserves to be named.

This is the version of the inflation hypothesis I most want you to take seriously. It is not that the demand for AI is fake. There is real demand for some AI applications, and that demand is growing. The inflation is in the gap between *how much compute is being built* and *how much useful AI* that compute will actually produce. The buildout is being justified by demand projections that conflate useful inference with slop inference. The neighborhoods in those earlier chapters are being asked to host data centers on the assumption that the inference those data centers will run is all valuable. A meaningful fraction of it will not be.



The Inflation Hypothesis, Stated Plainly

Let me state the inflation hypothesis in its strongest form, because it is the hypothesis I most want this chapter to put on the table.

The hypothesis is this. A substantial fraction of the current AI data center buildout is being constructed on the basis of demand projections that assume:

1. *All AI inference will continue to route through frontier-class models*, when in fact the trajectory of model capability improvements at the small-model end means that most inference will migrate to edge devices over the next three to five years. Each unit of inference that migrates is a unit of

demand that hyperscale infrastructure no longer needs to serve.

2. *Consumer engagement with AI-generated content will continue to support the slop economy*, when in fact engagement metrics for AI-generated content are already declining, regulatory pressure is increasing, and the underlying advertising economy is shifting away from impression-based monetization toward engagement-based monetization that slop will not survive. The compute being built to support slop production will increasingly produce output that has no economic outlet.
3. *Enterprise AI features will continue to be priced on usage-based billing that supports current infrastructure ROI*, when in fact the unit economics of frontier API calls in production enterprise software are already breaking down, and enterprises are migrating to cheaper alternatives (smaller models, local inference, batch processing) that consume dramatically less hyperscale capacity per dollar of value delivered.
4. *Training demand will continue scaling with model capability improvements*, when in fact the major labs are increasingly relying on architectural improvements, data quality improvements, and reinforcement learning techniques that do not require ever-larger training runs. The marginal training compute required to deliver the next generation of frontier capability may be smaller than the previous generation, not larger.
5. *Specialized applications (drug discovery, scientific computing, robotics) will scale fast enough to absorb infrastructure overhang from consumer/enterprise demand softening*, when in fact the specialized applications are growing from a small base and will not, on any realistic timeline, scale to fill the capacity being built.

If these assumptions are wrong — and the available evidence suggests that several of them are at least partially wrong — then the buildout is, in a literal financial sense, *overbuilding*. The capacity will exist; the demand will not arrive to fill it; the assets will be written down; some of the operators will fail; the residents of the affected neighborhoods will be left hosting infrastructure that was built for a demand pattern that never materialized.

This is the hypothesis. I want to be clear about how confident I am in it, because the chapter would be dishonest if it overstated.

My confidence level: I think the inflation hypothesis is *probably* mostly right, but with substantial uncertainty. The available evidence — the utilization numbers, the engagement metrics, the strategic shifts among major labs, the migration toward edge inference, the breakdown of frontier API unit economics in enterprise applications — all points in the direction of the buildout being substantially ahead of sustainable demand. The counter-evidence — the genuine demand from scientific applications, the real productivity gains from some enterprise tools, the continuing growth in consumer engagement with AI products — is real but appears insufficient to fill the capacity being built.

The honest summary is something like this: *the buildout will probably turn out to have been roughly 50–70% justified by real durable demand, and 30–50% built on speculative assumptions that will not be validated*. The 30–50% overhang will be absorbed in some combination of asset write-downs, capacity repurposing, and unit-economic deterioration. The residents of the neighborhoods hosting the overbuilt portion of the buildout will not be compensated for their share of the cost.

These percentages are imprecise. They are my best honest estimate, based on the available data. Industry analysts who are betting their careers on this question disagree with each other by margins much wider than the precision of any individual estimate. The point is not that the inflation is precisely 40% or precisely 30%. The point is that *meaningful inflation almost certainly exists*, and that the stand-

ard industry narrative — that the buildout is fully justified by demand — does not survive contact with the utilization data.



The Physical Version of the Same Question

Everything to this point has been about compute — how much inference and training the buildout is sized for, and how much of that demand is real. There is a second axis the compute framing leaves out, and the chapter’s title question — *how much do we actually need?* — applies to it just as hard. The buildout is not abstract. It is physical: chips, racks, buildings, transformers, cooling, and, in the most expansive versions of the vision, the machines the compute is meant to enable.

Those expansive versions are worth taking seriously, because they are doing a lot of the work of justifying construction far ahead of present demand. The justification is a projected future of physical abundance — humanoid robots, re-industrialization, machines building machines. The genre has a characteristic artifact: the confident viral figure for how quickly a swarm of humanoid robots could rebuild a city, passed around and amplified, approvingly, by the people building the robots. Set the number aside. Ask what it assumes. Every robot, every chip, every data center, every megawatt of new generation begins as mined material — copper, lithium, cobalt, nickel, rare earths, silica, steel, concrete — and ends, eventually, as physical waste. The chapter on mining and e-waste followed both ends of that line.

This is not hypothetical, and there is one case where the dynamic has already run all the way to completion. Beginning around 2016, China’s bike-sharing companies — Ofo, Mobike, Bluegogo, and at the peak nearly seventy others — flooded the cities with shared bikes. The idea was sound and the early numbers were real: ridership rose, car trips fell, registered users ran into the hundreds of millions. But the build was paced to a race for market share, funded

by venture capital — Ofo alone raised more than two billion dollars — and by early 2018 the firms had put some twenty-three million bikes on the street, faster than any plausible accounting of how many were needed, with rides priced below cost to grab users before the money ran out.

Then the funding turned. Bluegogo went first, then a couple of dozen others, then Ofo itself, with something like sixteen million customers queued for deposits that mostly never came back. And the bikes — the steel, the aluminum, the physical objects — had to go somewhere. They went into what the photographs of the period made famous as “bicycle graveyards”: fields of hundreds of thousands of bikes, in more than twenty cities, stacked in heaps the size of stadiums and left to rust. A useful idea, built on a genuine thesis, overshot real demand by close to an order of magnitude, and the overshoot became one of the defining images of industrial waste of the decade.

A shared bike weighs about twenty kilograms. A data center, a chip fab, a fleet of humanoid robots, and the power plants to run them are the same dynamic with the mass turned up by orders of magnitude — and the most expansive versions don’t only add, they subtract first. To rebuild a city you have to take down the one already standing, and that city is also material: millions of tons of concrete and steel and glass that become demolition debris the instant the rebuild begins. The vision counts the construction. It almost never counts where the old city goes.

The fair rebuttal is the one the buildout’s defenders always make, and it has real force: efficiency improves, recycling improves, designs dematerialize, some material may eventually come from off the planet, and the things being built have enormous value. All true. But the material accounting is precisely the part the projections leave out, and leaving it out does not make it stop applying — and the bike graveyards are a reminder that the recycling is supposed to happen and frequently doesn’t, or doesn’t keep pace with the building. “How much do we actually need” is a question about atoms as

much as about compute, and almost no one making the expansive case has done that arithmetic.

Where This Leaves Us, And What Comes Next

I have laid out, in this chapter, the empirical case that the AI buildout is being constructed substantially ahead of any durable demand signal that justifies it. The utilization data points to capital deployment running ahead of usage. The industry-by-industry tour shows that the categories most often invoked to justify hyperscale construction — robotics, science, healthcare, education, finance, defense — are either smaller than the rhetoric implies, served better by other architectures, or growing from bases small enough that they cannot absorb the capacity being built. The slop economy reveals that a meaningful share of the inference traffic currently flowing through hyperscale infrastructure is producing output that, by every available measure of engagement, people do not actually want.

Beneath all of it sit two limits the demand models tend to skip: a human attention span that caps how much output can ever be consumed, and a material ledger — every machine mined into being and discarded out of it — that the most expansive justifications never total. The inflation hypothesis follows directly from these observations: a substantial fraction of what is being built will not, on any realistic timeline, be filled by useful demand.

If the chapter ended here, it would leave the reader with the diagnosis but not the prescription, and with the diagnosis but not the engagement with the strongest counter-arguments. Both deserve their own treatment, and both will get it.

The next chapter — *The Architecture That Works* — addresses the practical question that follows naturally from this one. If the buildout is substantially overbuilt, *what would the right buildout look like?* That question has a surprisingly concrete answer, and the answer is being demonstrated in one specific application area where physical constraints forced the industry to get it right. The next

chapter walks through what honest sizing would look like, resolves the apparent contradiction between idle enterprise capacity and user-experience scarcity, examines the specific case of xAI's Colossus facility as an illustration of how the inflation works in practice, names the structure of the financial system around the buildout as an *anticipation economy*, and uses self-driving as the case study for what compute used well can actually accomplish.

The chapter after that — *The Bill* — engages with the two arguments most often deployed by the buildout's defenders. The financial markets are voting on the buildout, and the votes are coming in at trillion-dollar valuations. The United States and its allies face a real geopolitical contest with authoritarian competitors, and the Ukraine war has demonstrated that AI on the battlefield has decisive consequences. Both arguments are serious. Both deserve direct engagement rather than dismissal. The third chapter takes them on, examines what they imply and what they don't, and ends on the question that runs underneath the entire arc: *at what cost*, and *to whom*?

For now, hold the diagnosis. The buildout is largely ahead of demand. The demand that does exist could be served by architecture that the next chapter will describe. The costs being borne now are being borne by people who have not been adequately represented in the decisions. And the question of how this resolves — whether the bet pays off for the bettors, whether the buildout finds its eventual demand, whether the host communities recover — is one that the next two chapters can illuminate but not resolve.



CHAPTER 13

The Architecture That Works

What Compute Used Well Would Actually Look Like



The previous chapter ended with a diagnosis: the buildout is running ahead of demand, much of the inference flowing through hyperscale infrastructure doesn't need to be there, and the gap between what's being built and what's justified is what I called *inflation*. That diagnosis raises an obvious question. If the current pattern is wrong, what would the right one look like?

This chapter tries to answer that concretely. Then it takes on the objection that anyone who actually uses this stuff will be shouting at the page by now — *if the buildout is overprovisioned, why can't I rent an H100, and why does Google keep saying it can't meet demand?* — because the chapter would be a fraud if it ducked that. As it turns out, resolving the apparent contradiction is one of the more revealing things in the whole argument.

From there I'll use xAI's Colossus as the cleanest worked example of how the inflation actually operates, give a name to the financial machinery around the buildout — the *anticipation economy* — and end on self-driving, which is the best evidence we have for what compute used well looks like.

What Honest Sizing Would Look Like

One distinction first, because the rest of this section leans on it and the earlier draft blurred it. Almost none of the buildout I'm about to size is *built and sitting idle*. It's **announced** — capex targets, signed power contracts, cluster counts in press releases, projections out to 2030. So what follows isn't measured capacity against measured demand. It's **the plan against the possible reality**: what the industry says it intends to build, set beside what today's demand would justify. Both sides of that are soft, and I'd rather show you the softness than hand you a clean number and ask you to trust it.

With that flag raised — suppose you took the inflation hypothesis seriously and asked what a buildout matched to durable demand would actually look like.

Frontier training. Watch how fast the arithmetic turns into assumption. Ten or so organizations do frontier training. If each needs one or two hyperscale-class clusters — with slack for failed runs, redundancy, and experimentation — you get roughly twenty to thirty. The announced plans point at several hundred by 2030. Divide one by the other and out pops “5–10x too much.”

I've quoted ratios like that, including in earlier drafts of this chapter, as if they were findings. They aren't. That's one soft number — my “one or two clusters per lab” — divided by another soft number — capacity announcements, which are half marketing — with no error bars on either. Read it as an order-of-magnitude sketch, not a measurement.

And the assumption holding the whole sketch up — that training demand stays roughly what it is now — is exactly where it's most likely to break, for a reason worth taking seriously: *world models*. The frontier so far has mostly trained on text, and text is small. A growing and well-funded camp is betting the next leap is models that learn the physical world — space, motion, cause and effect, the sensorimotor understanding a toddler or a self-driving car needs. This is not a fringe wager. Yann LeCun left Meta in late 2025 to build

it at AMI Labs; Fei-Fei Li's World Labs raised a billion dollars for it; the venture Jeff Bezos is co-running, Project Prometheus, is pointed squarely at the physical economy with more than six billion behind it.

If that pivot is real — and it's a real *if* — training demand could climb steeply, because the inputs are enormous. NVIDIA's Cosmos world model was reportedly trained on something like twenty million hours of real-world video, a different order of scale than the text corpora behind today's language models, and training at that scale is genuinely compute-hungry. A buildout that looks like 5–10x overkill against text-model demand could look right-sized, or even short, against world-model demand.

I'll give the other side of even that, though, because it isn't settled. The most prominent researcher in the camp, LeCun, is betting on an architecture whose entire point is to need *less* brute-force compute, not more — his March 2026 result trained a working version on a single GPU. So world models might detonate training demand, or might arrive on a leaner path that doesn't. Nobody knows yet. This is the piece of the inflation thesis I hold most loosely, and it's where the boom-side argument is strongest.

Consumer inference. Harder to pin down, but it comes down to three things: how many active users, how many queries each sends a day, and what fraction of those queries route to hyperscale rather than to the edge. If the edge migration proceeds — and the trajectory of small-model capability says it will — then by 2028 the consumer inference actually landing on hyperscale is perhaps 30–50% of what today's projections assume. Overbuilt by 2–3x.

Enterprise inference. Same shape of argument. Enterprises are already drifting off the frontier APIs toward cheaper options, and the drift is speeding up. Overbuilt by perhaps 2x.

Scientific and specialized compute. Here is the one category that is *not* overbuilt. Drug discovery, climate modeling, protein folding, materials science — real demand, growing, justified by the work it does, and if anything *underserved* by the current buildout. A sanely-

sized buildout would shift capacity toward these uses even as the total came down.

Robotics and sovereign AI. Both real, both growing, both meaningful — and both still modest shares of the whole. The interest a lot of governments have shown in sovereign compute, so as not to depend on American hyperscalers, is durable and has been building since 2024.

Add all of that up and a justified buildout would be substantially smaller, more geographically spread out, tilted toward scientific and government work, with far less consumer and enterprise inference at hyperscale and far more of it at the edge — and with a much smaller footprint on the grid and on the neighborhoods.

What's actually being built looks more like the opposite instruction set: maximize total capacity, maximize the hyperscale share, push everything through frontier APIs because that's what the business model needs, build wherever the legal tactics from the neighborhood chapters make it cheapest, treat residential proximity as a feature rather than a problem, and trust that the demand will show up to justify it.

That distance — between what's planned and what today's demand would justify — is the inflation. The inference pieces are where I'd put the most weight: the edge migration and the enterprise awakening are already visible, and they point at a hyperscale inference buildout sized well above where demand is heading. The training piece is softer, for the world-models reason above. But even setting training aside entirely, the inference overhang alone is large.

And the cost of it gets paid in the present by the residents in the earlier chapters and by every ratepayer across the PJM region. It gets paid in the future by the write-downs that follow if the demand never arrives, and by the neighborhood cascades that grind on whether or not anyone ever writes the assets down.

But Wait — Why Can't I Rent an H100?

If you've tried to actually use this infrastructure in 2026, you have an objection ready, and it's a good one. *If there's so much idle capacity, why is it so hard to rent an H100? Why does Google's leadership keep telling earnings calls they can't keep up with demand? Why does Claude Code crawl at peak hours, sometimes for days? Why does ChatGPT fall over during a spike?* The numbers from the last chapter — 5% enterprise utilization, 11% at Colossus — say *abundance*. Your lived experience says *scarcity*.

Both are right. And the reason they're both right is the most useful thing in this section, because it shows what's actually going on under the hood. The short version is that there is no such thing as “the” AI compute market. There are at least three of them, with genuinely different supply-and-demand dynamics, and the apparent contradiction is just what you get from mashing them together.

The Three Markets

One: stranded enterprise capacity. This is where the 5% comes from. Companies that bought H100s in the 2023–2024 panic — banks, insurers, manufacturers, agencies — are sitting on piles of idle silicon. It's real, it's racked, it's drawing power, and it's idle most of the time because the applications that were supposed to run on it either never showed up or turned out to need a fraction of the compute the purchase order assumed.

And it's *stranded* in a precise way: it can't be redeployed to serve the demand that does exist. An idle H100 in a regional bank's data center cannot serve Claude inference for a developer in Seattle. It's on the bank's network, behind the bank's firewall, plugged into none of the orchestration any public cloud uses. The capacity is real; the *market* for it isn't, because the hardware is in the wrong account, wired for the wrong job, and the cost of repurposing it swamps whatever value it could throw off.

We've seen this before. The 1990s telecom boom laid down enormous quantities of "dark fiber" — cable physically buried but never lit for traffic. Real, expensive, and basically useless for years, until demand finally grew into it and the cost of lighting it fell.

The enterprise GPU build has produced the same thing one layer up: dark silicon. Real, expensive, and largely useless to the broader market because the integration cost is more than it's worth. NVIDIA booked the revenue. The enterprises booked the depreciation. The analysts count the chips in their "total deployed GPU" tallies. But none of it is part of the *usable* supply for the market that's currently begging for inference.

Two: hyperscale frontier inference, deliberately under-provisioned. This is where your bad afternoon with Claude Code comes from. The hyperscalers running Claude, ChatGPT, and Gemini do hold a lot of compute, and they run it harder than the enterprise market — probably 30–50%, though none of them will tell you. But they don't provision for the *peak*. They provision for something like the 70th or 80th percentile of load. When demand runs above that, the service degrades, and you experience it as "Claude Code is unusable this morning."

The thing to see is that this is a unit-economics decision, not a physics problem. Provisioning for peak means holding capacity idle most of the time to catch the occasional spike — and the GPUs depreciate and the power meter spins whether or not anything is computing. To make the economics work at today's prices, the providers run hot and eat the degraded peaks.

Which flips the usual story on its head. The industry says, "we can't add capacity, we're constrained." The honest sentence is, "we *won't* add capacity, because doing so would make unit economics we're already losing money on even worse." The first is a claim about physics. The second is a claim about money — and that second claim is itself evidence for the inflation. If provisioning for peak would torch the economics, then the price per token is *already* too low to support the infrastructure being run, before you add a

single chip for the real demand curve. They're losing money on the average query and choosing to ration the marginal one rather than make the loss bigger.

Three: newest-generation chips. When Google's CEO says he can't meet demand, he's mostly talking about the *latest* chips — Blackwell in 2025, the next thing in 2026. That supply genuinely is tight: TSMC's production was spoken for years ahead, the advanced packaging (CoWoS) is a real bottleneck, the high-bandwidth memory is supply-constrained, and NVIDIA hands out the scarce allocation on strategic grounds, not pure willingness to pay. So yes, Google can't get all the *Blackwell-class* chips it wants on the schedule it wants.

That is not the same as total compute being constrained. The previous generation — H100s, which by 2026 standards are last year's model — is not tight at all; rental prices for older hardware are softening. The "can't rent an H100" feeling is partly a question of *where you're shopping*. Hard, if you're trying to rent from AWS or GCP in the configurations they'd rather sell you. Easy, if you'll take older silicon from the specialty shops — Lambda, CoreWeave, RunPod, Vast.ai, Together, Modal, and the dozens of "neoclouds" that have sprung up since 2023 — often at lower prices. For anything short of the frontier, the hyperscalers have quietly become the most expensive option, not the cheapest.

Lay the three side by side and the contradiction evaporates. Aggregate capacity is overbuilt; frontier inference is under-provisioned on purpose; the newest chips are genuinely scarce. All true at once, all pointing the same way. None of them — alone or together — justifies a buildout sized on the bet that total compute demand keeps climbing at today's rate forever.

The Subsidy, and the Awakening

The under-provisioning tells you something about pricing, if you sit with it. When Anthropic throttles Claude Code at the peak instead of adding capacity, it's making a precise statement: at current prices,

the marginal query at the peak isn't worth enough to pay for the hardware to serve it. If frontier inference were actually profitable per unit, they'd provision for peak, because every extra query would be extra margin. They don't, because the extra queries aren't margin — they're loss, and serving more of them makes the math worse. So they ration by degradation. The slowdown isn't a supply problem wearing a costume. It's a pricing problem.

This is the Uber and Lyft playbook, pointed at inference: subsidize the ride, grab the market, kick the unit-economics question down the road to "later, when we have scale." The frontier labs underprice on purpose, because their valuations are priced on *growth* — they want you habituated, they want enterprises building on their APIs, they want the world leaning on them. The Uber version did eventually produce profitable rides, after a decade of losses and a fair amount of quiet erosion in what it was actually like to be a driver or a rider. The inference version is still in the subsidy phase, the economics still upside down on most workloads, the rescue — better capability, cheaper inference, pricing power — still parked in the future tense.

Enterprises are starting to do the arithmetic. Picture the mid-sized company that wired Claude or GPT-4-class APIs into customer service, document processing, and a dozen internal tools back in 2023, when the pricing felt cheap and the capability felt essential. By year-end the bill is tens to hundreds of thousands a month, big enough that the CFO notices, and the AI team gets asked to defend it. They struggle — not because the features are useless, but because a lot of the money is going to queries a much cheaper model would have handled just fine.

The autocomplete in the support tool calling out to GPT-4 when a 7B model running locally would've produced output nobody could tell apart. The document summarizer reaching for Claude when a small open model would do. The internal Q&A bot firing frontier inference at every question, including the ones where the person already knew the answer and was just stalling.

Each query is pennies. The pile, measured against the value it actually delivered, is wildly wasteful. The survey I cited earlier caught the leading edge of this: cost-per-inference jumped from 34% to 41% of IT decision-makers' top priorities in a single quarter. Through 2027, big chunks of enterprise work will slide toward smaller models, local inference, and batch processing that burn a fraction of the hyperscale capacity per dollar of value. The infrastructure going up right now is sized for the 2024–2025 demand pattern, projected forward. The 2027–2028 pattern will be smaller, and the difference is more inflation.

I want to be careful here, though, and not claim more than the evidence supports — because this is exactly the kind of argument that gets carried too far. The inflation hypothesis is my read, and I think it's the better-supported read, but it's a hypothesis, not a verdict, and honesty means saying so out loud. The other side is real. It is entirely possible that demand catches up to the capacity — that the productivity gains compound faster than the skeptics expect, that enterprise adoption deepens instead of stalling, that uses nobody has thought of yet show up and fill the buildout. Serious, well-informed people believe that version, and they are not obviously wrong. If they're right, what looks like overbuilding today will look like foresight in hindsight.

So the honest framing is conditional. *If* the inflation hypothesis holds, the open questions are how much of the buildout gets finished before the market figures it out, and how much damage lands on the host neighborhoods in the meantime. If it doesn't hold, a lot more of the buildout turns out to have been warranted. Either way, one thing doesn't depend on which story wins: the residents in the neighborhood chapters pay their costs whether the boom sustains or breaks.

And for everyone else — including a reader sizing up what to do with their own savings — the smart posture is not to bet the house on either story before the evidence has settled it. A reader who closes this chapter convinced the crash is imminent and rearranges

their financial life around that conviction is running a real risk of their own, because the crash everyone can see coming sometimes takes years to show up, and sometimes never shows up at all.

The Colossus Lesson

Come back to Colossus 1, because it's the cleanest example of the inflation working in real life.

It went up at absurd speed — nineteen days from concept to the first operational cluster, the full build trailing over the next eighteen months — at a cost somewhere around \$40 billion, to train Grok. The cooling was sized for full training load. The power was sized for full training load. Memphis was picked for fast permitting and grid access. The gas turbines went in without permits because the company did the math and decided the legal exposure was cheaper than the wait. All of it rested on one assumption: that xAI's own frontier training would fill the building, and that filling it was do-or-die for xAI's standing as a frontier lab.

Eighteen months on, with the thing running, it sat at roughly 11% MFU. The training xAI was actually doing didn't need the capacity — maybe Grok's requirements came in smaller than projected, maybe the training was less efficient than hoped, maybe the shifting market simply made more Grok capability worth less. Whatever the cause, the facility had been built ahead of its own creator's demand. Eighteen months after *that*, in May 2026, xAI leased the whole thing to Anthropic and Google for something like \$26 billion a year against the \$40 billion build — paying it off in about a year and a half.

On a spreadsheet, the rental is fine. But look at what it does *not* prove. It doesn't prove there was demand for the capacity *before* xAI *built it*. It proves that once the capacity existed, renters turned up. Their demand — Anthropic's appetite for Claude inference, Google's for Gemini Enterprise capacity — was real, but it wasn't the demand that justified pouring the foundation. xAI bet its own

training would fill the place. The bet was wrong. The building stands anyway, and other labs fill it.

That's the inflation in miniature: capacity built on a thesis about one kind of demand, the demand failing to arrive in that form, the capacity repurposed to a different and usually thinner use that's enough to keep the lights on. The cost gets recovered. The story that was pitched doesn't survive. And the gap between the pitch and the reality is the inflation. The people in Boxtown and Whitehaven get nothing from the distinction — the air is the same whether the turbines are powering Grok's training or Anthropic's inference, and the tactics used to build the thing never depended on the demand thesis being true. They only depended on overcoming the neighbors.

The Anticipation Economy

There's one more layer to the Colossus story, and any honest reading has to include it. Taken purely as a transaction, the rental is *excellent* for SpaceX. It pays the build off in eighteen months, the margins are reportedly strong, and the cash flow steadied the balance sheet right when it needed steadying. The financial press read it as Musk, once again, finding a way to monetize an asset that looked like it was about to become a liability — and called the move visionary: a pivot from “win the model race” to “become a hyperscaler,” which in 2026 is a posture the market will value at multi-trillion-dollar levels.

All of that is true *and* none of it contradicts the analysis. The rental is simultaneously a tactical financial win and a confession that the strategy behind the original build failed. The fact that both are true at the same time is the most important thing the deal reveals, because it's how the whole technology economy currently runs.

We are in an *anticipation economy* — a thing the standard vocabulary doesn't have a clean word for. In it, an asset's price is set not by the cash it produces but by *what the next round of buyers will expect it to produce*. The valuation floats up on the meta-level, expectations

about expectations, several stories above the actual productive activity — and the operators who are best at conjuring assets whose anticipated value runs ahead of their realized value get rewarded most, even, or especially, when the thing underneath isn't making money.

Musk is the clean case because both things are true of him at once. He has delivered genuinely enormous contributions to several frontier technologies — reusable rockets, satellite internet, mass-market electric cars, real progress on self-driving — *and* he has become exceptionally good at the financial-anticipation game. The technology isn't fake because the financial layer runs on anticipation, and the financial layer isn't erased by the real engineering underneath. The honest reading holds both, and the Colossus rental is a textbook instance of the second sitting on top of the first.

What Keeps It Going

The anticipation economy keeps running because a handful of conditions prop it up:

Index and venture participation. Big institutions hold SpaceX and xAI through index funds and venture allocations, where any single position is a sliver of a portfolio and never gets judged on its own. It's sized to catch upside with the downside diversified into oblivion. Lose a little if the thesis fails; lose a *lot*, in relative terms, if you sit out a thesis that works. That asymmetry pushes everyone toward participating no matter what the underlying analysis says.

Capital with nowhere to go. The long stretch of cheap money produced more capital hunting for returns than there are technologies generating returns to match it. The surplus flows toward whatever can absorb it — which means the assets with the most credible-sounding stories about transformational future cash flow. Anticipation becomes the asset class.

Storytelling as a craft. The operators who specialize in this have honed it: stories about transformation rather than optimization, exponentials rather than lines, physical infrastructure — rockets, data

centers, tunnels — that signals seriousness, real technical substance laced through the speculative parts, all delivered with enough confidence that being the one to hedge in the room costs you socially.

The cost of walking away. Once enough big institutions are in a trade, getting out while everyone else stays in makes you the one who missed the upside if it works, with no reward for prudence if it doesn't. The incentives in asset management favor going along over exercising judgment, which is exactly why the people who could puncture the thing don't.

Tactical wins that launder strategic failures. This is the feature that makes Colossus such a clean example. Each win along the way — a launch, a contract, a partnership — gets presented as proof the big thesis is working, even when, looked at squarely, it's proof the original thesis failed and is being swapped out. The Colossus rental is a real win: real cash, real margin, real customer. And it quietly papers over the fact that the rental only exists because the Grok-training thesis never produced the demand that was supposed to fill the building. Win and failure, reported as a single piece of good news.

What It Produces

An economy organized around anticipation rather than cash flow throws off a few predictable things, and you can see every one of them in the AI buildout.

It produces buildouts bigger than realized demand can justify, because the capital is there, the narrative blesses the spending, and the running tally of tactical wins provides cover. It produces operators who are spectacular at managing the story and merely adequate at running the business, because the story is what gets rewarded. It produces externalities that land on people outside the game entirely — the neighborhood residents, who hold none of the equity and collect none of the fees and eat the costs whether the game rolls on or falls apart.

And it produces a public conversation that misreads the whole thing, because the business press's instruments can't tell the difference between an operator building something durable and an operator stacking tactical wins on a tower of failed strategic bets. From the outside the two look identical, right up until the chain snaps.

When does it snap? That's one of the hardest questions in economics to answer with a straight face. Previous anticipation economies ran a surprisingly long time — the dotcom era roughly 1995 to 2001, the mortgage-securities era roughly 2002 to 2008, call it six years each. The AI era got going in late 2022; by that yardstick it has years left in it. Or it cracks next quarter, when some lab misses an earnings number, or a rate case rejects a hyperscaler's interconnect, or a court rules the wrong way on one of the cases from the neighborhood chapters. Nobody knows. The honest answer to "when does this break" is: we don't know, but the shape is familiar.

And when it does break, the damage falls on the late investors who bought at the top, the employees whose options go to zero, the public infrastructure committed to support it, and the communities that hosted the buildings — while the senior operators, paid the whole way through, generally walk away with their fortunes intact and go find the next thing.

What Compute Used Well Looks Like: Self-Driving

After all of that, the hopeful part — because the architecture that works isn't a thought experiment. It's already on the road.

Driving for a living is, for most of the people who do it, brutal. Long-haul truckers spend weeks at a stretch on the road, eating badly, sleeping in the cab, away from their families, soaking up the cardiovascular damage of fifteen-hour days in a seat, with crash-fatality rates among the worst of any job in the country and a life expectancy to match. It has been a real path into the middle class, especially for immigrants and for Black drivers, and the human cost of it has been enormous and mostly uncounted by the economy that depends on the freight.

Self-driving is trying to end that job, and after a couple of decades of grinding it has reached the point where, in the cities where it runs, the autonomous cars crash less often than the humans. The compute behind that has a very particular shape. The *training* is a legitimate hyperscale workload — huge, centralized, done by the people who need it. The *inference* lives on the edge, on the car, because a vehicle deciding whether to brake can't wait on a round trip to a data center; the latency and the reliability requirements force the computing to be local. Heavy centralized training, distributed edge inference. The physics of the problem left the engineers no choice, so they built the right thing.

That's the model. If the rest of the AI buildout were shaped like this, most of the inflation in this chapter would simply not exist. The training compute would still get built, in the volumes real training demand justifies, by the labs that need it. The inference would move to the edge — onto phones and laptops and enterprise hardware — for the large majority of workloads. And the hyperscale inference buildout currently rising in residential neighborhoods would be maybe a third of its size, or less.

The detail I find most telling is that some of the same operators who got self-driving's architecture *right* are getting general AI's architecture *wrong* — Tesla under Musk on the first, xAI under Musk on the second. That tells you the architecture isn't a question of capability. It's a question of incentive. When the physics forces the right answer, the operators build it. When the physics doesn't force it, they build whatever maximizes their own returns, even when a different design would be better for everyone else — which isn't a moral failing peculiar to any one person, just what happens in a market that doesn't price externalities.

General AI's architecture is wrong mostly because the people who'd benefit from the right one — everyone paying for inference, the communities hosting needless data centers, the planet absorbing the extra power — aren't the people who design it. Self-driving worked because the physics wouldn't let it not. The lesson for the

rest of the buildout is that the physics has to be reasserted some other way: by regulation, by consumer demand for edge-default designs, by enterprises voting with their workloads.

And the good news is that the convergence has, slowly, begun — the edge migration, the enterprise awakening, the steadily improving small models are all early signs of it. Self-driving has shown one path. The rest of the industry will, most likely, find some version of it. To the degree the current buildout runs ahead of where that path leads — which is what the inflation hypothesis predicts, and what I think the evidence favors, while admitting it isn't certain — the question is how much of the excess gets built before the path goes mainstream, and how much damage lands on the host communities along the way.

What Comes Next

So that's the case: what an honestly-sized buildout would look like, why idle capacity and a frustrating user experience can be true at the same time, how the inflation plays out on the ground at Colossus, what kind of financial machine is paying for it, and what compute used well looks like when somebody is forced to build it right.

What I haven't touched is the question any reader is now holding: who's actually paying for all this, and what happens to them if the inflation resolves the way I've suggested it might — and, alongside it, what about national security, what about China, what about the world this buildout is supposedly defending. That's the next chapter, *The Bill*.



CHAPTER 14

The Bill

Who's Actually Paying for All of This



The last two chapters were about the machine — how big the buildout is, how it should be sized, what the right architecture looks like. This one is about the money. Specifically, about a question the technical chapters never quite reach: *who is paying for all of this, and what happens to them if it doesn't work out the way it's being sold?*

Two arguments stand in the way of taking the costs seriously, and the buildout's defenders reach for both. The first is that the smart money has spoken — the capital markets are pouring trillions into this, and people who manage money for a living don't do that on a whim. The second is that it doesn't matter whether it pencils out commercially, because it's a matter of national survival: the war for the twenty-first century will be fought partly in compute, and falling behind is not an option. Both arguments are serious. Both are partly right. And the best way into both is to start with a single morning in June.

The Largest IPO in History

On the morning of June 12, 2026, about an hour before the opening bell, a Falcon 9 lifted off from Florida carrying a load of Starlink satellites. It was the 650th flight of the rocket. In Starbase, Texas, Elon Musk stood behind a Nasdaq-branded podium, and in New

York his executives rang the bell. SpaceX — recently merged with Musk’s AI company, xAI — was going public, and by the end of the day it had made history.

The numbers are worth stating plainly, because they are real and they are staggering. The company sold roughly 555 million shares at \$135 and raised \$75 billion in the offering itself — the largest IPO ever, nearly triple the previous record, Saudi Aramco’s \$25.6 billion in 2019. The stock opened at \$150, surged past \$160, and at one point touched a valuation north of \$2.2 trillion, one of the ten most valuable companies on Earth. Musk, on paper, became the world’s first trillionaire. By any conventional measure, it was a triumph.

Here is the part the triumph tends to crowd out. SpaceX has never turned a profit. It lost close to \$5 billion in 2025, and \$4.28 billion in the first quarter of 2026 alone — more than four times its loss a year earlier. Strip out the rockets and Starlink, and the AI division it had just absorbed was the wound: roughly \$6.4 billion in losses on \$3.2 billion of revenue. Morningstar, running the kind of discounted-cash-flow model that has valued companies for a century, put the whole enterprise at \$780 billion — call it overvalued by more than half.

And six days after the IPO, on June 18, the company’s bankers were back in the market arranging a bond sale of at least \$20 billion, the bulk of it to refinance a bridge loan SpaceX had taken on to buy xAI in the first place.

Sit with the shape of that. A company losing billions a quarter raises \$75 billion from the public, then borrows \$20 billion more within the week, while its founder keeps roughly 85 percent of the voting power and the analysts who model these things for a living call it overvalued. Michael Burry — the investor who famously shorted the 2008 housing market — took a public position against the trade. None of this means SpaceX is a fraud, or that it won’t deliver. It means we are watching a particular kind of moment: the one where paper valuations get converted into real, banked capital while the window is open.

There is a name for it, and it isn't flattering, but the name matters less than the pattern, and the pattern is everywhere right now. OpenAI and Anthropic have both filed to go public. Oracle, Nvidia, Alphabet, Amazon — all have rushed into the debt markets to fund data centers, raising hundreds of billions between them.

So the capital markets are, as the defenders say, voting. The votes are coming in at trillion-dollar valuations. The only question is *what*, exactly, they're voting on — and *whose* money is being voted with.

Whose Money Is It

Follow the \$75 billion backward.

The buyers of a mega-IPO, the buyers of investment-grade bonds, the institutions whose appetite makes these deals possible — Vanguard, BlackRock, Fidelity, State Street, the big pension funds, the insurers, the university endowments, the public retirement systems — are not spending their own money. They are pooled vehicles. They hold the savings of tens of millions of ordinary people. The teacher with a state pension, the firefighter with a municipal plan, the family with an IRA full of index funds, the worker whose 401(k) defaults into a target-date fund — those are the people whose money is actually deployed in the buildout, channeled through intermediaries they will never meet and mostly do not think about.

And the exposure has quietly become enormous. The largest handful of tech companies — the ones an earlier era nicknamed the Magnificent Seven — have grown to roughly a third of the entire S&P 500 by value, with Nvidia alone around seven or eight percent of it. When SpaceX enters the indices, it will climb that list fast; if OpenAI and Anthropic follow, faster still.

Which means the default retirement investment for most working Americans — the broad-market index fund, the thing they were told was the safe, boring, responsible choice — has, without anyone deciding it, become something close to an AI-buildout index. The bond side is the same story through a different door: the invest-

ment-grade bonds funding the data centers sit in pension portfolios and the bond sleeves of those same target-date funds.

So when someone says the “sophisticated investors” are confident, the sentence is true and also misleading. Underneath the sophistication, the capital is overwhelmingly ordinary people’s retirement money, and most of those people have no idea what it’s funding and no real way to opt out.

That changes what’s at stake if the bet goes wrong. We have a sense of the scale from the last two times. By common estimates, the dotcom collapse erased something like \$5 trillion in equity wealth, and the 2008 crisis took roughly \$2.7 trillion in retirement assets at the bottom. Both times, the heaviest losses landed on the same cohort: the people closest to retirement, who had shifted toward stocks as their date approached and who could not afford to wait out a recovery that, as we’ll see, can take a very long time. Millions who had planned to retire between 2008 and 2012 worked five or ten years longer than they meant to.

If the AI buildout resolves the way speculative buildouts usually have, that bill falls hardest on the Americans now between fifty and sixty-five — people who taught school, fixed pipes, kept books, ran small businesses, and saved exactly what the retirement industry told them to save, into exactly the vehicles it told them were safe. They did everything right, by the rules as written, and they are the ones most exposed.

I want to be careful not to state that as a prophecy, because it isn’t one. The bubble might deflate gently. The technology might deliver enough real productivity, fast enough, to grow into the valuations — the boom-that-validates is a live possibility, and serious people are betting on it. But the historical current runs the other way, toward real losses for retirement-aged savers, and nothing in the structure of this market gives strong reason to expect an exception.

The Railroad

Musk, to his credit, named the historical comparison himself. In SpaceX's pre-IPO pitch, he likened the company to Union Pacific — the transcontinental railroad of the 1860s and '70s, the project that physically stitched the American economy together. The implied argument is seductive: trust the trajectory, never mind the present losses, because the transformation will be vast enough to redeem the bet. And as history, it's accurate — more accurate, maybe, than he intended.

Because the honest story of Union Pacific is not a clean story. The railroad was the site of one of the largest corruption scandals in American history. Its own insiders set up a separate construction company, Crédit Mobilier, awarded it wildly inflated contracts paid for with federal subsidies, and handed shares to members of Congress to keep the favorable legislation coming. Early investors were largely wiped out in the company's first decade. The path to durable profit ran through bankruptcies, restructurings, federal bailouts, and political careers destroyed. The communities along the route were transformed, often violently — Native populations displaced, and thousands of Chinese laborers worked to death laying the track.

And here is the thing: the transformation was still real. The railroad did eventually become profitable. It did remake the economy, more or less as its boosters promised. Both halves are true at once — the project justified itself in the long run, and the specific deals that financed it were a machine for moving wealth from the public and from retail investors to the insiders who structured them. The transformation was real. The cost was real. The people who paid the cost were, by and large, not the people who collected the returns.

That is the template. The eventual technology may well justify *some* version of this buildout. It is far less likely to justify the specific transactions, the specific siting, and the specific cost allocation we're watching in 2026. The gap between the project's long-run worth and

the present deal's specific costs is the inflation — and the inflation is being absorbed, in the main, by the host communities, the ratepayers, and the late investors buying in near the top.

The Compute War

Now the harder argument, because it's the strongest case for building anyway, and because it's true.

The claim is that the world is in a real contest over which kind of order runs the century — open and democratic, or authoritarian and state-controlled — that AI will be decisive in that contest, and that a world where authoritarian states hold the better AI is unacceptable on both economic and moral grounds. This is not a fig leaf. It's the argument behind the CHIPS Act, the export controls, the federal money for domestic fabs, and much of the regulatory permissiveness that let the buildout proceed with so little federal friction. The people making it largely believe it, and some of them are reading classified assessments the rest of us aren't.

The clearest evidence sits in Ukraine, which over four years became the most data-rich war in history. Both sides flew millions of drones; the character of combat changed under everyone's feet. In one record month, Ukraine's military reported that drones accounted for 96 percent of Russia's battlefield casualties — unmanned systems outpacing every other weapon combined. Over the course of 2025, Ukraine's defense minister, Mykhailo Fedorov, said drones killed or seriously wounded more than 240,000 Russian soldiers across some 820,000 video-confirmed strikes. And in the summer of 2025, for the first time in the history of warfare, a position was stormed and prisoners taken by ground robots and drones with no human soldier present — “the position was taken without a single shot being fired,” the Ukrainian commander told CNN.

Mass armor and massed infantry, the dominant forms of twentieth-century war, turned out to be vulnerable to swarms of machines costing a few hundred to a few thousand dollars apiece. And that kind of war runs on compute, at several layers: on the drone itself,

for real-time perception and guidance under jamming — edge inference, the same architecture as a self-driving car; in field command posts, for sensor fusion and swarm coordination; in rear-area data centers, for training the perception models and grinding through the sensor data; and across the alliance, for shared intelligence and the rapid distribution of improvements.

The side with all four layers, integrated and at scale, has a decisive edge. That is a genuine argument for compute infrastructure, and it has nothing to do with consumer chatbots.

Which is exactly the catch. The military-relevant compute is a *subset* — heavy on edge silicon, focused training, and intelligence processing. It is not heavy on consumer inference, enterprise SaaS features, or slop generation, and those are most of what's actually being built. Treating the whole buildout as a national-security project, when the bulk of it is consumer products subsidized to grab market share, is a category error — a convenient one for the operators, who get to drape commercial infrastructure in the flag, and an expensive one for the communities hosting facilities that mostly serve neither defense nor them.

Worse, the misallocation is self-defeating on its own terms: every dollar poured into hyperscale slop capacity is a dollar not spent on the drone silicon and ISR compute the actual contest requires. And the right architecture — distributed, edge-heavy, resilient — is more survivable in wartime than centralized hyperscale, which builds the enemy a concentrated target. A buildout sized for the security argument would look quite different from the one being built in its name.

The deeper point survives even if you set the war aside, and it is the one the “compute superiority” slogan tends to obscure. Strip the strategic argument down and what is actually scarce is not data centers. It is the ability to design and fabricate advanced chips — the fabs, the lithography, the packaging, the high-bandwidth memory, the supply chain that runs from a TSMC cleanroom in Taiwan through ASML in the Netherlands to the lines that turn finished

wafers into parts. That capability is genuinely contested and genuinely worth a national effort to hold; it is what the CHIPS Act is actually about. A data center is a building full of those chips. The chips are the asset. The building is just where some of them happen to sit.

And the demand for the chips is durable in a way the demand for the buildings is not, because the future that needs them is mostly an *edge* future. The bet worth making is not that hundreds of millions of people will route their daily lives through frontier APIs in distant data centers. It is that AI moves into the physical world — robots in factories and, in time, homes; autonomous vehicles; on-device assistants; intelligence embedded in machines that act in real time and cannot wait on a round trip to Virginia. The timing is the real uncertainty there, not the direction. If that world arrives, it runs on staggering quantities of silicon, almost all of it sitting on the device and drawing the device's own power.

So the honest version of the strategic case is narrower, and more defensible, than the one usually made: build the fabs, secure the supply chain, fund the research, keep the capacity to make the best chips in the world on the right side of the contest. That is the part of “compute superiority” worth defending. The hyperscale data center buildout is riding the same flag without the same warrant — sized for a centralized-inference present that the edge is already migrating away from, not for the distributed future the strategic argument itself points toward.

At What Cost, and to Whom

The two arguments don't cancel. They're answers to different questions. The financial one is about *how the buildout is financed and who collects* — and it points to late-cycle extraction that will hurt late investors and host communities. The geopolitical one is about *whether to build at all* — and it says yes, some of it, urgently. Both can be entirely true at the same time. What they jointly imply is not *stop*. It's *build it differently*: site it away from homes, make it pay for the

grid it needs, aim the capital at the uses that actually matter, and stop pretending the current pattern is the only one available.

But that's the prescription. The question this chapter promised to answer is the cost, and the cost is real and itemizable. It falls on the host communities from the earlier chapters — the noise, the air, the bills, the cascade. On the ratepayers across PJM and beyond, tens of billions a year in transfers from households to the hyperscalers whose infrastructure their rates subsidize. On the late investors who bought SpaceX at \$135, or will buy the next IPO, or hold the bonds.

On everything the capital and political attention *didn't* go to — the housing, the healthcare, the climate infrastructure, the other problems no one had bandwidth for. On the air and water and land around the facilities. And on democratic legitimacy itself — the slow corrosion, across thousands of towns, of the belief that ordinary people can shape the decisions that reshape their lives.

The benefits are real too — the genuine capabilities, the security, the productivity gains. But they flow to a different set of people than the costs do. They accrue to the operators, to the investors who exit at the right moment, to the users paying subsidized prices for premium tools, to the broad economy capturing the macro gains. The costs accrue to the communities, the ratepayers, the late investors, and the institutions whose legitimacy this pattern erodes.

So the buildout is not, on its face, bad for the world. Summed across everyone, it's probably net positive — the technology likely justifies some scale of it, the security benefit is real, the useful applications will last. The problem is the *distribution*. The people bearing the costs are not the people getting the benefits, they are not being compensated, and the legal machinery that might have ensured a fairer split was, as the neighborhood chapters showed, deliberately dismantled. An outcome can be positive in aggregate and still be one that decent people, asked honestly in advance, would not have chosen — and the people paying for it were never asked.

It helps to put two faces on it. A woman in Brittany Heights watches her home lose its value to a hum that never stops. A retired

teacher in Phoenix watches a third of her 401(k) evaporate when the bubble resolves. A child in Boxtown grows up breathing turbine exhaust. Different people, different states, different mechanisms — and the same buildout, with the benefits flowing somewhere none of them live. They will never know they're characters in the same story.

Summed across all of them — the host communities, the retirees, the ratepayers, the late investors — what you get is a generational transfer of wealth from a broad base of ordinary Americans to a narrow tier of operators, founders, and the intermediaries who structured the deals. Economists have a clinical phrase for this: late-stage extractive financialization. It isn't a slur; it's a description, and it puts the 2026 AI buildout in a lineage with the railroad bubbles of the 1870s, the dotcom bust, and the mortgage crisis of the 2000s. The specifics will differ. The structure — overbuild, inflate, write down, transfer — tends to rhyme.

Nobody Knows

I've laid out the case as I see it, and I want to be honest about the size of what I don't know, because it sits underneath everything above.

I don't know whether this is a bubble. I don't know whether the demand projections will hold. I don't know whether the host communities will recover or hollow out, whether the affected retirees will lose five percent of their savings or fifty, whether the figures at the center of this era will read in hindsight as visionaries or cautionary tales or — most likely — as complicated people whose real contributions and real damage were inseparable. Nobody knows these things in 2026. Anyone who tells you otherwise is selling something.

What *is* operating, right now, is a narrative — a story about why the buildout is necessary, why the costs are acceptable, why the line goes up. It has many authors whose interests happen to align: the hyperscalers, the chipmakers, the financial press, the defense establishment, the investment industry, the trade groups. It coheres not

because anyone coordinates it but because everyone benefits from it cohering. And like the dotcom narrative and the housing narrative before it, it will keep doing its job long after the facts stop cooperating, because the job — mobilizing capital, justifying permissiveness, holding valuations up — is still needed.

The narrative asks to be swallowed whole, because it hangs together better as a whole than its parts do separately. The honest move is to decline, and take it claim by claim.

How many drones has Ukraine used? Probably accurate; the evidence is open. Are several hundred new data centers needed by 2030? Partly true, partly inflated, resting on demand projections the narrative itself shapes. Is the buildout essential to national security? Partly — a real subset is; most of it isn't, and the "all of it" version is doing PR work. Are data centers good for the neighborhoods that host them? No — the evidence on that is overwhelming, and the claim survives only as narrative. Are the operators visionaries? Partly, and partly hagiography; they've built real things and done things that, in other industries, would be called predatory. Each claim has to be weighed on its own evidence. Some survive. Some don't.

Never Bet Against America

There's a line Warren Buffett has repeated across sixty years in public life: *Never bet against America*. He has closed crisis-year letters with it. It anchored him through the 1973 oil shock, the 1987 crash, the dotcom collapse, and 2008. And it has, every single time, eventually been right. The country absorbed the losses. The institutions restructured. The technology compounded. The next generation built on what the last one overbuilt — often on the very infrastructure the bubble had left behind.

I want to end on that line, because at the level of the country and the century, I think it's correct. The AI buildout will almost certainly leave real, durable infrastructure. The technology will almost certainly produce real, durable gains. The security objectives will, in

some form, be served. The country will, in some form, absorb the losses and move on — writing down what needs writing down, restructuring what needs restructuring, the way it always has. In 2080, the historians will likely look back on the 2020s as a period of enormous transformation, at enormous cost, that America came through in characteristic fashion.

But the qualifications matter, because they're where the people reading this actually live, and the load-bearing word in Buffett's line is *eventually*.

The country recovers eventually. The recoveries take time. The Dow's 1929 peak was not recovered, in real inflation-adjusted terms, until 1954 — twenty-five years. The Nasdaq's 2000 peak took fifteen years. Japan's Nikkei peaked in 1989 and did not return to that level until 2024 — thirty-five years; an investor who bought the top in their forties was in their late seventies before their money came back, if they lived to see it. National resilience is real across generations. It is not a promise to any particular person on any particular timeline.

And "America" is a unit measured at the scale of the whole. The country compounds; specific people absorb specific losses, and many of them never personally recover. The retiree drawing down a crushed portfolio in 2009 was a real person whose story didn't recover on the country's schedule. That's the gap Buffett's line quietly contains: the recovery's timing and the original capital's fate are two different stories.

The infrastructure survives the wreck — the dark fiber of 1999 carried the streaming boom a decade later, the over-laid railroads of the 1870s carried the next century's industry — but in every case the original equity holders were wiped out and the eventual owners bought the assets for cents on the dollar from the ruins. The infrastructure survived. The capital that built it did not.

So Buffett is right, and he is the kind of right that should be read carefully. The country recovers, the infrastructure persists, the long arc bends upward. The affected retirees will not, in the main, be

made whole by that arc. The host communities will not be repaired by the eventual repurposing of the facilities that hollowed them out. The bill is being presented, in 2026, to people who did not order what's being charged to them — and the long-run resilience of the American economy, real as it is, is not the same thing as a refund.



CHAPTER 15

The Other Ledger

What All This Infrastructure Is Actually For



Earlier chapters laid out what chips and data centers cost — the energy, the water, the mining, the disposal. I tried to be honest about each of those, including the parts that are genuinely bad.

This chapter is going to do the same thing in the opposite direction. What does all this infrastructure actually *do*? What gets made, possible, better, cheaper, safer, or more widely available because the chips exist? Because that list is also real, and leaving it out would be as dishonest as pretending the costs don't exist.

I want to warn you about the tonal trap in this chapter, because it's the one that every tech-friendly book falls into and which I am going to try to avoid. The trap is that once you start listing good things, you start *selling*. The writing gets warm, enthusiastic, a little breathless. Cool inventions! World-changing breakthroughs! Better future! By the end of the chapter, you are not reading a ledger; you are reading a brochure.

I will try not to do that. This chapter is a ledger. The items on it are real, some of them are extraordinary, and none of them erase what was in the preceding chapters. You should finish this chapter feeling that the costs *and* the benefits are both large and both real, and that the interesting questions — which the rest of the book is

about — are how to get more of the second while paying less of the first.

Fair?



Medicine

Let me start with the thing that is, in raw human terms, the biggest item on the ledger.

A CT scanner is a chip-dependent machine. So is an MRI, an ultrasound, a PET scanner, a modern blood analyzer, a pulse oximeter, a continuous glucose monitor, a pacemaker, an insulin pump, a cochlear implant, a hospital telemetry system, and an electronic health record. If you walked into a 1970 hospital and a 2026 hospital, the biggest visible difference would not be the architecture; it would be that the 2026 hospital has computers and screens attached to essentially everything, and the 1970 hospital does not.

The consequences of this, averaged across the population, are gigantic.

Breast cancer mortality in the United States has dropped by roughly 40% since 1990. A major reason is mammography screening and, more recently, AI-assisted mammogram reading that catches tumors earlier and with fewer false positives than human radiologists working alone. Earlier detection means earlier treatment, which means more survival.

Heart attack mortality has dropped by more than half since 1980. Chip-enabled improvements are not the only cause — drugs matter, and surgery matters, and anti-smoking campaigns matter — but continuous ECG monitoring, automated defibrillators in public spaces (all of them chip-driven), and better imaging that lets cardiologists see exactly where an artery is blocked are all part of the story.

Childhood cancer survival has gone from around 10% in 1960 to around 85% today. Almost no one over 50 has not lost a relative to a

cancer that today's children would survive. The reasons are many, but precision imaging, targeted radiation (which requires chip-controlled delivery), and treatment protocols generated from enormous patient databases are major pieces of the pie.

In 2020, a team at DeepMind — the AI research lab owned by Google — released a system called **AlphaFold** that predicts the three-dimensional shape of proteins from their genetic sequences. This doesn't sound dramatic unless you know the backstory, which is that predicting protein shape had been one of the great unsolved problems of biology for fifty years. Tens of thousands of researchers had spent entire careers on individual proteins. AlphaFold solved the general problem, published its predicted structures for essentially every known protein — more than 200 million of them — and released the database for free. This has accelerated drug discovery, enzyme design, and basic biology research in ways that are still playing out, but already visible. AlphaFold runs on GPUs in a data center. Without the chips, the solution does not exist.

None of this means medicine is perfect or fair. Care is still distributed unevenly, access is still gated by money, and plenty of medical AI is overhyped and oversold. But on average, across the population, the chip-dependent part of medicine is saving an enormous number of lives, and the ones it is saving include, almost certainly, people you love.



The Environment (Yes, Really)

Here is the item that surprises people.

The biggest single reason we *know* climate change is happening is that we have been monitoring the Earth, continuously and in detail, from satellites, for fifty years. The satellites have chips in them. The data comes back to ground stations that have chips in them. The data is processed in data centers that have chips in them. The climate models that interpret the data run in data centers on GPUs.

Every major climate insight of the last fifty years — the rising global temperature, the melting Arctic ice, the accelerating sea level rise, the Amazon deforestation rates, the ocean circulation changes — came out of this infrastructure. The scientific case for climate change, which is what led to every policy response and every activist movement about it, *exists* because the computational infrastructure exists to gather and process the data.

If you are a 14-year-old who is worried about climate change, you are worried about it in significant part because chips and data centers are telling us, moment by moment, what the atmosphere and the oceans and the forests are doing. Take those chips away and you don't have a calmer, cleaner planet. You have the same planet, doing the same bad things, and almost no one knows.

This is worth sitting with, because a lot of environmental discourse has a weird implicit assumption that if we could somehow roll back industrial technology, we would be fine. We would not be fine. We would just be in the same trouble with less information about it.

There is more. The integration of wind and solar into the electrical grid — which is the actual mechanism by which the world's electricity is getting cleaner — is only possible because of chip-dependent grid management. Wind and solar are variable. A grid running on wind and solar has to *balance supply and demand, second by second*, across hundreds of different sources, billions of different loads, and thousands of miles of transmission lines. This is impossible by hand. It requires continuous computation. Every renewable-heavy grid in the world — Denmark, the UK, California, Texas — runs on chip-based grid management software that didn't exist thirty years ago.

Precision agriculture, which reduces fertilizer and water use in farming by a significant fraction, depends on chips (in soil sensors, GPS systems, automated irrigation controllers). Satellite monitoring of illegal deforestation and overfishing, which is one of the only ways we enforce environmental law across borders, depends on

chips. Modeling the climate impact of different policy choices — which is how we figure out what to do — depends on enormous data-center simulations running on GPUs.

The infrastructure the book has been criticizing is, at the same time, the infrastructure making environmental policy possible. This is genuinely the case. It deserves to be stated clearly, even in a book that takes the environmental costs seriously.



Safety

The list of things that don't kill you because a chip is watching is long.

Car airbags deploy because a small chip — an accelerometer, the same technology as the one in your phone — detects a crash and fires the airbag in a few thousandths of a second. Anti-lock brakes work because a chip monitors each wheel's speed fifty times a second and adjusts brake pressure to prevent skidding. Electronic stability control, now mandatory in most countries, works the same way. Traffic light timing, which reduces accidents and congestion, is handled by chip-based controllers that coordinate across whole cities. The lines on modern highways that buzz when you drift across them — chip-activated. Commercial aircraft autopilot, which has made flying safer per mile than almost any other form of transportation — chip-dependent.

Earthquake early warning systems, which exist in Japan, Mexico, California, and a few other places, use networks of chip-based seismometers to detect the initial shock wave of an earthquake and send warnings to cities that haven't felt it yet. The warnings arrive seconds to tens of seconds before the shaking. It sounds small. It has saved a lot of lives. Trains are stopped automatically. Surgeons pause. People brace.

Tsunami warning systems, wildfire detection, hurricane tracking, flood prediction — all of these run on chip-based sensor networks

feeding into data centers that run prediction models. Weather forecasting accuracy has roughly doubled every 30 years since 1950, and the doubling is driven primarily by compute: bigger models, more data, more detailed simulations of the atmosphere. A five-day forecast in 2026 is about as accurate as a one-day forecast was in 1970. A reliable three-day hurricane track forecast in 2026 means cities evacuate with enough lead time to actually get people out. In 1970 it didn't, and hurricanes killed far more people.

Hospital monitoring systems alarm when a patient's vitals cross a threshold. Continuous glucose monitors for diabetics warn of low blood sugar before symptoms. Home smoke detectors that networked into chip-based hubs now detect slow-smoldering fires hours earlier than the old standalone ones.

Each of these is small on its own. Together, they are one of the largest quiet revolutions in how humans die, which is to say: we die less, and later, and less randomly, than our grandparents did, and a surprising fraction of the reason is that chips are constantly watching for the things that would hurt us.



Access

There is a story about the second half of the 20th century that goes like this: rich countries invented things, poor countries eventually got them, but always later and always more expensively, because the infrastructure of the rich countries was what the things depended on.

In the 21st century, that story has broken in a specific place, which is mobile technology. A poor country does not need to build out the twentieth-century infrastructure (landline phones, bank branches, physical retail, physical libraries) to get most of the benefits. It can skip straight to the twenty-first. And it has.

In Kenya, most adults have a bank account — not because Kenya built out a bank branch network, but because Kenya pioneered

mobile payments through a system called **M-Pesa**, running on chips in cell phones. People in rural Kenya who previously had no way to save money or transfer it to family in other villages now have, essentially, a bank in their pocket. The consequences have been enormous; multiple studies have shown measurable reductions in poverty tied to M-Pesa adoption.

In India, Aadhaar — a national biometric ID system — has given a billion-plus people formal identification for the first time in their lives. Aadhaar has plenty of problems (privacy concerns, exclusion errors, misuse), and critics have documented them thoroughly. It has also given hundreds of millions of poor people the ability to receive government benefits, open bank accounts, and access services they could not previously access.

Across sub-Saharan Africa, millions of children learn to read on cheap tablets running free educational apps. The tablets have chips made in fabs those children will never see, loaded with software written by engineers they will never meet, distributing curriculum prepared by teachers they will never talk to. The children learn anyway. The outcomes, as measured by standardized literacy tests, are not as good as a great classroom teacher, and they are much better than no classroom teacher, which is the actual alternative in a lot of places.

Accessibility technology for people with disabilities has, in the last 20 years, transitioned from “specialized, expensive, often bad” to “built into every phone and laptop.” Screen readers for the blind, captions for the deaf, dictation software for people who can’t type, eye-tracking input for people who can’t use their hands, hearing aids that connect directly to phones and are much cheaper than the old specialized ones, navigation systems for people with visual impairments. These are not specialized products anymore; they are features you can turn on in your phone. A blind person with a 2026 smartphone has access to a set of capabilities that a blind person in 1990 could not have dreamed of.

Translation has gone, in the last decade, from “often garbled machine translation, useful for the gist” to “good enough that two people speaking different languages can have a real conversation through a phone.” This is a small-looking change with surprisingly large consequences. Travel is different. Immigration is different. Scientific collaboration across languages is different. A Kenyan farmer and a Finnish engineer can now, routinely, read each other’s text in their own languages, in real time.

These are goods. Some of them are extraordinary goods. None of them are perfect. All of them depend, at the bottom, on the chip-and-data-center infrastructure this book is about.



Knowledge

The stock of accessible human knowledge has grown, over the last thirty years, more than in any comparable period in human history.

Before 1995, if you wanted to learn something, you needed to be near a library that had a book about it, or near a person who knew it, or enrolled in a school that taught it. All three of those gates excluded enormous numbers of people. A smart child in rural Indonesia, in 1985, had essentially no path from “curiosity about how the human immune system works” to “reliable information about how the human immune system works.” A smart child in rural Indonesia in 2026 has Wikipedia in Indonesian, plus YouTube explainer videos, plus, increasingly, AI tutors that can explain the material at any level of depth the child wants.

The degree to which this has flattened access to information is hard to state without sounding like a brochure. So let me state it bluntly: never before in human history has so much of what humanity knows been accessible to so many humans, regardless of where they live, how much money their family has, or what schools they can attend. The gap between a rich-country education and a poor-

country education is still real, but it is smaller — for the motivated student — than it has been at any previous point.

AI tutoring is still early, and the evidence on whether it actually helps kids learn is mixed. But it is getting better quickly, and the underlying technology — good AI, available over the internet, accessible from a cheap phone — is already in the hands of several billion people. The outcomes five or ten years from now are likely to be substantially better than they are today, if the technology develops even approximately as expected.

None of this solves unequal education by itself. The child who wakes up without breakfast, in a home where no one expects her to go to college, still faces almost everything that's hard about being that child. But the information gate — the specific barrier of “I cannot find out what I want to know” — is lower than it has been in any previous generation, and the primary reason is the chip-and-data-center stack.



Science

The scientific enterprise is, increasingly, a computational enterprise.

The first black hole image, released in 2019, was produced by a globe-spanning radio telescope array whose data was integrated and processed by data centers. The detection of gravitational waves by LIGO in 2015 required computational analysis of signals a thousand times weaker than the noise. The James Webb Space Telescope's images depend on enormous amounts of processing before they become the pictures you see. The Human Genome Project, which took 13 years and \$3 billion to produce the first human genome in 2003, now takes a day and about \$200 per person, because sequencing machines have gotten better and the software has gotten enormously better.

Drug discovery is increasingly AI-assisted. The initial stage — identifying which molecules might work against a given disease —

is being massively sped up by AI models that can screen millions of candidate molecules in the time a traditional lab could test a few hundred. Whether this translates into a faster pipeline for approved drugs is still being worked out (late-stage trials are still slow and human-dependent), but the early-stage productivity gain is real and documented.

Materials science, climate science, high-energy physics, cosmology, neuroscience, condensed matter physics, genomics, proteomics, epidemiology — every one of these fields has been rebuilt, over the last twenty years, around computational methods that weren't possible before. Some of them have seen order-of-magnitude increases in the rate of new discoveries as a result.

The boring version of this story is: science is getting faster. The less-boring version is: science is making discoveries right now that would have been impossible at any previous point in human history, and this is happening across essentially every field, and the cause is the same cause — chips got fast enough, and data centers got big enough, that problems that used to be intractable are now tractable.

If you enjoy the fruits of modern medicine, modern engineering, modern clean energy, and modern agriculture, you are enjoying the downstream effects of this acceleration. The downstream effects will keep coming, for your whole life, assuming we don't wreck the upstream.



And the Buildout Itself

Everything on the ledger so far is about chips and computing in general — the long, accumulated good of fifty years of getting better at building and using them. But the rest of this book is about something more specific and more recent: the enormous, expensive, controversial buildout of AI data centers happening right now. It would be easy to read the cost chapters and conclude that *that* buildout, specifically, is all cost and no benefit. It isn't, and honesty requires

putting its real upside on the ledger too — even though, as the chapters ahead will argue at length, the benefit is almost certainly smaller than the scale of what is being built.

Some of the good is already here. The same AI models that produce the slop the later chapters will complain about also do genuinely useful work. They have measurably sped up software development — and not at the margin. A large enterprise software system that used to take hundreds of engineers most of a decade can now be specified, generated, and tested with a quantity of compute that costs, in raw terms, almost nothing. The expensive parts of building software — knowing what to build, integrating it, verifying it works — remain stubbornly human, but the part that used to be slow and costly, the writing of the code itself, has collapsed in price by a factor that is genuinely hard to overstate. Whatever else the buildout has produced, it has produced that, and it is a real economic good that will compound for decades.

Some of the good is in the assistive applications that the “Access” section above only began to describe. A blind person can now point a phone at a room and hear it described. Two people with no language in common can hold a real conversation through a translation app that, five years ago, would have produced garble. A student anywhere can ask an AI tutor to explain a hard idea at exactly her level, as many times as she needs, without embarrassment or judgment. These are not the headline uses of AI, and they do not generate the revenue that justifies the buildout to its investors, but they are real, and for the specific people who use them they are not small.

And some of the good is in the use case the chapters ahead will hold up as the clearest model of what compute used well actually looks like: self-driving. After decades of effort, autonomous vehicles have reached the point where, in the cities where they operate, they crash less often than human drivers. The compute behind that — heavy training in the data center, fast inference on the car itself — is genuinely saving lives, and will save more as it spreads. It is the

clearest existing proof that the infrastructure this book criticizes can, pointed at the right problem, do something unambiguously worth doing.

I want to be careful not to let this section do more than it should, because the rest of the book is going to spend a great deal of effort arguing that the buildout is substantially larger than these genuine goods require. Both things are true at once, and holding them together is the entire point. The buildout is producing real value — in software, in science, in assistance, in safety. It is also being built well ahead of that value, on a bet that the value will grow to fill the capacity. The benefit is real. The question the rest of the book asks is not whether the benefit exists, but whether it is large enough, and well-enough distributed, to justify what is being spent and who is being asked to pay for it.



What This Chapter Doesn't Say

I want to close by being clear about what this chapter is *not* arguing.

It is not arguing that technology is net-good and therefore we should stop worrying about the costs. You have just spent three chapters on the costs. They are real. The energy, the water, the mining, the e-waste, the attention economy, the concentrations of power that come with concentrations of infrastructure — all of it is real, and none of it is magically cancelled out by the benefits on this ledger.

It is not arguing that every new technology is good. Some technologies are bad and should be banned or regulated. Some technologies are mixed and need careful stewardship. Some technologies are so transformative that we can't even yet tell what they'll turn out to be.

It is not arguing that you should be grateful. Gratitude is a strange emotion to direct at infrastructure. The people who built the infrastructure are not asking for your gratitude; they are asking to

be paid, which is different. Your relationship with the infrastructure is simpler than gratitude: you live inside it, it shapes your life, and you have some say (small but non-zero) in how it evolves.

What the chapter is arguing is this. The infrastructure you have been worried about — the chips, the fabs, the data centers, the AI models, the whole stack — is, among other things, the infrastructure that feeds you, cures you, warns you, educates you, and tells you the planet is in trouble. The cost ledger is real. The benefits ledger is also real. Neither one erases the other.

The interesting question — the grown-up question, which is the question this book has been building toward — is not whether to have the infrastructure. It's how to shape it. Who decides what it does, who pays for it, who benefits, who bears the costs, what it's allowed to be used for, what it's not. Those decisions are being made now, by specific people, in specific institutions, through specific processes.

The rest of this book is about that question, applied to the buildout happening right now: who lives next to it, how much of it we actually need, what it would look like if it were built well, who pays the bill, and who decides.



A note on sources: the medical statistics are from the CDC, NIH, and the American Cancer Society. The AlphaFold figures are from DeepMind's published papers and the EBI AlphaFold Protein Structure Database. The weather forecasting accuracy figures are from the European Centre for Medium-Range Weather Forecasts (ECMWF). The M-Pesa and Aadhaar impact studies are from academic work; William Jack and Tavneet Suri's work on M-Pesa poverty reduction is a good starting point if you want to follow up. The claims about AI-accelerated software development and the self-driving safety record are developed, with their caveats, in the buildout chapters that follow. Where specific percentages appear, I've used the middle of credible estimates; individual studies vary.



CHAPTER 16

The Attention Chapter

Why the Scarce Resource Isn't What You Think It Is



Here is something I noticed recently, talking to a friend. When she throws away food, she feels it. The plate is heavy. She can see the rice she didn't eat, the half of a sandwich she let go cold. There is a whole human ritual around scraping a plate into a garbage can — the weight, the sound, the slight guilt. Food waste is embodied. You cannot pretend it didn't happen, because there is literal, physical evidence in your hand.

Now think about what happens when you ask an AI a question you don't really need to ask.

Nothing. The tokens stream. You glance at the answer. You close the tab. There is no weight, no sound, no scraping. The electricity was spent in a building five states away, by a chip you will never see, routed through wires and switches you cannot picture. The “waste,” if there was any, is completely invisible to you. There is no plate to scrape.

This is the first half of a really important asymmetry. Food waste is visible and embodied. Intelligence waste — if that's even the right word — is not. Which means the feedback loop that makes humans frugal with food is, for computation, absent.

But here is where it gets interesting. Because the second half of the asymmetry cuts the other way, and once you see it, the whole argument about AI and attention looks different.

The Bottleneck Isn't Supply

Stop for a second and think about how much information you can actually absorb in a day.

You have sixteen waking hours, give or take. You have two eyes that can track one thing at a time. You have two ears, and while you can hear several things at once, you can only really parse one voice at a time. Your working memory holds maybe seven items, plus or minus two. Your reading speed tops out somewhere around 300 words per minute, and that's if you're actually comprehending rather than just scanning. By 10 PM you are tired, and by midnight you are basically useless.

In a full day of intense effort, you might genuinely, deeply absorb — the kind of absorbing where you could explain it to someone else a week later — maybe 20,000 to 30,000 words of new, substantive content. Call it the length of a short book, or a long article, or a serious conversation.

That's your ceiling. That's how much intelligence, in the form of ideas and information, you can actually consume per day, no matter how much is available.

And here is the thing: you are already saturated. Before AI existed, you were already getting more information than you could absorb — from school, from books, from the internet, from people talking to you, from videos, from your phone. The bottleneck in your life is not the supply of intelligence. The bottleneck is you.

This is important, because it means the economics of intelligence are very different from the economics of food.

Food has diminishing returns to the eater. One sandwich: hungry no more. Two sandwiches: full. Three: sick. Four: actively harmful. This is why personal frugality with food is self-enforcing — your

body tells you to stop. Waste comes from producing too much, not from consuming too much.

Intelligence works the opposite way. Supply can be essentially infinite and cheap. But your capacity to receive is fixed, and already fully spoken for. What changes when more intelligence becomes available is not how much you consume — that number is roughly constant — but what you consume instead of what you used to consume.

Displacement, Not Addition

Let me make this concrete.

Say you have a homework question about photosynthesis. Before AI, you would have done some combination of: asked a parent, asked a friend, looked it up in your textbook, done a Google search and clicked through three or four results until one made sense, watched a YouTube video, or given up and guessed. Let's say, conservatively, this used to take you 20 minutes and involve reading roughly 3,000 words of mediocre internet content, watching five minutes of a video that was 30% ads, and maybe sending two text messages.

Now you ask an AI. It takes you three minutes. You read 400 words that were actually tailored to your question.

In terms of total energy cost: the old version involved your laptop for 20 minutes (17 Wh), your phone for a few minutes (2 Wh), the data-center energy to serve you the Google results and the YouTube video (maybe 30 Wh of streaming), and the collective energy of all the ad-tech that tracked you through the process (surprisingly large, maybe another 10 Wh). Total: around 60 Wh.

The new version: your laptop for three minutes (3 Wh), and the AI query (maybe 5 Wh of data-center energy). Total: around 8 Wh.

The AI version used less energy than the non-AI version. Not because AI is magical, but because the old version involved a lot of wasted motion — pages loaded you didn't read, ads you skipped, results that didn't answer your question. When you replaced that

with a more targeted interaction, the total energy went down, even though the new interaction was with a supposedly “expensive” AI.

This is the thing almost nobody factors in when they talk about AI’s energy use. AI is not being added to the top of your existing information diet. It is replacing parts of it, and the parts it replaces were not free.

Now, this is not always true. Sometimes AI adds a new thing that wasn’t happening before — you ask it to summarize something you would never have read, or generate a picture you would never have commissioned, or brainstorm ideas you wouldn’t have bothered to think about. That’s real, additional compute, not replacement. And sometimes AI is just lazier and worse than the thing it replaces — you ask it a question you could have answered yourself, and you get an answer you don’t actually absorb, and you’ve used a little electricity to generate a small amount of nothing.

But the blanket claim that “AI adds to the world’s energy use” assumes that nothing gets displaced. Almost always, something does.

The Actually Scarce Resource

So if the bottleneck is you — your attention, your time, your capacity to absorb — then the question “am I wasting AI?” is actually less interesting than the question “am I wasting myself?”

The electricity for a ChatGPT query costs maybe a tenth of a cent. Your time costs vastly more. Your attention — the stuff you use to understand things, make decisions, notice when people you love are sad — is the most precious resource you have, and nobody makes more of it.

If you ask AI a lazy question and skim the answer, the kilowatt-hours were not the waste. You were the waste. The electricity will be replaced by the next spin of a wind turbine. The ten minutes of your life and the erosion of your attention span? Those don’t come back.

This is the frame shift this chapter is trying to make.

The old frame: *compute is expensive, I should use less of it to save the planet.*

The new frame: *compute is cheap and getting cheaper, but my attention is finite and my life is short, so I should use AI well.*

“Well” means different things in different situations, but most of them cash out in one idea: ask better questions, and actually metabolize the answers.

What Good Consumption Looks Like

If you wanted to be a good consumer of intelligence — AI or otherwise — here are the things that seem to matter.

Know what you’re trying to learn. The difference between a useless AI conversation and a useful one is almost always in the question, not the answer. Vague questions get vague answers. “Tell me about the French Revolution” is a worse prompt than “I don’t understand why the Reign of Terror happened after the moderate phase; what changed?” The second one produces an answer you can actually absorb, because you already have a hook in your brain for it to hang on. The first produces a wall of text that slides off.

Read the answer. Really read it. Don’t skim. If the answer is long enough to scroll, you are not engaging with it; you are comforting yourself with the presence of information. Close the tab. Ask a shorter question next time.

Notice when you’re using AI to avoid thinking. This is the part that will make adults look sanctimonious, so I’ll try not to. But honestly: there is a difference between “I’m stuck, AI help me see the next step” and “I don’t want to think about this, AI do it for me.” The first builds your brain. The second atrophies it. You can feel the difference if you pay attention. The problem is that paying attention is exactly the skill you’re giving up.

Use AI for the parts where it’s actually better than you. Summarizing long documents, translating languages, generating first drafts you then edit, explaining concepts in different ways until one

sticks, doing calculations, checking your reasoning. These are things AI is great at and you can verify.

Don't use AI for the parts where it's worse. Judgment calls about your own life. Emotional conversations that need a real human. Final decisions about things that matter. Creative work you actually want to be yours. Learning that requires struggle — the struggle is the point.

Maintain the ability to do things without AI. This is worth stating clearly because nobody else will. There will be times in your life when the internet doesn't work, or the AI is down, or you're somewhere it's inappropriate, or you just want to know that you can. Maintain the muscle. Write things by hand sometimes. Do math in your head sometimes. Read a whole book without notes sometimes. The fact that you can do these things without help is part of what makes you you, and it is worth keeping even when there are easier options.

The Environmental Connection

Now, finally, the loop back to where this book started.

The reader who started this book was worried about data centers and the planet. That worry is real and legitimate, and the earlier chapters laid out the actual numbers.

But I want to say something that probably wouldn't have made sense in the energy chapter. Most of what is environmentally wrong with AI isn't the AI. It's the attention economy it's plugged into.

The biggest use of data centers, by a wide margin, is not AI. It is streaming video, social media feeds, and advertising technology. TikTok, Instagram, YouTube, and the ad-tech that follows you around the internet use vastly more data-center energy than every AI chatbot combined, and they do it in a way specifically designed to use as much of your attention as possible.

The algorithm that served you the upsetting video about data centers using too much water was itself hosted on a data center. The ads that interrupted the video were served by a data center. The app

that notified you about the video was running on a data center. The comments you scrolled through were stored in a data center. The hours of your life that the app extracted — those were extracted using a lot of electricity, and they were extracted from you, which is the only non-replenishable resource in the whole system.

If you actually wanted to reduce the environmental impact of your personal computing, the single highest-leverage thing you could do is not to stop using AI. It is to use less social media. A day of intensive AI use, even for a power user, might involve 50 queries and a few hundred Wh of data-center energy. A day of scrolling short-form video involves closer to 50,000 Wh of streamed content served by data centers, plus all the ad infrastructure, plus the manufacturing amortization of the device you're doing it on.

But nobody is going to tell you to stop scrolling, because the feed is what you use to be told things.

This is not a moralizing speech about social media — there are enough of those, and you have probably tuned them all out. This is an observation about where the energy actually goes, and where the attention actually goes, and why the two are related.

You are not being asked to care about data-center energy because data centers are the biggest environmental problem. (They are not. Cars, buildings, agriculture, and industrial heat are all vastly bigger.) You are being asked to care about data-center energy because caring about data-center energy feels actionable, and caring about industrial heat processes does not. Someone has gotten you to worry about the thing that's most visibly tech-shaped, which is probably not the thing that would most matter if you fixed it.

This is also true of the relationship between you and your attention. The apps that extract your attention do so because your attention is the product they sell to advertisers. The electricity cost of running the app is a rounding error in that business model. The cost you pay, in hours of your life and in the shape of your own thinking, is not a rounding error. It is the entire point.

What This Chapter Is Actually About

I started this chapter with my friend's observation about food waste. Here is the version of her insight that goes one more step.

When food waste is visible, people feel bad about it, and they use less food. Fine. But food is the thing you physically eat, and it becomes you, and there is a limit. When something is infinite in supply and invisible in waste — like compute, like digital content, like the information streaming at you from every screen — the question “am I wasting it?” stops being useful. You cannot run out of it. It does not become you. The feedback loop is gone.

What replaces that feedback loop is a different question, which you have to ask yourself because nothing in the environment will ask it for you: *Is this making me better?*

Is this conversation with an AI making me smarter, or am I using it to avoid being smart? Is this video teaching me something, or am I watching it to keep from noticing that I'm bored? Is this article changing how I think, or am I nodding along because agreeing feels good? Is my life becoming more the life I want, or is it becoming the life that's optimized to keep me on the app?

These are hard questions. Most people do not ask them well, which is part of why the world has gotten the answers wrong in a lot of specific ways. But they are the real questions, and the environmental ones are — honestly — downstream of them.

If you are careful with your own attention, the environmental questions mostly take care of themselves. Not because your individual queries matter much either way, but because careful people build careful institutions, and careful institutions make better decisions about where data centers go and what they run on and who pays for them.

If you are not careful with your own attention — if you let it be spent for you, in tiny increments, by algorithms that do not love you — then no amount of worrying about data centers is going to fix the part of the problem that is actually wrecking things.

The chip your phone is running right now is doing one of two things. It is helping you be more of who you want to be, or it is helping someone else turn you into a product. You get to decide which one, every time you pick it up.

That is the real question this book is about.



*A note on sources: the food-waste-and-invisible-compute framing in this chapter originated in a conversation with a friend who saw it clearly before I did. The data-center-use breakdowns — streaming vs. social vs. AI — are from the International Energy Agency and from utility disclosures in markets where they're required. The attention-economy framing draws on Tim Wu's *The Attention Merchants* and Tristan Harris's work at the Center for Humane Technology, both worth reading if this chapter interested you.*



CHAPTER 17

Who Decides

Where the Decisions Actually Get Made



If you drew an organization chart of the people who decide what the chip and AI industry does — what gets built, where, powered by what, sold to whom, under what restrictions — it would fit on a single page. That is one of the strangest facts about the modern world. The infrastructure that carries most of the global economy, runs most scientific research, and touches the daily life of nearly everyone alive is shaped by the decisions of maybe a few hundred people, most of whom have never been elected to anything.

The sentence is true, and on its own it's misleading, because it suggests that *naming* those people is the same as understanding how the decisions get made. It isn't. Knowing the CEOs is the civics-class version of the story. The version you need — if you ever want to predict an outcome, or change one — is about the *mechanisms* the decisions travel through: the legal instruments, the financial structures, the approval processes that determine which choices reach which people, and which people have any say at all.

It's worth having the cast in your head first, though, because they're real and they recur.

A handful of companies make the chips. TSMC in Taiwan makes most of the world's advanced ones; Samsung in South Korea is the nearest rival; Intel in the United States is trying to climb back after

falling behind. Between them they produce essentially every leading-edge logic chip on Earth, which means their chief executives have more influence over what computing looks like in 2030 than all but a few elected officials anywhere.

None of them can operate without ASML, the Dutch company that makes the extreme-ultraviolet lithography machines the advanced fabs depend on — there's a waiting list, and who gets the next machines is its own quiet decision. Nvidia, under Jensen Huang, designs the GPUs most AI training runs on, one of the most powerful commercial positions in any industry on the planet.

Above the silicon sit the buyers and builders. The hyperscalers — Microsoft, Google, Amazon, Meta — own most of the world's cloud and AI infrastructure, and each has committed to capital-spending programs in the tens or hundreds of billions a year, a scale that used to belong to governments. The frontier labs — OpenAI, Anthropic, Google DeepMind, xAI, a few others — actually train the models; they're small in headcount and enormous in influence. And a newer camp, described earlier in the book, is betting on a different architecture entirely: Fei-Fei Li's World Labs, Yann LeCun's AMI Labs, Bezos and Bajaj's Project Prometheus.

Then the governments. The U.S. Commerce Department's Bureau of Industry and Security writes the export controls. The FTC and DOJ handle antitrust. The EU's AI Act is the world's most detailed AI regulation, with reach beyond Europe's borders. China is running a multi-year, multi-hundred-billion-dollar effort to build a chip industry free of American and Dutch technology. The Dutch, Taiwanese, South Korean, Japanese, Emirati, and Saudi governments all hold instruments in play.

And — easy to miss, and the most important for most of what this book is about — the local governments. County commissions approve zoning. Utilities negotiate power contracts. Municipalities grant or withhold water. State utility commissions decide rate cases. A county commissioner in rural Virginia has more immediate say over whether a data center gets built there than the President of the

United States does, and that decision gets made in a meeting ordinary people can attend and almost never do.

All of these actors hold exactly the formal roles the textbook assigns them. None of them are villains; most are thoughtful people who'll discuss the implications of their work at length, in public. The point isn't that they're bad. It's that the way their decisions actually *become outcomes* runs on mechanisms the textbook doesn't name — and that the actors themselves tend to discuss only in carefully chosen rooms.

The Civics Version, and Where It Breaks

The textbook account is clean. In a democracy with a market economy, companies make commercial decisions under executives accountable to shareholders; governments make the rules under officials accountable to voters; courts settle disputes; the press informs the public; the public votes; the cycle turns. Messy in detail, coherent in shape.

What the account leaves out is that, for the decisions that matter most in this industry, the substantive choices are made before any of those public processes begin — and the public processes mostly ratify what's already settled. The clearest way to see it is to ask, for any given decision, a simple pair of questions: who was in the room when it was actually made, and who only found out afterward?

The answers rhyme across every mechanism, so it's worth walking the main ones.

The Secrecy

The first room is the one sealed by a non-disclosure agreement, and it's the single most consequential decision-making instrument in the industry. The earlier chapter on legal tactics described how NDAs work in data center siting; the thing to see here is what signing one does to a *decision-maker*.

When a developer approaches a town, a utility, an agency, or a landowner, the first document the lawyers produce is an NDA, and it binds the other party not to reveal who the developer is, what's proposed, or — often — that any discussion is happening at all. A municipal official who signs it can still vote on the eventual project. What the official can no longer do, until the formal announcement, is tell constituents what's being weighed, or ask whether they'd support it, or even confirm that talks are underway. By the time the gag lifts, the substantive terms are settled and the window for organizing has closed.

That is the whole mechanism. The decisions get made in rooms the NDA defines, with the participants it limits, on the timeline it sets; the council vote and the public-comment period come afterward, to bless what's already been decided. None of it is illegal — each NDA is a standard contract any competent lawyer would draft. It becomes consequential only when it's run routinely, by well-resourced parties against under-resourced ones, on decisions with large public stakes. When five elected officials across five counties told NBC News in 2025 that the agreements they'd signed barred them from informing their own constituents, they weren't failing their duty. They were the mechanism working exactly as designed — and NBC found that design in the majority of the proposals it reviewed across fourteen states.

The Money Decides

The second room is financial, and the Bill chapter described its plumbing at length. As a *decision-making* system, the shorthand is this: the major strategic choices in this industry are made less on what would produce the best operations than on what the next round of capital needs to believe.

In a mature business, you invest because the projected returns beat the cost of capital. In the AI and chip world of 2026, the dominant question is what story the next financing round will reward — and decisions about which products to launch, which mar-

kets to enter, which compute to commit, get weighed first on whether they reinforce that story.

When the operationally sensible move and the story-reinforcing move conflict — and they often do, because the efficient choice is usually boring and the exciting choice is usually expensive — the pressure from existing investors and the looming next round tends to win. A hyperscaler's headline capex commitment is partly a read on real demand and partly a requirement that the stock stay where the analysts have priced it, blended so the two can't be pulled apart from outside.

Sitting alongside that is the capital itself, which is its own kind of vote. The institutional allocators — the pension funds, index funds, sovereign wealth funds, insurers, endowments — decide every quarter which companies get cheap money and which don't, and that differential determines what each company can build, hire, and buy. In a real sense they're making strategic decisions for the companies they fund, without ever sitting on the board: a hyperscaler can't run a \$40 billion buildout unless the allocators keep supplying the capital structure under it.

They keep supplying it, for buildouts that — as the spine of this book has argued — are running ahead of any clear demand. Partly that's because the allocators are judged against benchmarks now so weighted toward the AI giants that *underweighting* them is the professionally risky move. Partly it's a knowledge problem: the allocators are sophisticated but not specialists, and the information they act on is filtered through the same narrative machinery that shapes the companies' own decisions. They aren't blind. They're just not seeing the situation any more clearly than it's being presented.

And the people whose money it actually is — the retirees, teachers, firefighters whose 401(k)s and pensions are the real source of the capital — are represented only by those intermediaries, several steps removed. Their consent is structural: they joined the retirement system, which bought the index, which weighted toward the AI companies, which deployed the money. It is not informed in any

meaningful sense. Most of them could not name the largest holding in their target-date fund. The mechanism works whether or not they could.

The Approval Gauntlet

The third room — really a sequence of rooms — is the permitting stack, the layered approval process that turns a plan into poured concrete. Each layer has its own officials, its own standard, its own scope: federal environmental review where federal land or money is involved; state review where states have their own version; utility-commission sign-off for grid interconnection; water-authority approval; air permits, where the Clean Air Act's line between "major source" and "temporary equipment" is exactly what the Memphis turbine fight turned on; then local zoning — the layer closest to residents and the one where opposition has historically worked best; then the largely ministerial building permits.

The part worth understanding is that the *order* is the developer's choice, and it's chosen for leverage. Approach a small town first, and it knows that saying no means forfeiting the tax base. Approach the utility first, lock in interconnection, *then* go to the town, and now a no means stranding commitments that are already approved. Sequenced well, the process produces approvals that look procedurally clean while being substantively settled in advance. By the time the most open layer — local zoning, where anyone can show up and speak — is reached, the prior layers have made refusal expensive, and the developer can wave those commitments at the local board.

So procedurally, almost no one is excluded; everyone has standing to comment somewhere. *Effectively*, most of the public is excluded most of the time, because participating across the whole stack takes technical knowledge, legal capacity, time, and persistence that ordinary residents don't have. The gauntlet is open in principle and closed in practice.

And when it fails — when a town learns what's coming too early, or organizes well, or votes no anyway — there's a backup, the legal

weaponry the earlier chapter detailed: the suit against the town, the state pre-emption bill, the eminent-domain claim, the move from local courts where opponents have standing to federal ones where they often don't. Its real power is the *threat*. A town that knows it might be sued for blocking a project thinks twice even on good grounds it would probably win, because the legal fees alone could exceed its budget. The weaponry mostly works by shaping the choices the formal decision-makers face, long before any lawyer files anything.

What It Looks Like in One Place

It's easier to see all of this at once in a single project, so take Saline Township, Michigan, which the Legal Weapons chapter described — this time watching which room each decision was actually made in.

The first decisions happened in a room sealed by confidentiality: a developer, Related Digital, assembling 575 acres of farmland through entities that didn't carry its name, with the officials who knew bound not to say much. The residents weren't in that room. By the time the proposal surfaced, the land was lined up and the terms were set.

The next room was the financial one, and neither the residents nor, really, the local officials were in that either. The project was one node in a national buildout financed by the capital machinery from the Bill chapter; it later emerged the site would be leased to OpenAI and Oracle under the federal "Stargate" program. The decision that there would *be* a data center here had effectively been made several rooms upstream, by parties the township would never meet.

What was left for the public room — the local zoning hearing, the one place residents could actually walk in and speak — was whether to ratify it. They showed up, they spoke, and the board voted four to one against. For about forty-eight hours, it looked like the open room had worked.

Then the backup opened. Related Digital sued, two days later, for “exclusionary zoning.” The township’s lawyers laid out the box: fight and probably drain the budget; fight and likely lose anyway, because the developer could route around local zoning through the University of Michigan; and a \$50 million judgment against another town hanging overhead as a warning. The township settled. The data center is being built.

That is the process end to end — the substantive decisions made in sealed and financial rooms the public never entered, the one open room reduced to ratification, and a legal backstop ready for the moment ratification failed.

It doesn’t always run to completion, and the exceptions are the proof of what the whole thing depends on. In Tucson, the same kind of project — a \$3.6 billion Amazon data center known throughout the process only as “Project Blue,” with local officials under NDA — broke open when a document reached a reporter early enough for the public to organize, and the city council reversed itself and killed it in 2025. The mechanisms in Tucson were identical to the ones in Saline. The only thing that differed was whether the public got into the room while the decision was still open.

The Quieter Rooms

Those are the rooms that produce most of the visible decisions. A few others operate in the background, less dramatically and just as durably.

There’s the standards room: the technical standards for networking, data formats, security, and interoperability set by bodies like the IEEE, IETF, ISO, and ITU, through slow consensus processes almost no one outside them follows. The people who show up to those meetings, and the companies paying them to, quietly shape what every product built on the standards can do.

There’s the research room, which may be the most powerful of all and the least structured. A breakthrough by a few dozen people can move the entire industry: AlphaFold came out of a team of

around thirty, and the paper that launched the modern transformer era had eight authors. What those researchers choose to work on, to publish, to hand to industry or keep in the lab, propagates through everything downstream — and a research community that organizes around safety produces different outcomes than one that organizes around capability alone.

There's the open-source room. Much of the stack — Linux, Python, PyTorch, thousands of other tools — is maintained by loose networks of volunteers and semi-professionals whose choices about what to build and what to abandon ripple through every company that depends on them. The rise of open-weight models — Llama, Mistral, DeepSeek, the Qwen family — is reshaping the competitive structure of AI in ways the closed providers can't fully control.

And there's the journalism room, which is why anyone outside the companies knows any of this. Nearly every specific fact in this book — the grid cost-shifting figures, the NDA pattern, the utilization rates, the turbine counts, the shell companies — came from investigative work, not corporate disclosure. The quality of public understanding depends on whether that work gets funded and published, which makes the collapse of local journalism — private-equity owners, ad revenue dependent on the very industries the papers should scrutinize — a mechanism in its own right, one that weakens the only layer producing visibility into the others.

Advocacy organizations belong here too, at their best and worst: the Southern Environmental Law Center won real results in Memphis by engaging the Clean Air Act's technical details and building an actual case, while plenty of groups run on little more than indignation. The difference between the two is whether they do the legal and technical work, and the ones that do have outsized effect.

Who's Not in Any of the Rooms

Turn the mechanisms inside out and the same names keep coming up as the parties who bear the consequences without a seat.

Residents near proposed infrastructure: the NDA keeps them ignorant while the decisions are live, the permitting stack lets them speak only after the leverage is gone, and the legal weaponry punishes their representatives for organizing well. Ratepayers across an affected utility region: the capital flows in, the grid expansion gets approved, and the costs are spread across them in commission proceedings most never knew were happening — the rate increases earlier chapters tallied, 94 percent in D.C., 74 in Maryland, 73 in Maine, landed on people with no realistic way to participate. The beneficial owners of the institutional capital: their retirement savings fund the buildout through fiduciaries whose incentives may or may not match their own.

A few groups sit even further out. The training-data subjects — anyone who ever wrote anything online — contributed to the models by default, through a collection process that included consent in no meaningful sense; the lawsuits and rules are still catching up to a thing that already happened. The supply-chain workers — the miners, refinery hands, fab technicians, and e-waste processors who make the hardware physically possible — sit behind very different labor laws depending on where they are, and the difference shows up in their working conditions and their life expectancies.

The general public, which polls consistently more cautious about the pace of AI deployment than the companies setting it, has no mechanism keeping pace to turn that caution into a binding constraint. And future generations, who'll live in the world these infrastructure and climate and capability decisions are shaping, are excluded by simple necessity — no one in any of the rooms represents them.

What Follows From This

The reason to lay the mechanisms out plainly isn't despair. It's the opposite. Decisions made in specific rooms, by specific people, under specific procedures, are decisions made under *rules* — and rules are built things, not natural ones. They can be opened, rewrit-

ten, contested. The people can be regulated, pressured, or replaced. None of this machinery is a law of physics; it's constructed, and constructed things are vulnerable to construction.

History bears that out. The environmental movement of the 1970s changed corporate behavior by changing the procedural rules that governed pollution decisions. The civil rights movement changed the law by changing who was in the room. The labor movement changed working conditions by changing the mechanisms through which labor disputes could be raised. None were inevitable; each required sustained attention from people who weren't, at the time, the ones in charge.

What's distinctive about this moment is the combination: the decisions are unusually large, the number of people making them is unusually small, the mechanisms are opaque to most outsiders, and the public is mostly not watching. The mechanisms run on that inattention — the NDA on constituents not asking in time, the financial machinery on no one separating performance from narrative, the permitting stack on residents skipping the early layers where leverage lives.

That's also the seam in them. It is genuinely hard to predict whether attention arrives in time, and the honest answer is that it usually arrives only once the costs are visible enough to force it. But the rooms are named, the procedures are public, and the rules were written by people. The next chapter is about what that leaves open.



CHAPTER 18

What You Can Do That Actually Matters

The Short List, Honestly Sorted by Leverage



Here is the chapter where books like this usually fail.

The writer, having spent two hundred pages laying out how large and complicated the problem is, suddenly needs to hand the reader something to do. So the writer makes a list. *Recycle your electronics. Turn off your devices. Don't buy a new phone every year. Educate yourself. Vote. Reduce, reuse, repair.*

You have read this list before, in twenty other books. You probably already know, without being told, that most of what's on it does almost nothing. And you probably know, because nobody writes this part honestly, that the gap between “what we're told to do” and “what actually matters” is one of the more infuriating parts of being young and trying to care about the world.

This chapter does it differently. I'm going to tell you, as honestly as I can, what's on each shelf — and I'm going to try not to sound more certain than I am, because the honest answer to most of the questions in this chapter is *it depends, and the part that's up to you is yours to figure out.*



Shelf One: Things That Are Pure Symbol

These are the actions that show up most often in lists like this and that matter the least. I want to name them clearly, because the worst thing that can happen to your attention — which is, as the attention chapter argued, the one genuinely scarce thing you have — is that you spend it on things that don't work.

Turning off your Wi-Fi router at night. Saves roughly 50 Wh a day. Over a year, about 18 kWh — roughly what a single long-haul flight burns per passenger in six minutes of cruising. If every person in America did this, it would cut national electricity use by about 0.02%. The action is almost exactly zero.

Unplugging chargers when not in use. Modern chargers draw a fraction of a watt idle. A year of aggressive unplugging saves you, optimistically, the cost of one cup of coffee.

Boycotting AI or specific tech companies. If you personally stopped using ChatGPT tomorrow, the world would have one fewer user out of several hundred million. Your decision would be invisible in the statistics, and the company's choices about what to build and how much power to draw would not change in any detectable way. (This doesn't mean you *shouldn't* stop using something you find troubling. It means the value of doing so, if it has value, is to you, not to the world.)

Switching search engines as a gesture. Moving from Google to DuckDuckGo doesn't change your search's energy use in any meaningful way. It may affect your privacy, which is a separate and valid reason — but the environmental claim is essentially nil.

Social media activism about the issue. A post about data center water use reaches the few people the algorithm shows it to, who mostly already agree with you. The companies are aware of the criticism; they have departments for it. Your post is, at best, a tiny contribution to a chorus whose effect on policy is real but diffuse and slow.

Buying the “eco-friendly” version of a device. Most of a device’s lifetime impact is in its manufacture and disposal, not in which materials are in its case. An “eco” phone with 10% recycled aluminum is maybe 1–3% better over its life — and often costs more, which means you’re buying about 1% of an improvement for about 20% more money.

What unites the category: these actions feel good, take little effort, are easy to talk about, and produce no measurable change.

I’m not telling you to do nothing. I’m telling you that if you do one of these, do it because *you* feel better — not because it’s a contribution. Telling yourself it’s a contribution is the small error that quietly damages your ability to think clearly about what actually is one.



Shelf Two: Things That Matter a Little

These are real. They don’t change the world; they change one person’s footprint by a measurable but modest amount. In aggregate, if a lot of people did them, they’d matter more.

Keep your devices longer. This is the one personal-consumption choice in the whole chapter worth real emphasis. If everyone used their phones for four years instead of two, the total impact of the phone industry would roughly halve — both upstream in mining and manufacturing and downstream in e-waste. Nothing else you do with your own device comes close. The chip is expensive to make; using it twice as long cuts that expense in half. It’s the boring answer, and it’s the right one.

A phone bought in 2026 will, for most people, handle everything you’d ask of it in 2030. Same for laptops. Batteries and screens can be replaced; most of what makes a device feel “old” is software nagging you to upgrade. The nag is designed to take your money. Resist it.

Recycle electronics properly when you do dispose. Find a certified recycler (R2 or e-Stewards in the US; the WEEE system in Europe). Don't put electronics in the trash, and don't let them sit in a drawer forever. The difference between proper recycling and improper disposal is real — it decides whether the materials are recovered in controlled facilities or processed in dangerous conditions somewhere else.

Eat less beef. I know this feels off-topic. It isn't. The embodied cost of your daily diet is much larger than the embodied cost of your daily computing, and beef is disproportionately responsible. If you wanted to make one significant personal environmental choice, the calorie mix of your diet is roughly a hundred times more consequential than anything you do with your devices. For most rich-country kids, the footprint of lunch is larger than the footprint of the phone.

Use less air conditioning, when you can. AC is the largest single energy expense in most American homes and the fastest-growing. A few degrees on the thermostat in summer is worth more, in energy, than your entire computing footprint — genuinely worth doing where it doesn't risk your health. (In Phoenix that safe range is small; in Boston it's larger.)

Pay attention to what you buy. Not in the “consume ethically” sense, which is mostly impossible given how supply chains work, but in the plain sense: don't buy what you don't need, don't throw away what works, don't replace things because of marketing. The consumption you *don't* do is the only kind with a clean conscience.

Useful, all of it. None of it world-changing. If the world needs to change, something else has to happen — and that's the next shelf.



Shelf Three: Things That Matter a Lot

The interesting leverage, for most people, isn't in personal consumption. It's in collective action and in what you make of yourself. The question is which versions of each actually work.

Vote, when you're old enough — and not just for president.

The decisions that shape whether the costs of technology grow or shrink, whether AI gets built carefully or recklessly, whether data centers land in water-stressed places or not, are made by people you can vote for or who are appointed by people you can vote for. The gap between a government that takes these questions seriously and one that doesn't is enormous, and you move it, a little, by voting.

Presidential races get the attention and often matter least for the specific issues in this book. *Local* races are where data center siting, water rights, zoning, and utility decisions get made. A county commissioner seat can decide whether a facility goes in the aquifer-draining spot or the closed-loop-cooled one. Most people skip those elections. You shouldn't.

And there's a kind of election most adults don't know they can vote in. *Public utility commission races* — in the states that elect commissioners rather than appoint them — decide who allocates the rate increases from Chapter 5 between data centers and the households living next to them. Eleven states elect their commissioners. Turnout is often under 20%, which means a small organized group can move the result. If you live in one of those states, that may be the most important vote you cast on anything in this book, and almost no one else is watching it.

Support the people who do the work you can't. Investigative journalism and public-interest legal groups are the actors that actually produce change in this space, and both run on money. When you have income, even a little, a subscription to one outlet that does real reporting, or a donation to one rigorous advocacy or legal organization, is worth more than any personal-consumption

change you could make. Pick carefully — some are rigorous, some are vibes — and I'll name specific ones a little later in the chapter.

Become genuinely good at something real. This is the most important thing on the page, and it's the hardest to feel at fourteen because the payoff is years away. Let me say it as plainly as I can.

The people who actually change how any of this works — who design a more efficient chip, who figure out how to cool a data center without draining an aquifer, who write a regulation that holds up, who report the story that shifts opinion, who price the risk everyone else is ignoring — are people who can *do something*. They earned a skill. They spent years getting good at a specific thing, and now the world pays them for it.

That last part matters more than it sounds. The goal isn't just expertise. It's expertise you can make an honest living from — a craft the world will actually pay for, that produces something you can point to and check.

Here's why I keep coming back to that. The surest protection against becoming a person you wouldn't respect is being able to feed yourself with real work. People who can do something valuable don't have to live off a cause, a brand, or a hustle. People who can't, sometimes do, and that's where the drift starts.

You can watch it happen: someone goes into an "issues" career sincere, learns that the visible part — the posting, the speaking, the being-known-for-caring — pays better and costs less than the actual work, and ten years on they're a personality with opinions and no craft. The cause became the career, and the career became the grift. What keeps the other kind of person honest is that they have a real skill to fall back on, and would rather use it than perform.

So pick something real. It almost doesn't matter which, because a world running on technology needs an enormous range of competence, and very little of it is the stuff that gets talked about.

Whichever of you likes building physical things: there's the whole world of hardware — the chips themselves, the power electronics, the cooling systems, the grid equipment, the batteries. The

people who make these things better have direct leverage over nearly every number in this book, and they're paid well, because the skill is rare and the work is real.

Whichever of you prefers the physical world to the screen: there are the trades. Electricians, HVAC technicians, the people who actually build and run the infrastructure this entire book is about. They're in short supply, and harder to offshore or automate than people like to claim. Someone who can wire a building or service a chiller has more practical leverage over how the physical world works than a hundred people with opinions about it.

Whichever of you likes numbers: there's a quiet, consequential world of people who understand the *money* — the capital structure behind the buildout, how utility rate increases get split between data centers and the people nearby, how retirement savings got concentrated into a handful of companies, how to tell a real bet from a bubble. Most of the public argument about the economics is made by people who can't actually do the math. If you can, you have leverage almost no one else has — the work is just being right about numbers other people would rather not check.

Whichever of you likes science: materials, battery chemistry, the measurement work that tells us whether the industry is doing what it says. Whichever of you likes computing: the architecture work from Chapter 13 — efficient models, edge inference, open-source tools, and the security of all of it. Whichever of you likes finding things out and writing them down: the craft of actual reporting, which is a different thing from the brand of commentary. And whichever of you likes teaching: every one of the people above had to learn from someone.

And whichever of you is drawn to law. There's real, unglamorous legal work here that almost no one does — the cases tedious, the clients under-resourced, the lawyers across the table among the best paid in the country. It's also the work that produces actual outcomes: the Southern Environmental Law Center beat xAI in Memphis because someone read the Clean Air Act's "major source" lan-

guage carefully and applied it to a strategy built to evade it. Environmental, utility, administrative, and municipal law are where most of this lives.

The same drift applies here as anywhere — the track that reins in bad behavior also rewards being *known* for fighting it, and the political-adjacent corners, the attorney-general path most of all, are as much a launching pad for ambition as a public-interest job. If it calls to you, it's a real path; hold it loosely, and watch what the work does to the people a decade ahead of you on it.

The common thread is not the field. It's the shape: get genuinely good at one real thing, good enough that the world will pay you for it, and you'll have both leverage and the freedom to keep your integrity — because you never had to sell it to eat.

One more, and it's the flip side. Some jobs, and some projects inside good companies, are making the world worse. You'll have to decide, eventually, whether to work on them. The easy move is not to think about it and take the paycheck. The right move, usually, is to think about it and sometimes walk away. A lot of competent people quietly declining to work on bad things is one of the few forces that actually reins in corporate behavior — and having a real skill is what makes walking away possible, because you have somewhere else to go.



The Specific Levers The Last Several Chapters Suggest

The middle of this book — the people who live next to the data centers, the demand picture, the architecture that would actually work, the financial bill, the mechanisms by which decisions get made — produces a few concrete levers I want to name plainly. Some you can act on this week; some you could choose as the focus of a career. None of them are the levers you'd have come up with from the standard environmental-impact discussions.

If you live near a proposed or under-construction data center, the window is roughly years zero through three of the Chapter 10 trajectory. Not later. By the time the bills get bad and the for-sale signs start sitting unsold, the people with options have left and taken the capacity to organize with them.

The window is while the neighborhood is still strong enough to fight: find the others who are paying attention, learn the names of the developers and the shell companies, demand the NDAs be disclosed, show up to the early planning-board meetings before most people know what's coming, and refuse the framing that the project is inevitable. Some communities that have done this have stopped projects. Most that haven't have ended up hosting them. I won't pretend the odds are even — they aren't — but the difference between trying early and not trying is most of what determines the outcome.

If your community is being approached, ask whether the boundary-site trick from Chapter 9 is in play. Developers often place a facility right on a jurisdictional line, so the impacts land on residents in one place while the approving body sits in another. If a proposed site is unusually close to a county or municipal line, ask why. The answer usually tells you something.

If you have any say over how your family's savings are invested, look at the concentration. I'm not telling you to sell anything — I'm telling you to look. As of 2026, the default target-date funds that hold most American retirement money are heavily weighted toward the AI-infrastructure companies, by way of being weighted toward the S&P 500, which is itself weighted toward them.

The history of speculative infrastructure buildouts is that the resolution tends to involve large write-downs, and those have historically fallen hardest on the people closest to retirement when the reckoning comes. Whether your exposure is appropriate for your timeline is a financial-planning question, not a political one, and reasonable people answer it differently. I'm not your advisor. I'm

only saying it's a question you can answer in fifteen minutes, and most people haven't asked it.

Evaluate the AI narrative piece by piece, not as a whole. When you read a news story or an industry pronouncement about the buildout, don't ask *is this whole story true*. Ask, for each specific claim, *what's the evidence for this one, and what does the answer tell me about which parts of the bigger story are tracking reality*.

Ukraine using AI-enabled drones at unprecedented scale: well-supported. The claim that the US must build at the current scale for national security: partly true, substantially exaggerated, functionally serving the narrative. The claim that data centers are good for host communities: false in a way the Chapter 8 evidence makes overwhelming. Piece-by-piece is the move that lets you keep thinking after the news cycle moves on, and it's a skill that compounds — the more you do it, the harder it gets for any single narrative to capture you.

If you have technical skills, or are building them, build for the architecture that actually works. Chapter 13 made the case that the right shape for general AI — heavy centralized training, distributed edge inference, occasional updates rather than constant round-trips — is already known and already demonstrated in self-driving, and is being slowed mostly by business-model preference.

Edge AI tooling, smaller and more capable open-source models, better distillation, privacy-preserving local inference, federated systems that actually scale: the tools are public and improving, the funding is thinner than the hyperscale side, and the work is genuinely consequential. (There's also the world-models bet from Chapter 12 — spatial and embodied intelligence rather than language. Earlier-stage, deeper open problems, fewer people, more direct path from competence to contribution. If you're drawn to robotics and the physical world, that's where a lot of the interesting work will be.)

Support the journalism, specifically. The local-news collapse from Chapter 10 — private-equity buyouts, ad-revenue dependence

on the very industries the papers should investigate — has weakened this exactly as it's gotten more important. The reporters doing the real work on siting, the NDA pattern, utility rate cases, and supply chains are mostly at understaffed outlets kept alive by subscribers. The marginal subscription — *Wisconsin Watch* on Beaver Dam, the *Minneapolis Star Tribune* on Minnesota's NDA pattern, ProPublica on supply chains, the capital-markets desks that actually cover the money — does more to constrain bad behavior than most consumption choices ever will.

Support the legal mechanism, specifically. The Southern Environmental Law Center brought the xAI Memphis case. Earthjustice, Public Citizen, the regional Riverkeepers, the state public-interest law clinics, the environmental-justice legal groups in affected regions — these are the entities that turn community organizing into actual consequences. They run on grants and donations, chronically out-resourced by the firms they face, and a single successful case can constrain industry behavior across many later decisions. The leverage per dollar is high.



Shelf Four: Things That Are Huge, If You Want Them To Be

A small number of people, in each generation, end up doing out-sized amounts of good. They're not necessarily smarter or luckier than average. They're usually people who decided — often in their teens or twenties — to take one specific problem seriously and stay with it.

You don't have to be one of them. Most people aren't, and that's fine. But if you might be, it helps to know the shape so you can choose it.

The shape is: pick a problem that matters, go deep over years until you're at the frontier of what's actually known about it — not a casual understanding, a real one — and then work on moving the

frontier. That's how every significant change in any technical field has ever been made. It isn't magic. It's time, attention, and the selection of what to spend both on.

For the issues in this book, the problems that could use more serious people include making AI systems safer and more interpretable, making chips more energy-efficient, cleaning up electronic-waste supply chains, designing data centers that don't harm local water, writing regulations for AI and for siting that actually hold, developing battery chemistries that don't depend on cobalt mined under bad conditions, improving grid integration so more of the computing runs on clean power, and doing the serious financial and economic analysis of the buildout's capital structure that almost no one with real numeracy is doing. There are real people working on each of these right now, and more are needed for all of them.

These aren't vague. The path from "fourteen-year-old who's interested in how chips are made" to "adult who works on chip efficiency" is a real path; people walk it every year. The tools are there, the questions are open, the mentors exist. The difference between "I wish someone would fix this" and "I'm someone who might" is mostly whether you put in the decade it takes to get good. A decade sounds long. You have several of them ahead of you.

Same caution as with law: being *known for caring* about a problem is not the same as being able to fix it. The version that matters is the quiet one — competence, not profile. The work is the point.



What This Chapter Refuses to Pretend

I want to close by saying clearly what this chapter has not said, because it matters.

It has not said that you personally are responsible for solving the environmental or political problems of the tech industry. You aren't. The companies making these decisions are making them; the governments regulating them are regulating them, or not; the infra-

structure is being built; the costs are being imposed. Individual-responsibility rhetoric has, historically, been used to pull attention from structural causes to personal habits while the structural causes kept going. Don't fall for it. Your job is not to be the last line of defense for a system that was never designed to be defended by individuals.

It has not said that everything is fine and tech will fix itself. The problems in this book are real, the solutions aren't inevitable, and good outcomes need specific people to do specific things. Some of those people will be you, eventually. Most will be others. Not all of them will succeed. The future is genuinely open, and pretending it's either doomed or saved is just an excuse not to engage with what's actually happening.

It has not said that you have to care about any of this. You don't. You get to care about whatever you care about. If it's art, music, biology, friendship, God, mathematics — those are legitimate and worth a life. I think these issues are important, and that some of you reading this should give them serious attention. Some of you shouldn't, and that's fine too.

What it has said is this. If you're going to engage, engage seriously. Don't get captured by the performative version. Don't confuse gesture with action. Notice who has leverage, and build the kind of self that can have leverage over decades — which mostly means getting genuinely good at one real thing. Protect your attention. Show up to the rooms where decisions are made. Evaluate claims piece by piece instead of swallowing narratives whole. Notice when you're being sold something. Keep caring about people.

This is the closing chapter of a book about infrastructure, and it's ending with advice that could go in almost any book. That's not an accident. The infrastructure is large and specific; the response to it, for most readers, is to become the kind of person who handles large and specific things well. There's no special move for AI and data centers. There's only the ordinary, difficult business of being a

thoughtful person in a world that rewards thoughtlessness, and doing it anyway, for a long time.



One Last Thing

I started this book by asking you to pick up the phone closest to you, feel its weight, and notice how strange it is that an object of such complexity sits in your pocket unremarked.

Do that one more time now.

Look at it. Since the beginning of this book you've learned what's inside it — the sand it came from, the fabs that made it, the cathedrals where its counterparts run, the energy and water and materials it took, the attention it was built to extract from you, the people who live next to the infrastructure that serves it, the financial structure that paid for the buildout, the small number of people whose decisions shape what it can do and what it does to others. You know more about this object than its designers ever expected its users to care to know.

That knowledge has value. Not because it lets you personally fix the world — it doesn't. Because the world is, increasingly, built of things like this: invisible, concentrated, consequential. And the people who shape that world well, from here out, will be the ones who understand what they're looking at.

You understand what you're looking at now. More than most adults. That's an asset. Get good at something real, and use it.

The book is done.

You are not.

Go.



A note on sources: this chapter reflects the author's judgment more than the others, and you should read it that way. The personal-action leverage estimates are rough — drawn from life-cycle assessments and energy

modeling where those exist, and from reasoning where they don't. The claim that "keep your devices longer" is the highest-leverage personal action is, I believe, well-supported by the literature on device-lifecycle impact; manufacturers' published assessments are a good place to check. The claims about civic action and careers are less quantifiable, and reasonable people disagree about them.

What I'm most confident of is the smallest and least quantifiable claim in the chapter: that real competence, the kind you can make an honest living from, is what gives a person both leverage and the freedom to keep their integrity. I've watched the alternative enough times to trust it.

